

Sector × Quintile Return Book

Average monthly returns of each factor's quintile portfolios, by GICS sector. 269 factors, ordered by in-sample L–S CAGR.

HOW TO READ

■ Q1 ■ Q2 ■ Q3 ■ Q4 ■ Q5

Each cluster of five bars is one sector; bars run Q1 (highest factor score) to Q5 (lowest), dark to light.

■ Grey clusters mark sectors excluded from the factor — the Q1–Q5 spread is negative there.

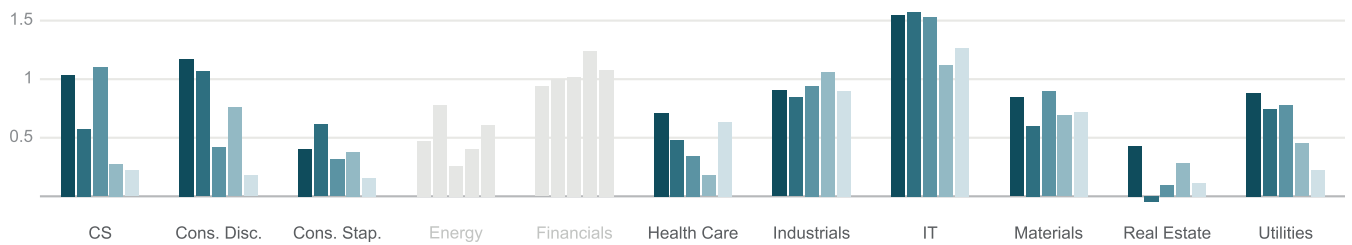
FACTOR STYLES

● Valuation	41	● Price Momentum	55
● Earnings Quality	55	● Size	1
● Analyst Expectations	24	● Historical Growth	55
● Capital Efficiency	28	● Volatility	10

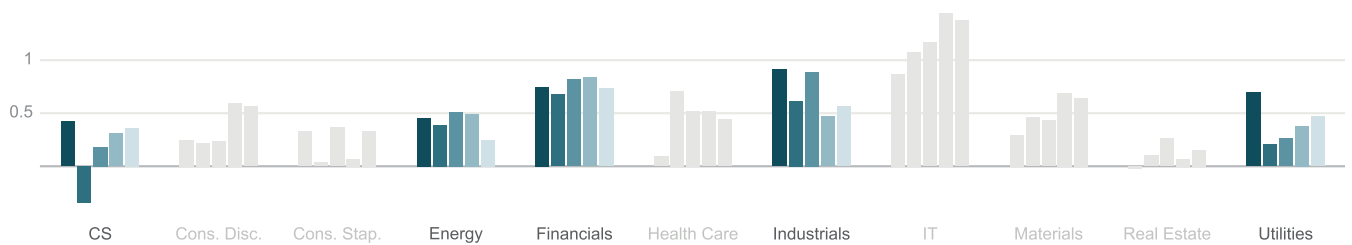
001 In current model portfolio (50 factors)

001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

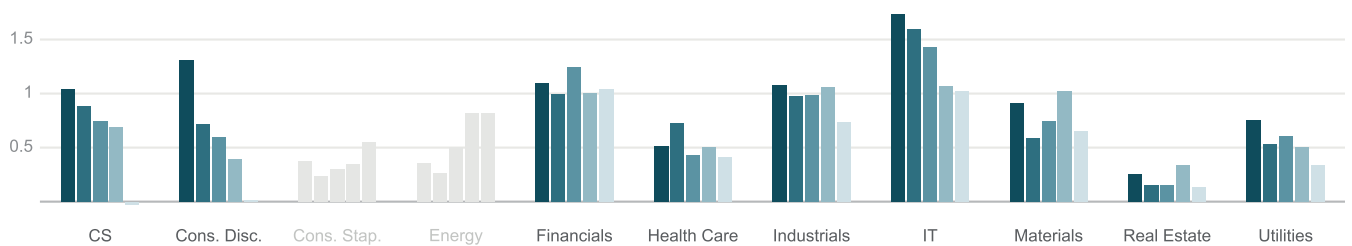
Historical Growth L-S CAGR +17.2%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

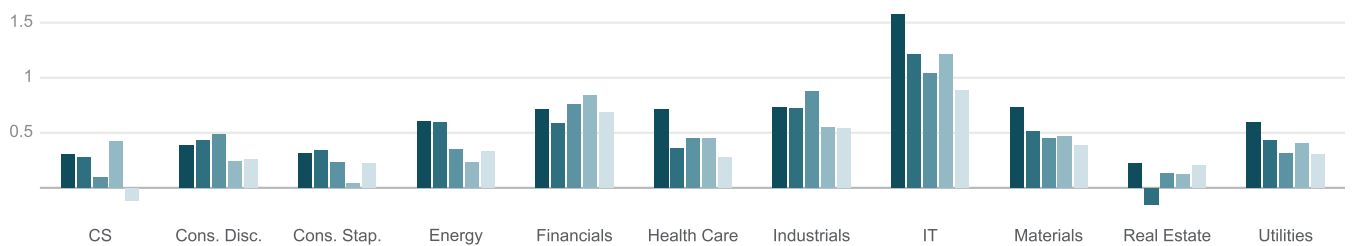
Historical Growth L-S CAGR +15.4%

**003 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

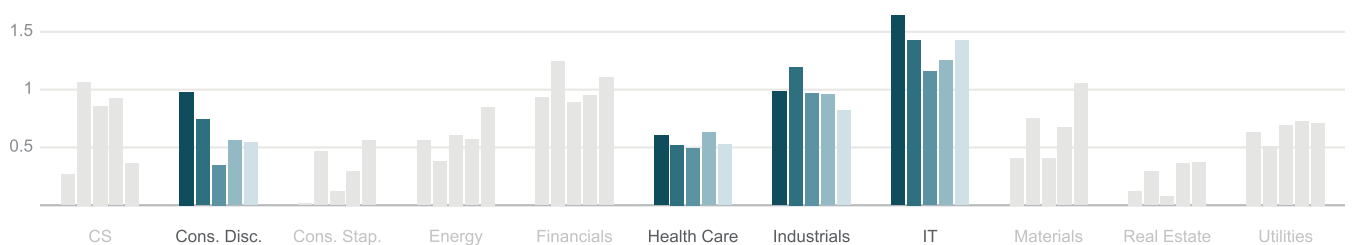
Historical Growth L-S CAGR +12.3%

**004 ● 50 Day to 200 Day Stock Price**

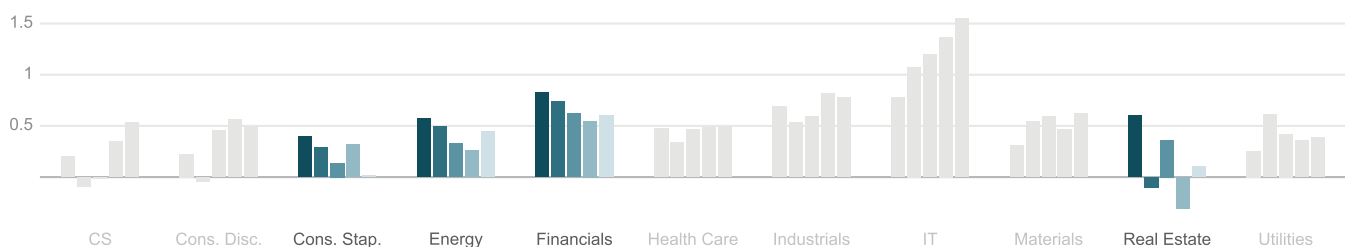
Price Momentum L-S CAGR +12.2%

**005 ● Change of Asset Turnover**

Historical Growth L-S CAGR +11.8%

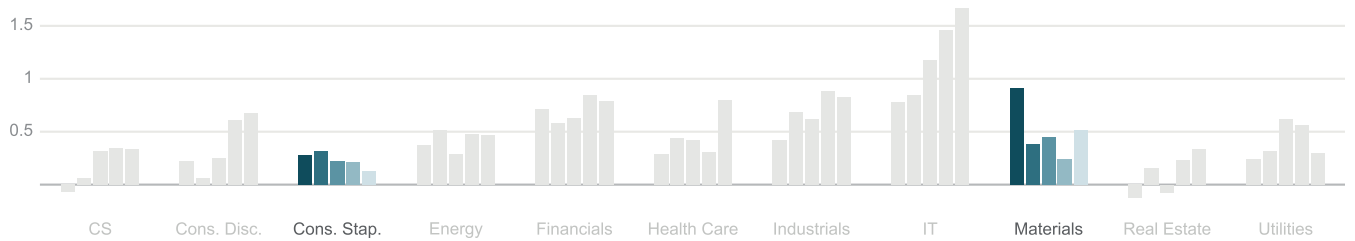
**006 ● Current Liability to Price**

Valuation L-S CAGR +11.7%

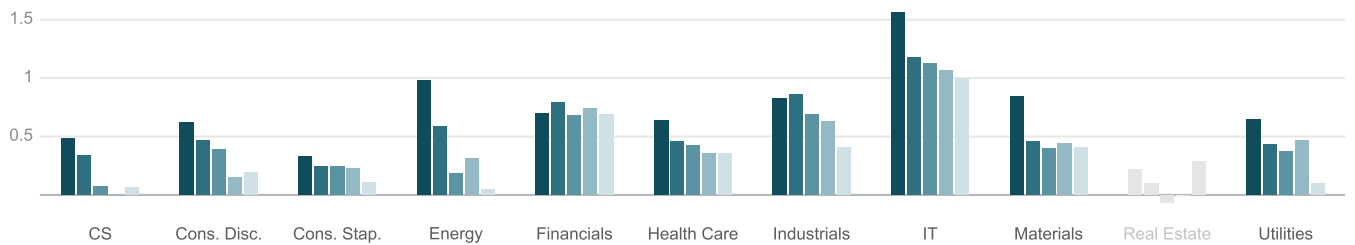


007 ● **Tobin's Q Ratio**

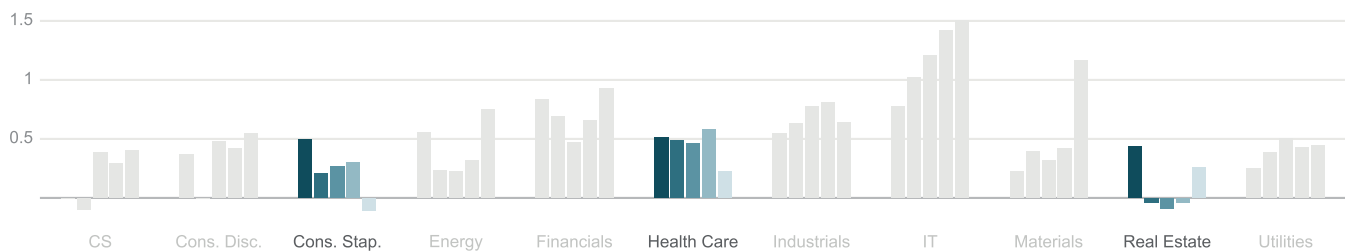
Capital Efficiency L-S CAGR +11.5%

008 ● **39W Lagged Return**

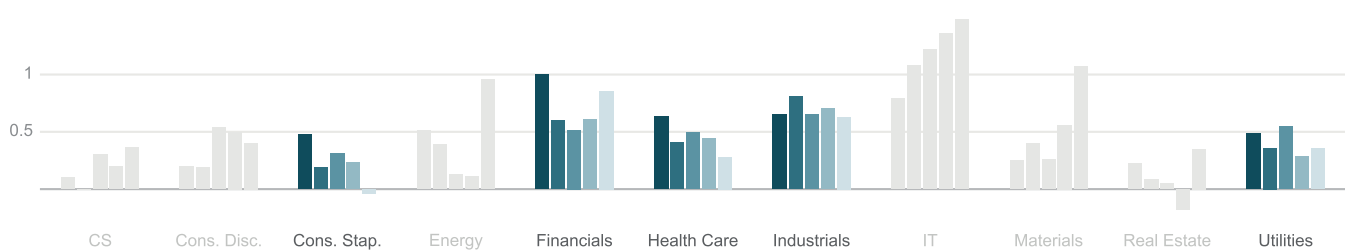
Price Momentum L-S CAGR +11.4%

009 ● **Pretax Income Per Share to Price**

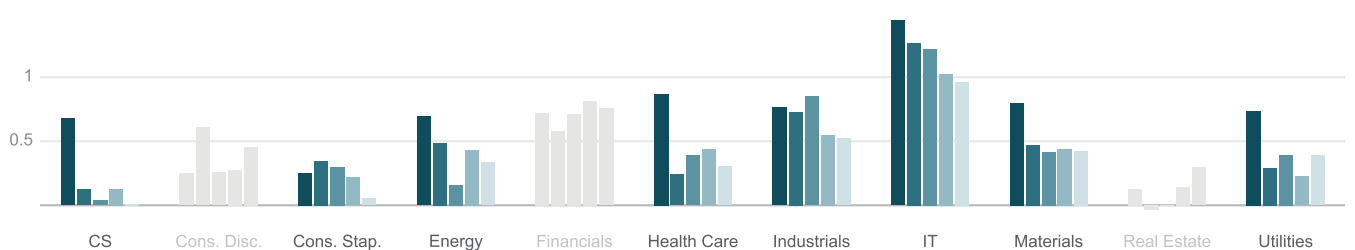
Valuation L-S CAGR +11.3%

010 ● **Operating Earnings to Price Ratio**

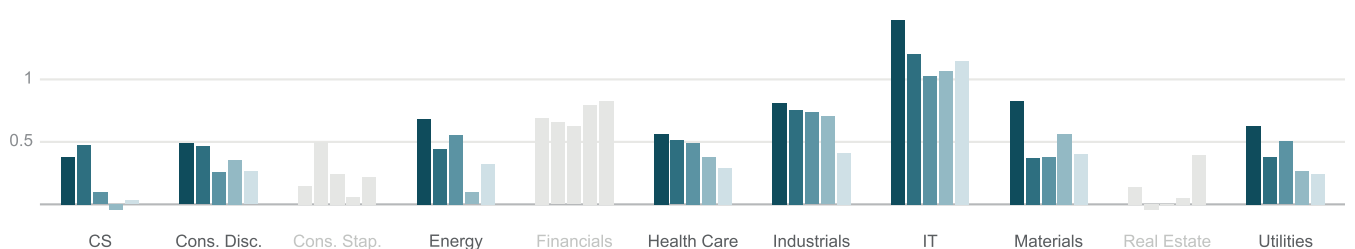
Valuation L-S CAGR +11.2%

011 ● **26W Relative Price Strength**

Price Momentum L-S CAGR +11.1%

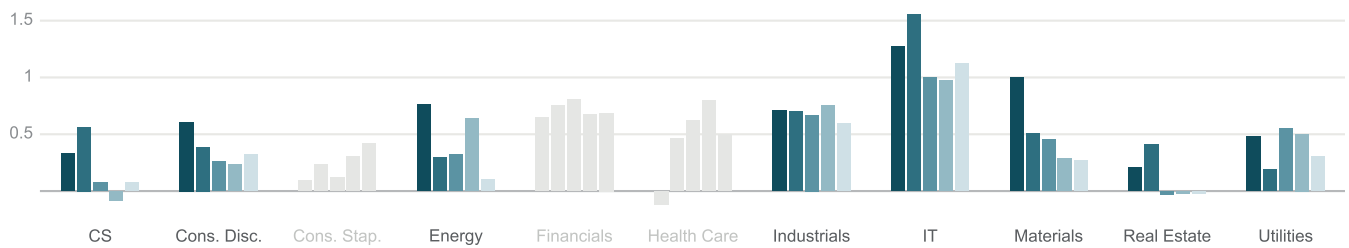
012 ● **9 Month Price Momentum**

Price Momentum L-S CAGR +11.1%



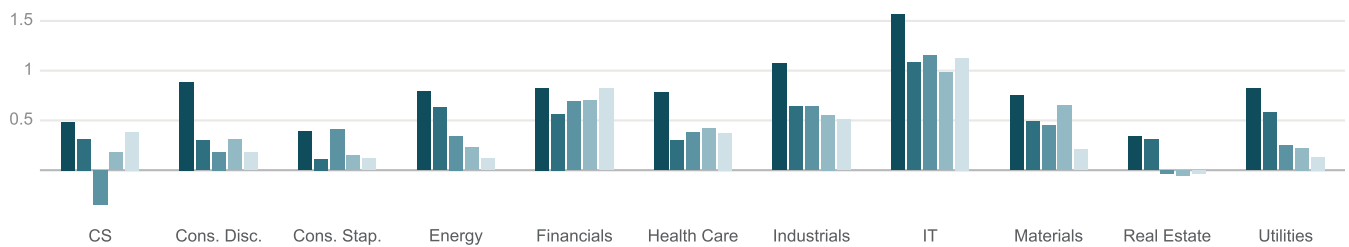
013 ● 12M Realized Price Volatility

Volatility L-S CAGR +11.0%



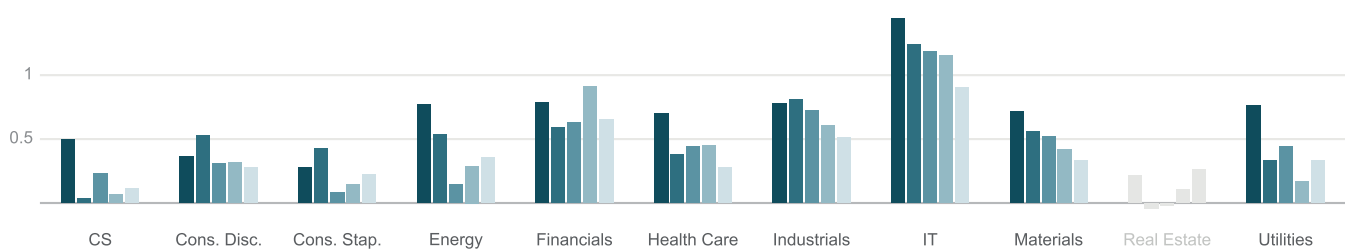
014 ● Slope of 52W Price Trend

Price Momentum L-S CAGR +11.0%



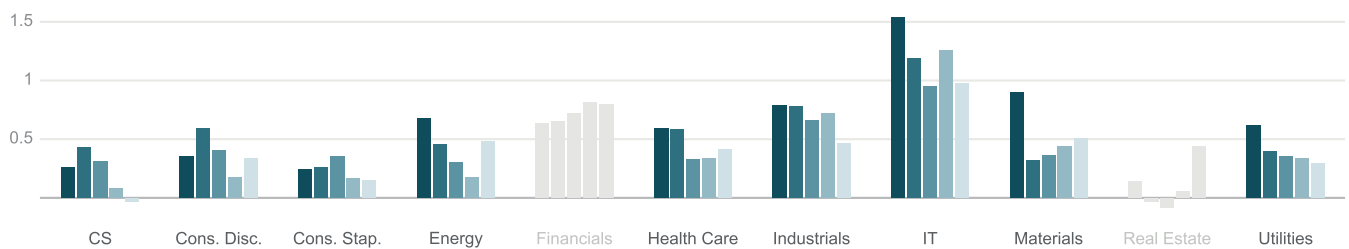
015 ● 15/36 Week Price Ratio

Price Momentum L-S CAGR +10.9%



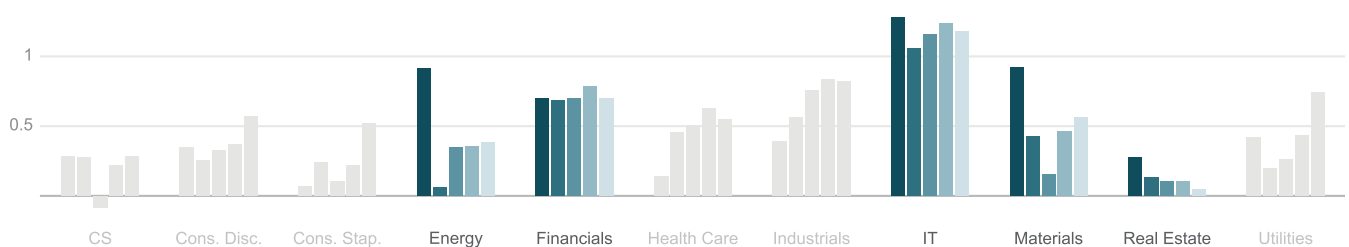
016 ● 4W to 52W Price Oscillator

Price Momentum L-S CAGR +10.8%



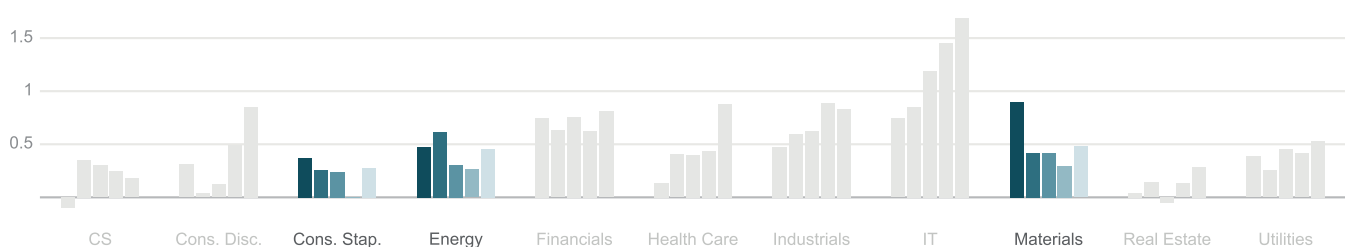
017 ● Analyst Dispersion for FY2 EPS

Analyst Expectations L-S CAGR +10.7%



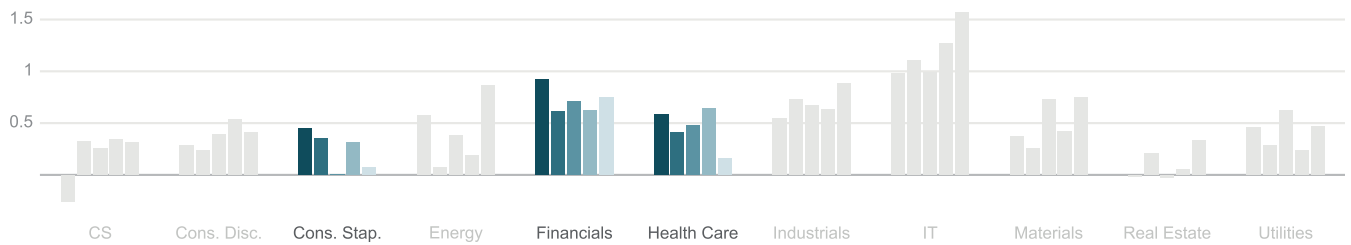
018 ● Book to Price

Valuation L-S CAGR +10.6%



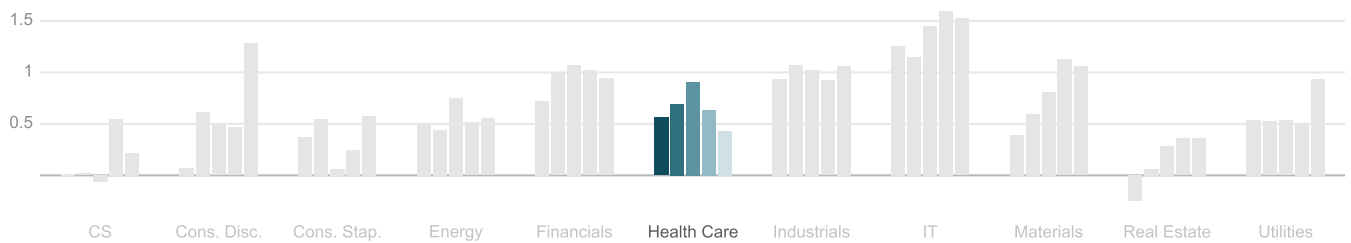
019 ● Normalized EP

Valuation L-S CAGR +10.4%



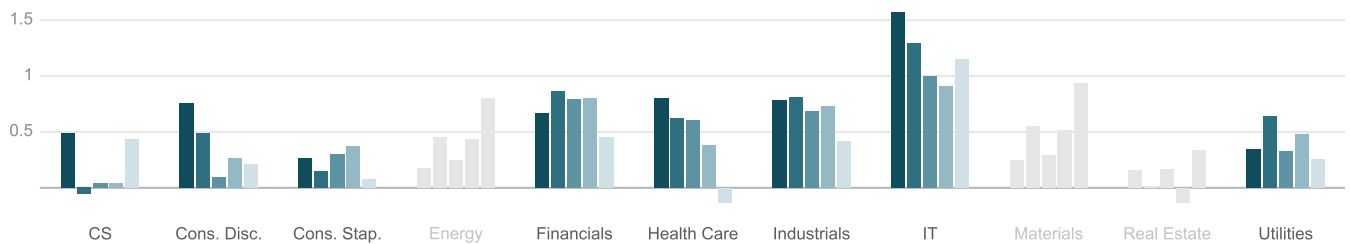
020 ● Indicated Annual Dividends to Price Ratio

Valuation L-S CAGR +10.2%



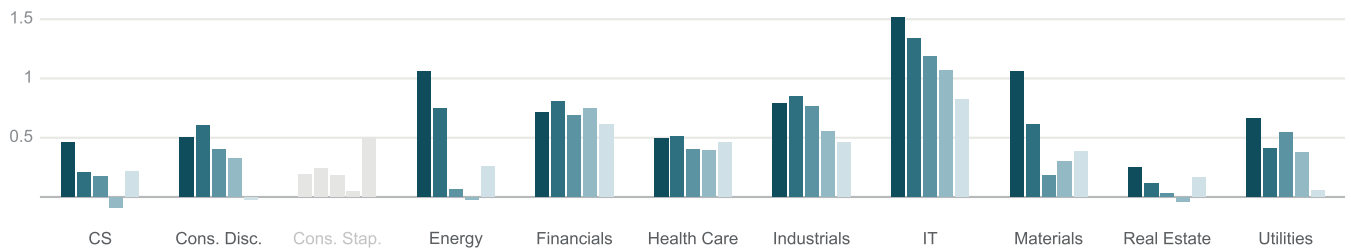
021 ● Operating Earnings to Assets Ratio

Earnings Quality L-S CAGR +10.2%



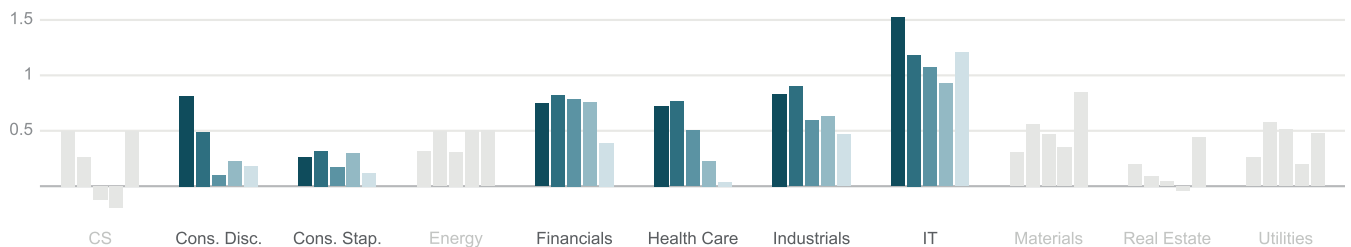
022 ● Percent Above 260-Day Low

Price Momentum L-S CAGR +10.2%



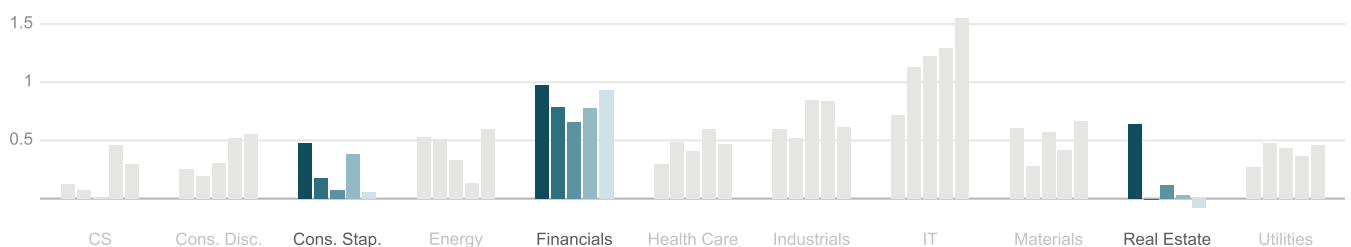
023 ● Return on Assets

Capital Efficiency L-S CAGR +10.1%



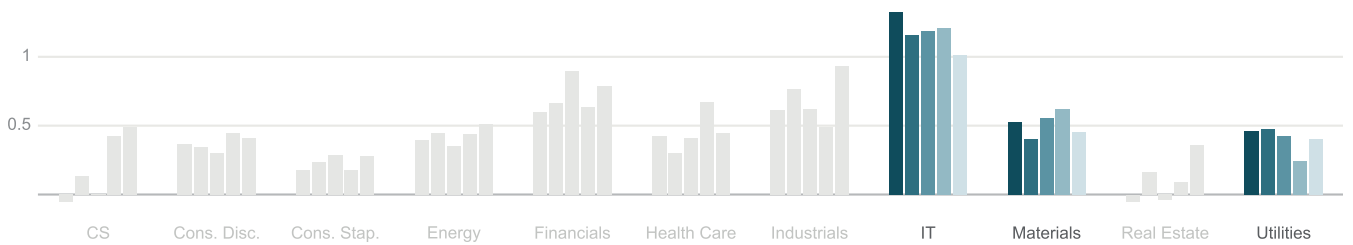
024 ● EBITDA to Price

Valuation L-S CAGR +10.0%

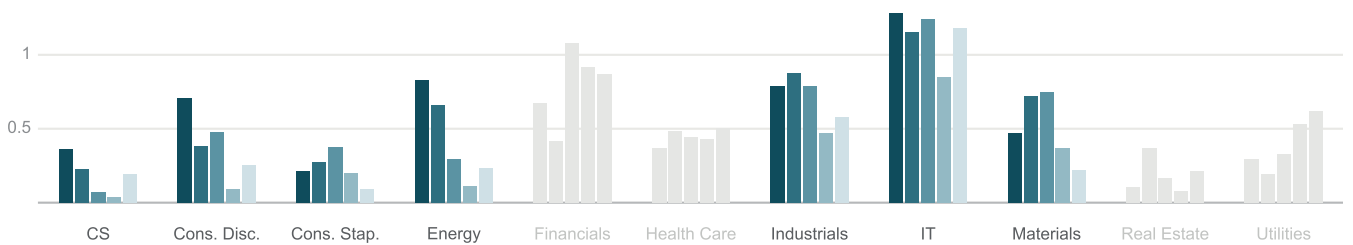


025 ● **Stability of Change in Number of Holders**

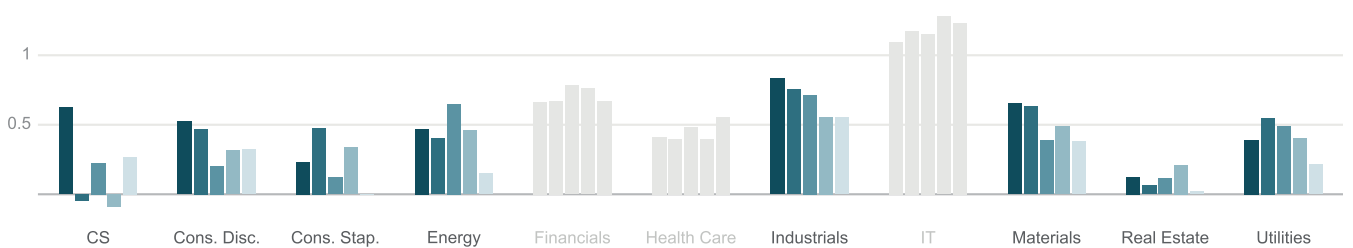
Price Momentum L-S CAGR +10.0%

026 ● **Unexpected Inventory Change**

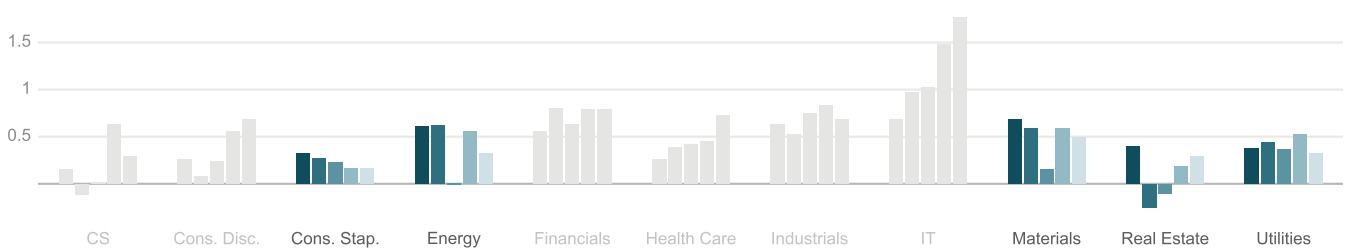
Earnings Quality L-S CAGR +9.9%

027 ● **Log of Total Last Quarter Assets**

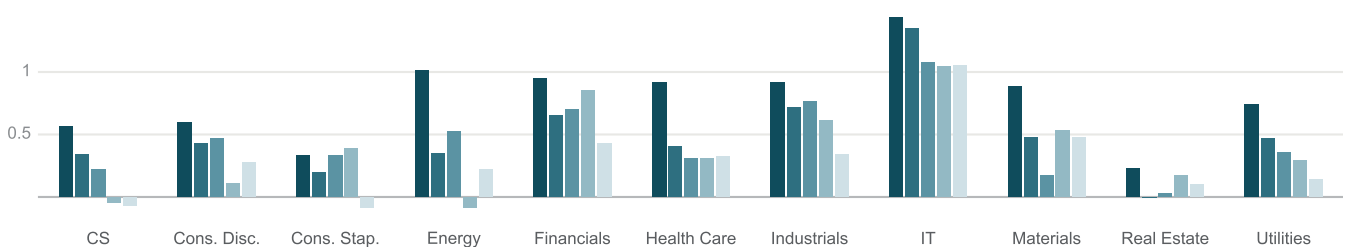
Size L-S CAGR +9.8%

028 ● **Assets to Price Ratio**

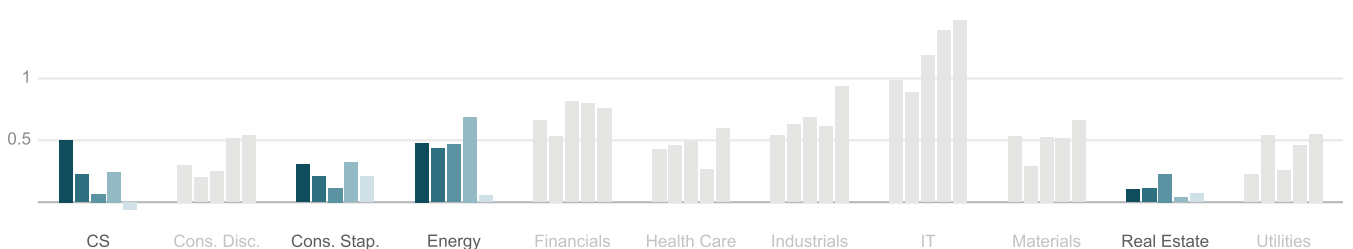
Valuation L-S CAGR +9.8%

029 ● **6M Chg in Target Price**

Analyst Expectations L-S CAGR +9.7%

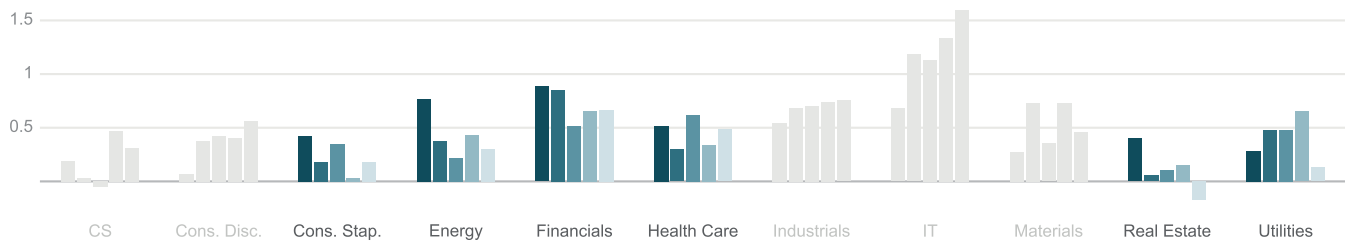
030 ● **Cash to Price**

Valuation L-S CAGR +9.7%



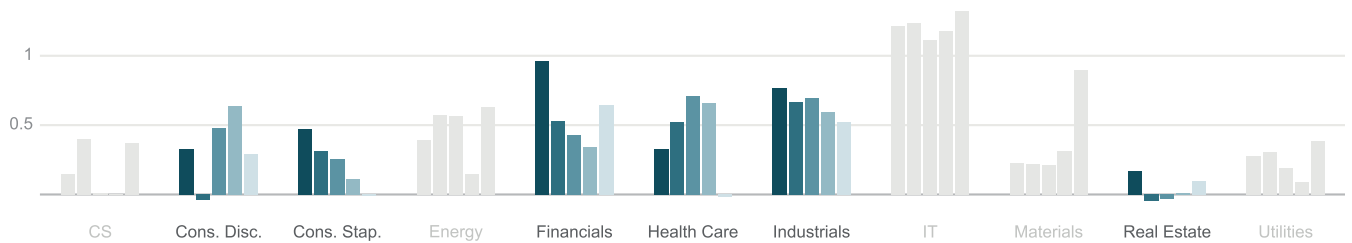
031 ● Sales to Price Ratio

Valuation L-S CAGR +9.7%



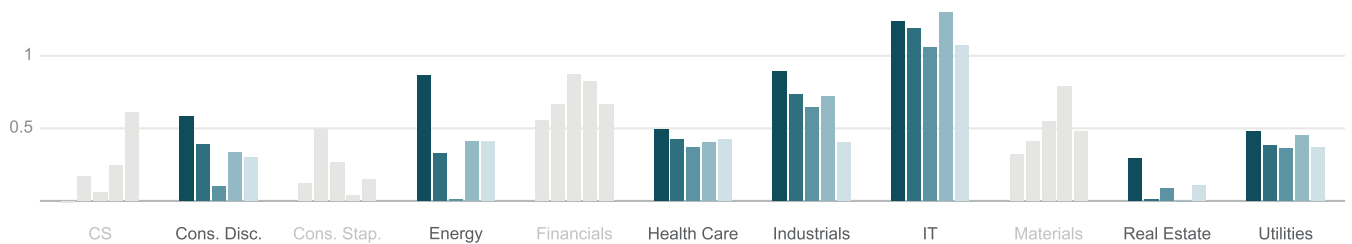
032 ● Growth Flow to Price

Valuation L-S CAGR +9.6%



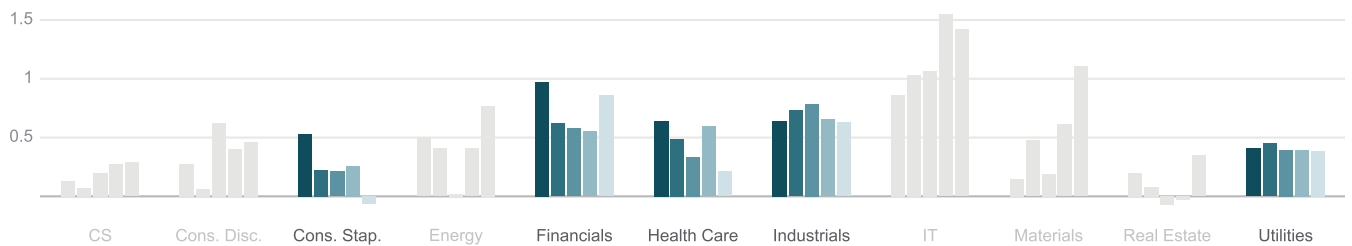
033 ● Book Leverage

Volatility L-S CAGR +9.6%



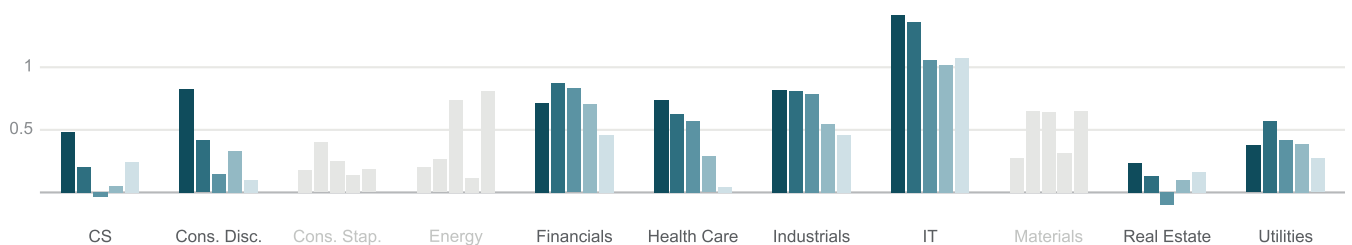
034 ● Earnings to Price

Valuation L-S CAGR +9.4%



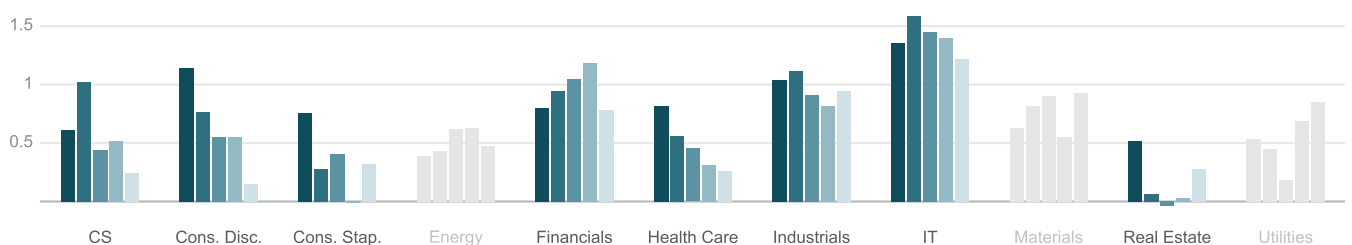
035 ● Cash Flow to Total Asset

Earnings Quality L-S CAGR +9.4%



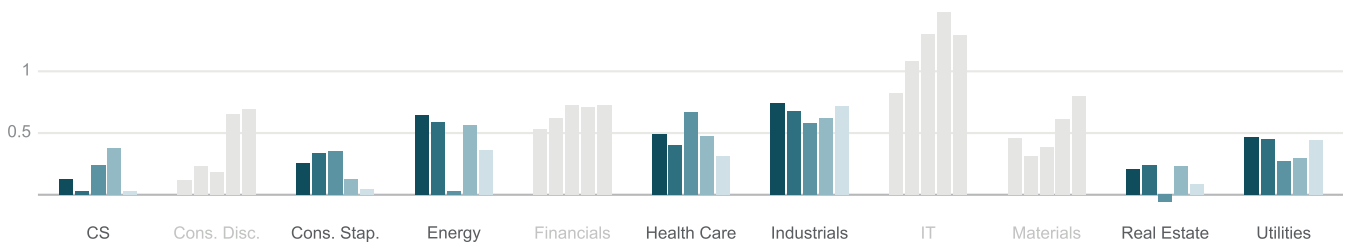
036 ● Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings

Historical Growth L-S CAGR +9.2%



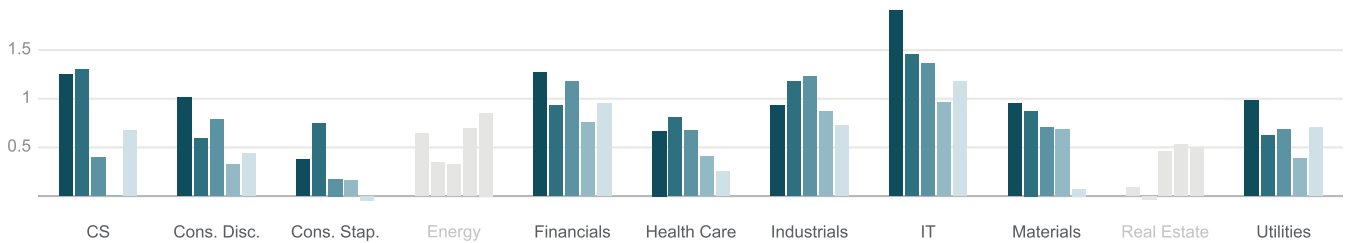
037 ● ROA 20 Qtr Standard Deviation

Earnings Quality L-S CAGR +9.2%



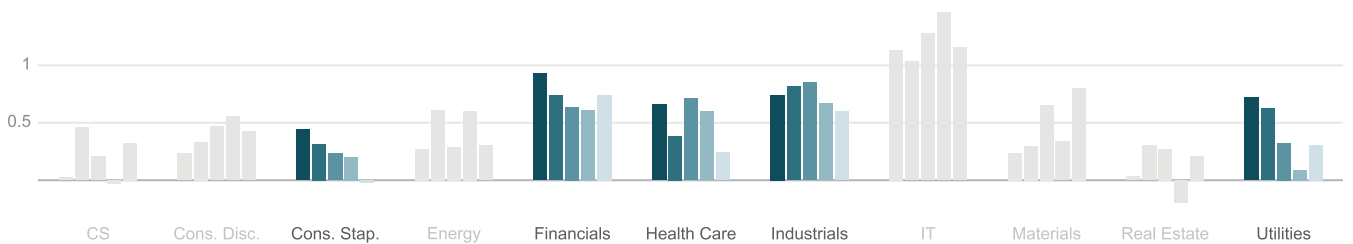
038 ● Adjusted 3M Revision in FY1 EPS Estimate

Analyst Expectations L-S CAGR +9.2%



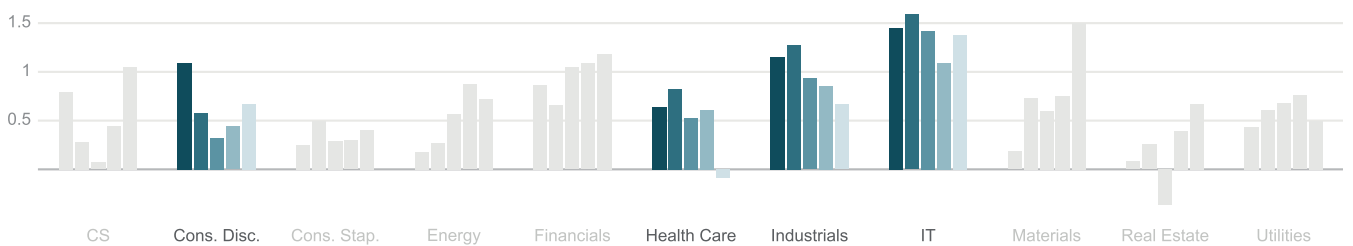
039 ● Inverse P/E Ratio Adj for Growth and Yield

Valuation L-S CAGR +9.1%



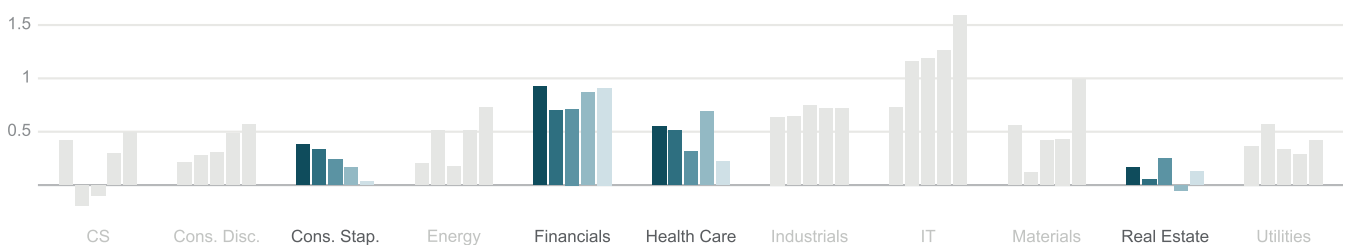
040 ● Trailing 12-Month Earnings/Trailing 12-Month Sales

Earnings Quality L-S CAGR +9.1%



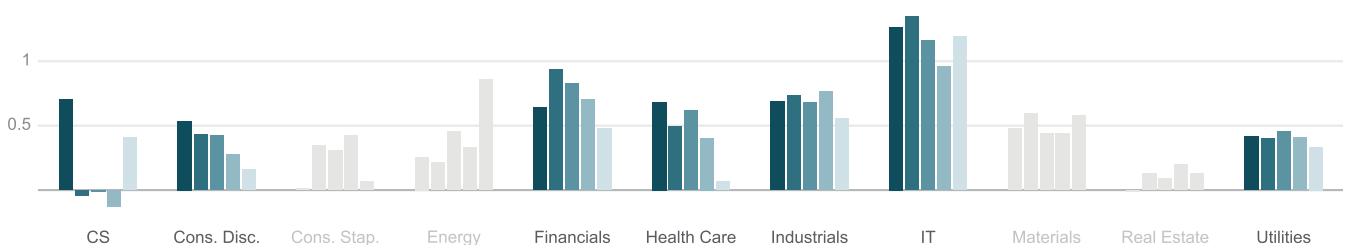
041 ● Net Cash Flow to Enterprise Value

Valuation L-S CAGR +9.1%



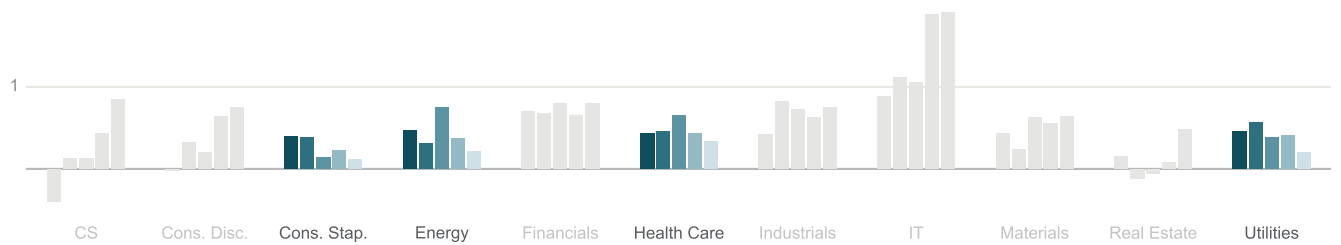
042 ● Solvency Ratio

Valuation L-S CAGR +9.1%



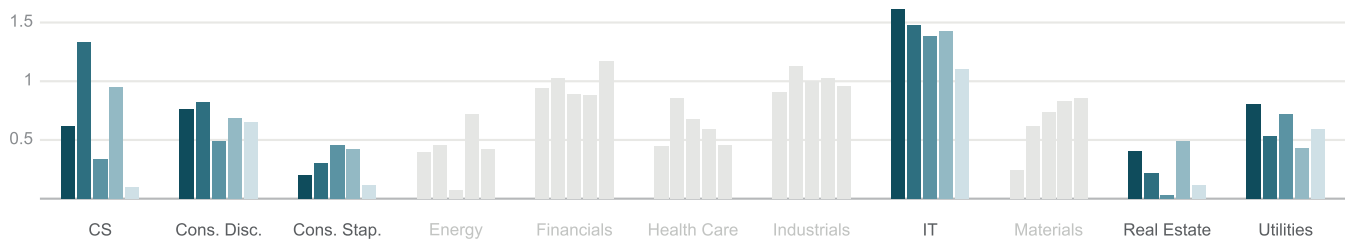
043 ● Dividends to Price Ratio

Valuation L-S CAGR +9.1%



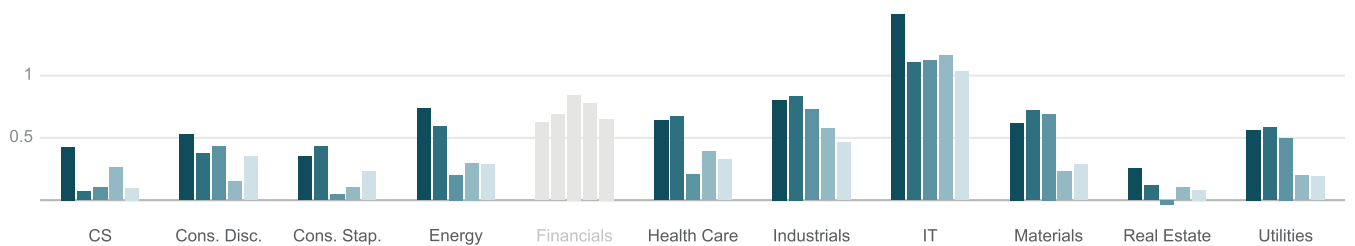
044 ● Adjusted Inverse PEG

Analyst Expectations L-S CAGR +9.1%



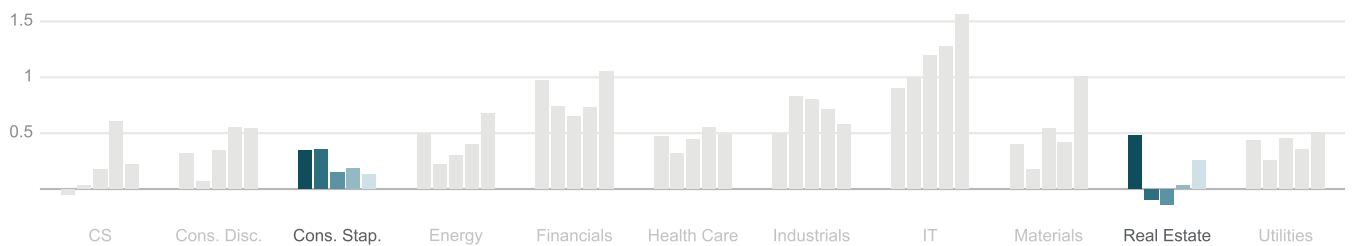
045 ● 180D Price TStat

Price Momentum L-S CAGR +8.9%



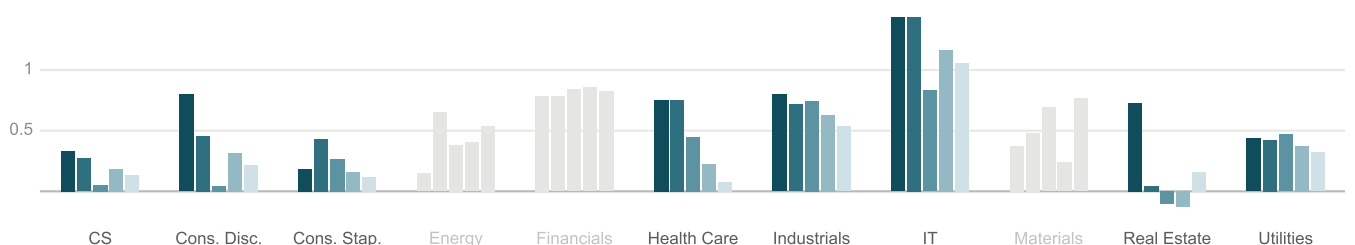
046 ● Adjusted EBIT to Price

Valuation L-S CAGR +8.9%



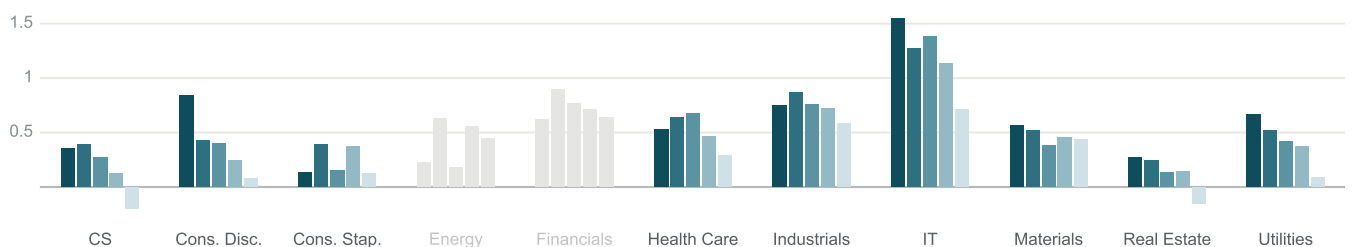
047 ● EBITDA to Assets

Capital Efficiency L-S CAGR +8.8%



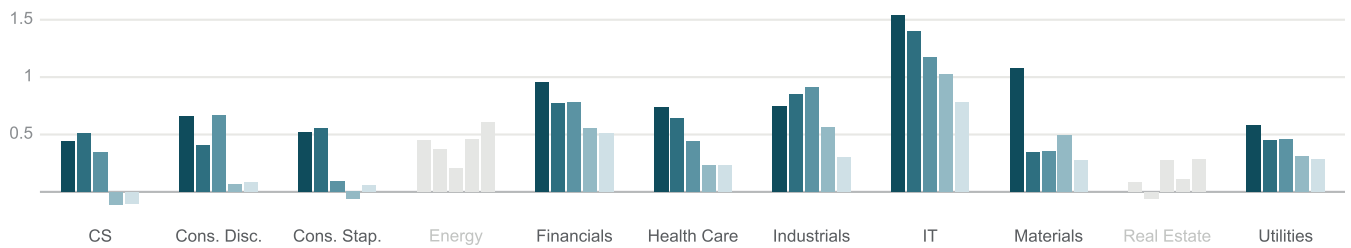
048 ● Expected LTG

Analyst Expectations L-S CAGR +8.7%

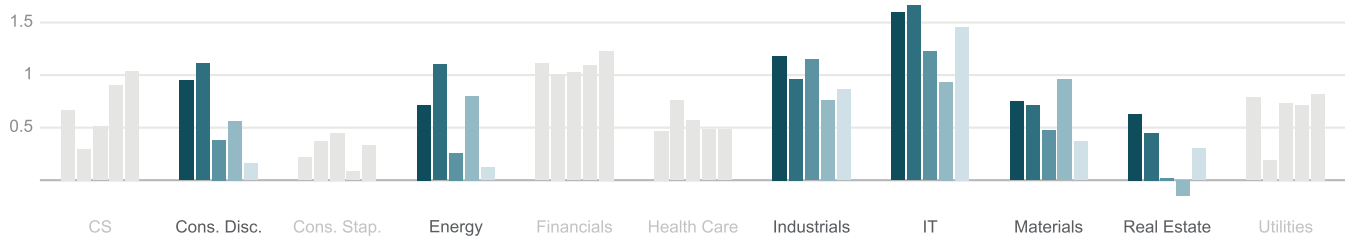


049 3M Revision in FY1 EPS Estimate

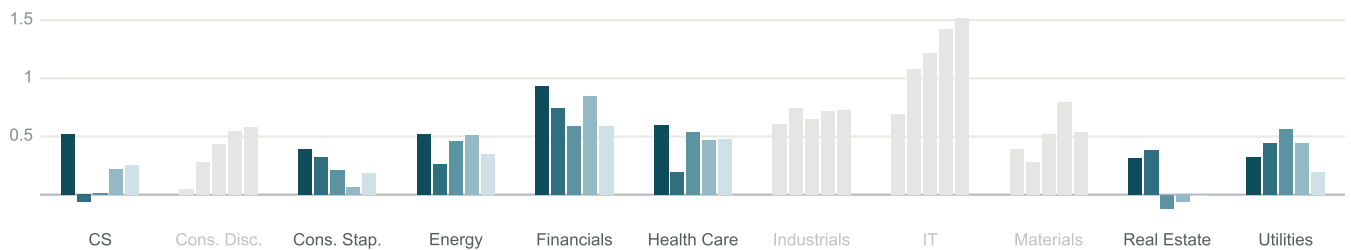
Analyst Expectations L-S CAGR +8.7%

**050 Adj 1Y Chg in Asset Adjusted FCF**

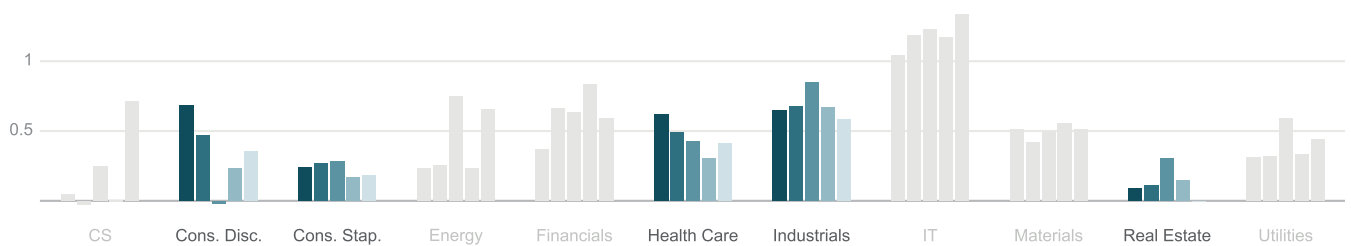
Historical Growth L-S CAGR +8.5%

**051 Sales to EV Ratio**

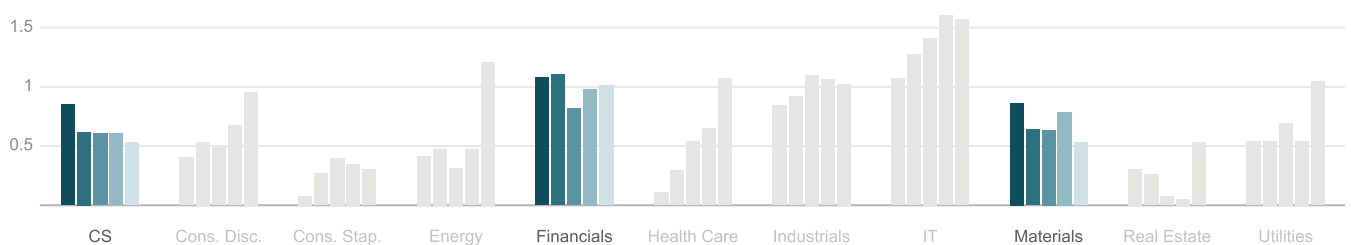
Valuation L-S CAGR +8.5%

**052 Working Capital to Sales**

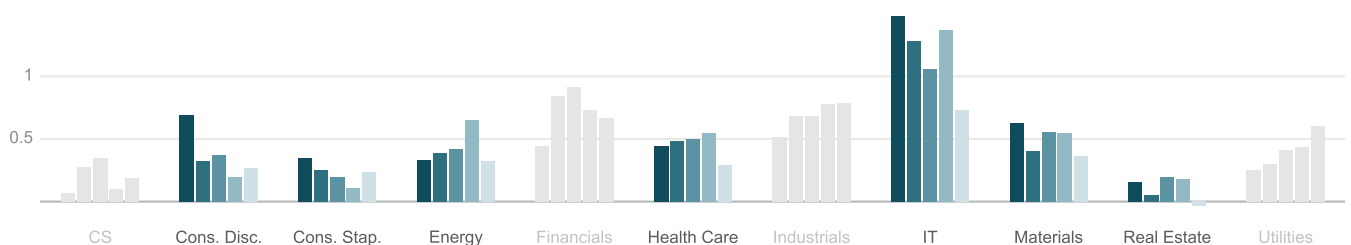
Earnings Quality L-S CAGR +8.5%

**053 Adjusted Book to Enterprise Value**

Valuation L-S CAGR +8.5%

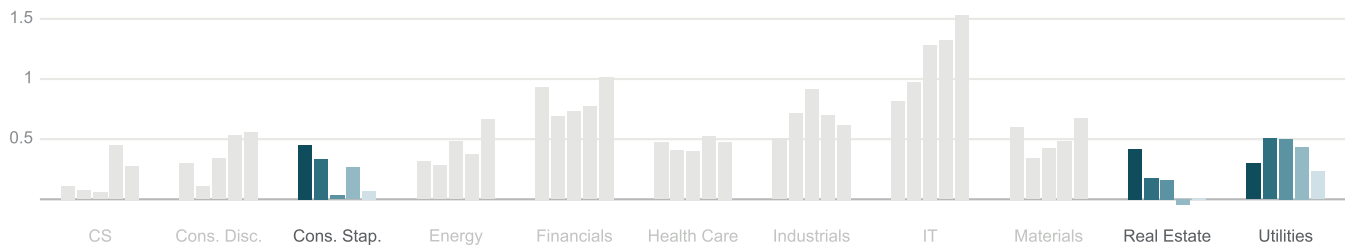
**054 Balance Sheet Accruals**

Earnings Quality L-S CAGR +8.4%



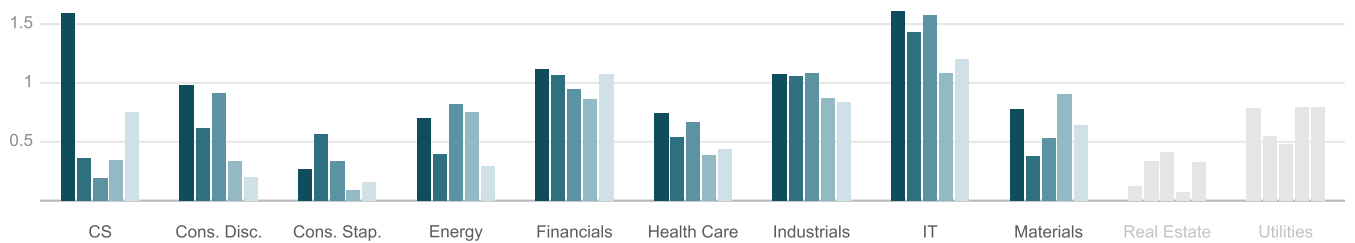
055 ● EBITDA to Enterprise Value

Valuation L-S CAGR +8.4%



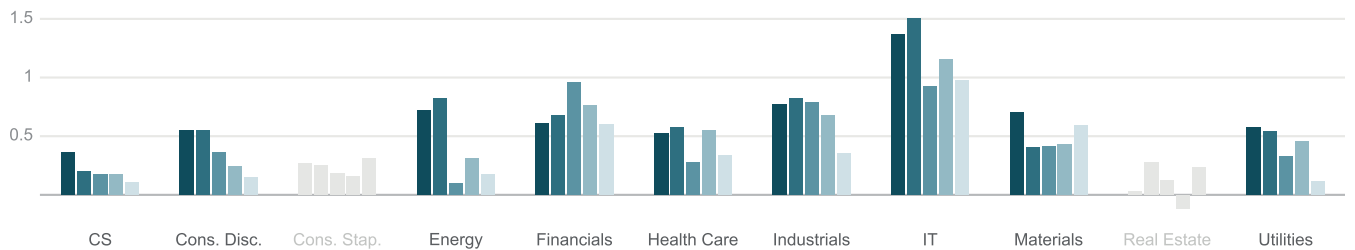
056 ● Short Interest Trading Volume

Price Momentum L-S CAGR +8.4%



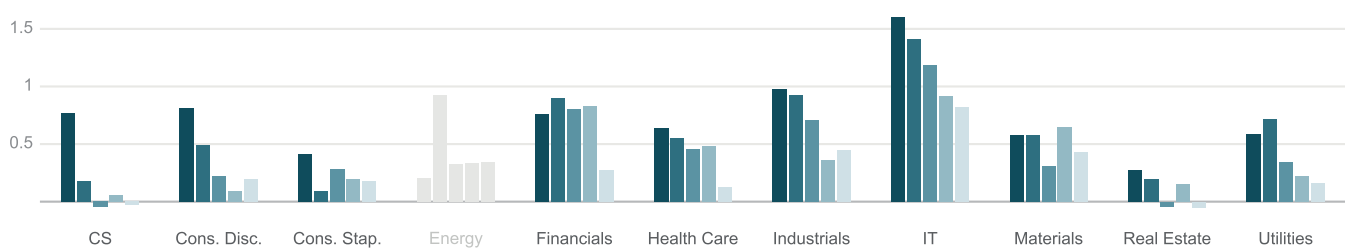
057 ● 12M - 1M Price Momentum

Price Momentum L-S CAGR +8.3%



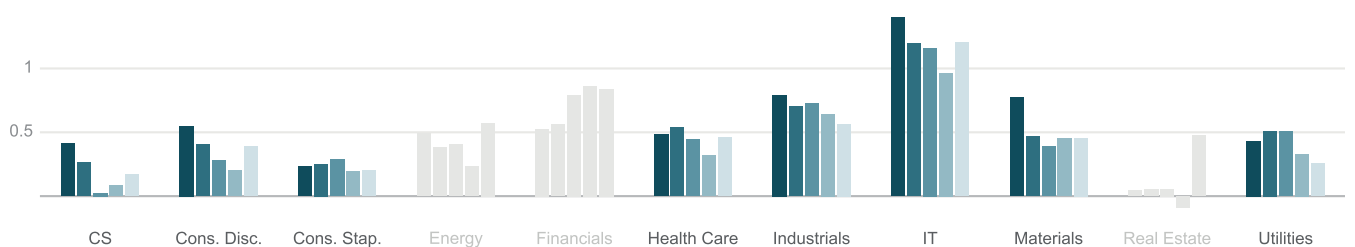
058 ● Amihud Illiquidity Measure

Volatility L-S CAGR +8.3%



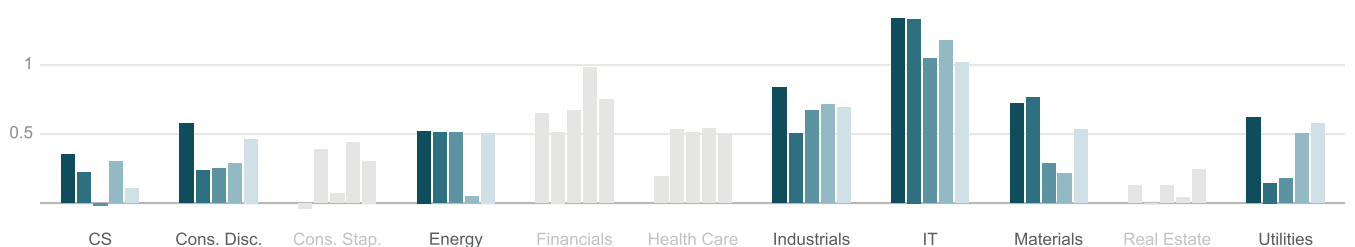
059 ● 52W High Low

Price Momentum L-S CAGR +8.3%



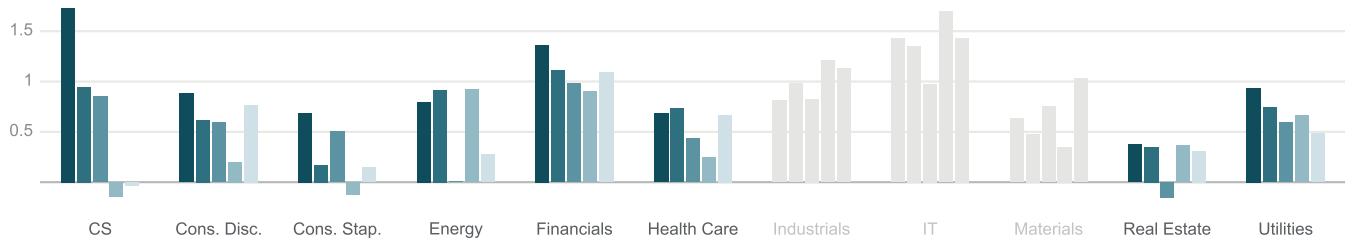
060 ● 52-Week Volume Price Trend with 20-Day Lag

Price Momentum L-S CAGR +8.2%

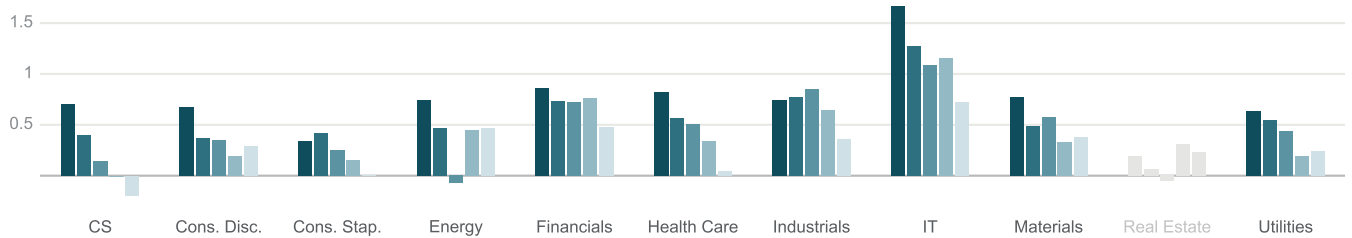


061 ● Cash to Adjusted Enterprise Value

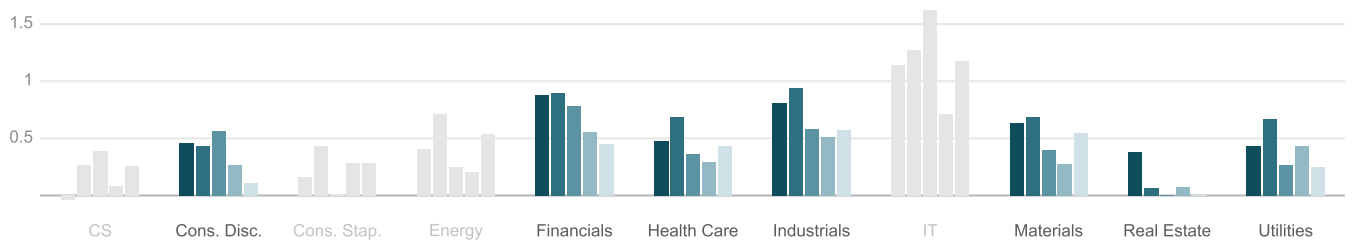
Valuation L-S CAGR +8.2%

**062 ● 3M Revision in FY2 EPS Estimate**

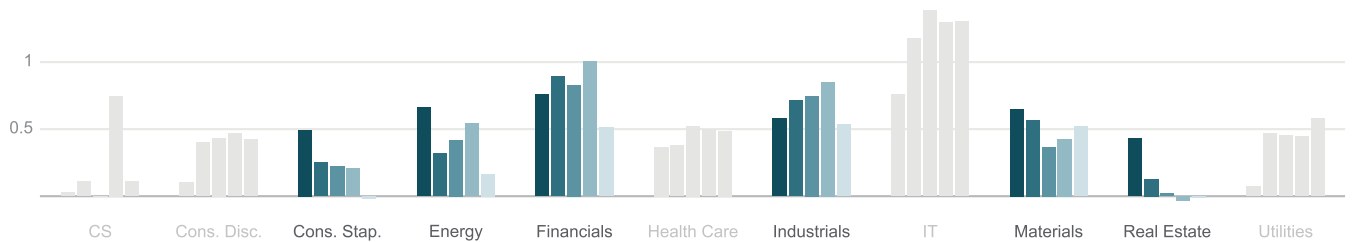
Analyst Expectations L-S CAGR +8.1%

**063 ● Institution Ownership Level - Active Manager**

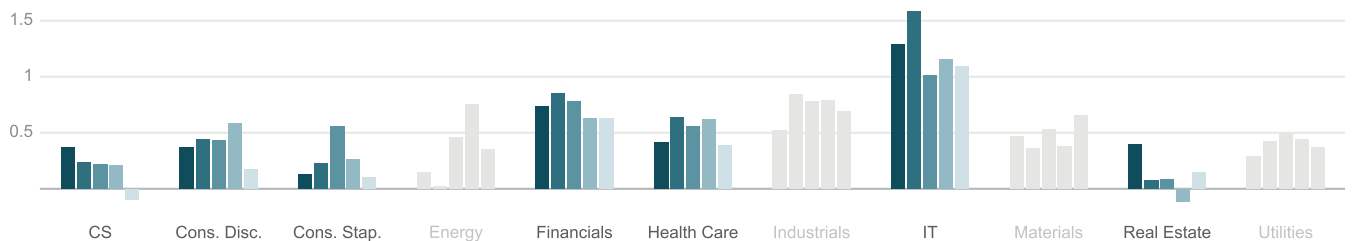
Price Momentum L-S CAGR +8.1%

**064 ● Operating Cash Flow to Price**

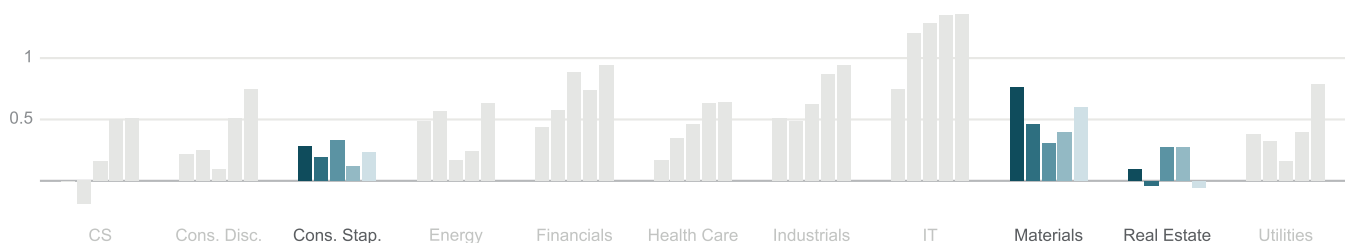
Valuation L-S CAGR +8.1%

**065 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

Historical Growth L-S CAGR +8.0%

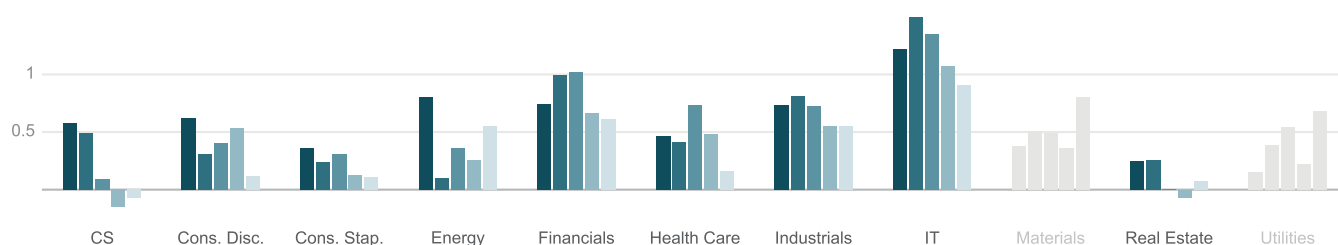
**066 ● Log of Unadjusted Stock Price**

Price Momentum L-S CAGR +8.0%

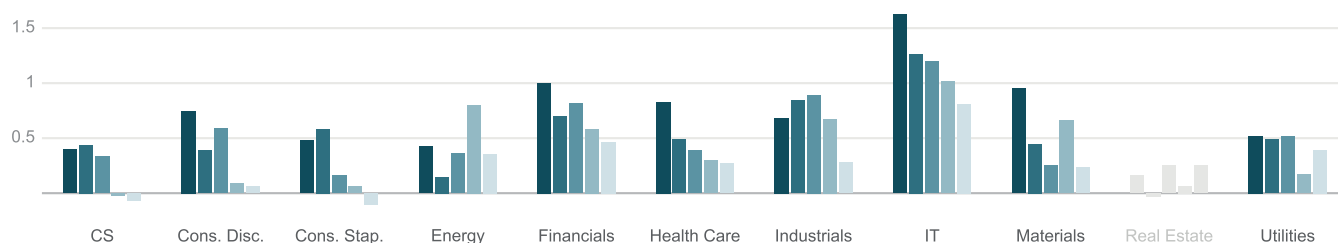


067 3Y Chg in Operating Cash Flow to Price

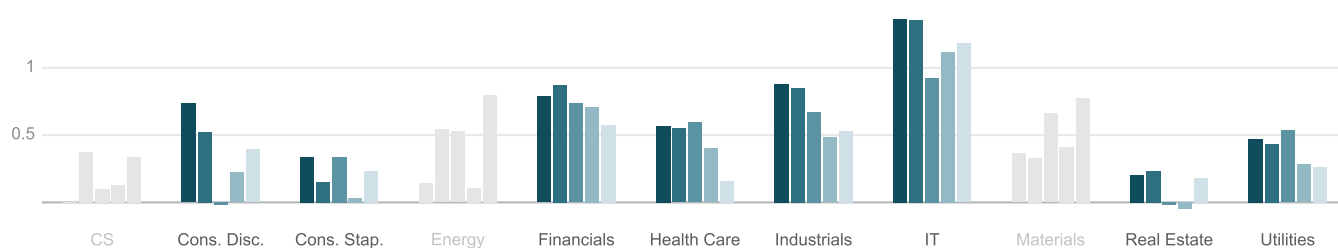
Historical Growth L-S CAGR +8.0%

**068 Street Revision Magnitude**

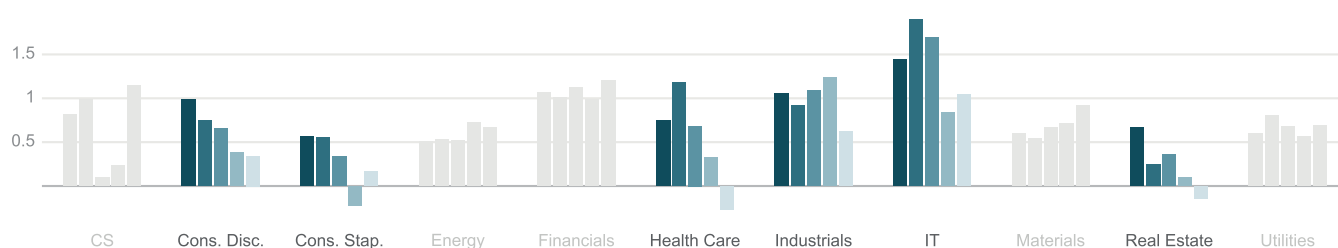
Analyst Expectations L-S CAGR +7.9%

**069 Cash Flow to Equity**

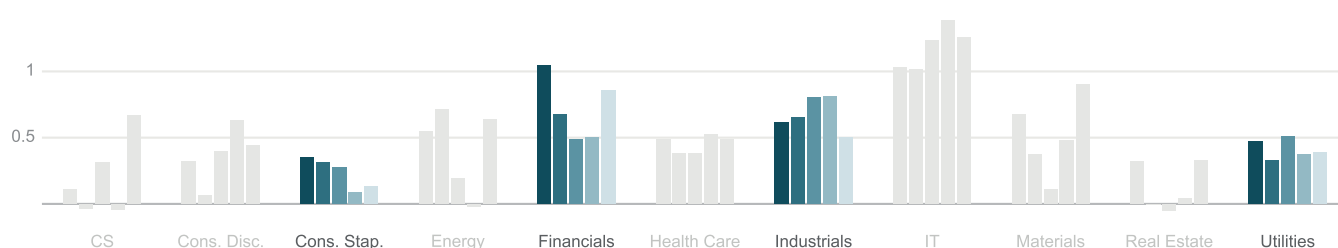
Valuation L-S CAGR +7.9%

**070 Operating Cash Flow to Invested Capital**

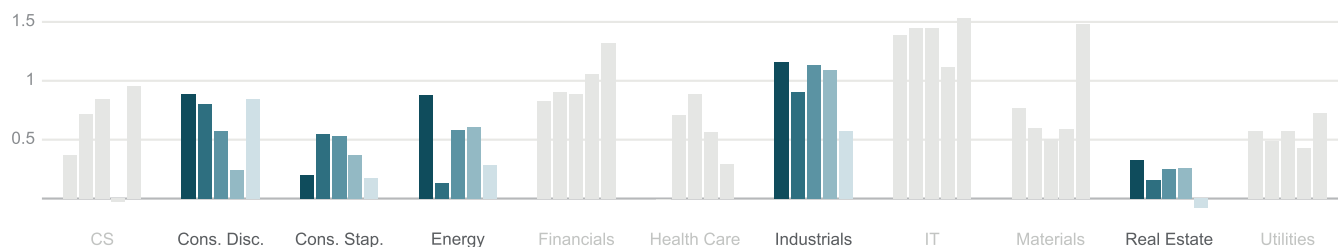
Earnings Quality L-S CAGR +7.9%

**071 Forward Earnings to Price**

Valuation L-S CAGR +7.9%

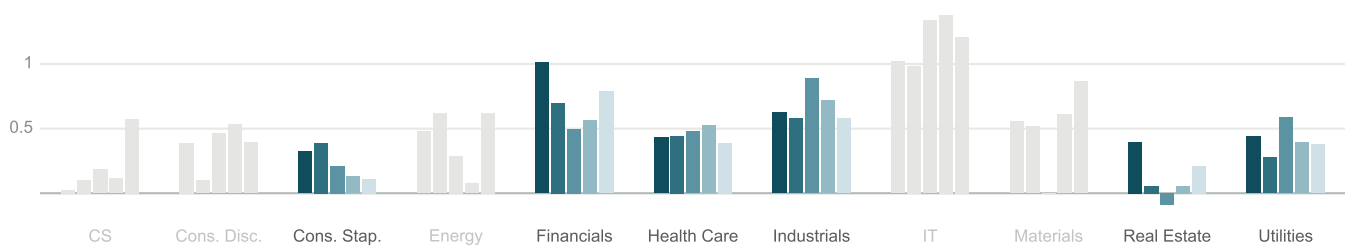
**072 12-Quarter Trendline in Trailing 12-Month Earnings**

Historical Growth L-S CAGR +7.7%



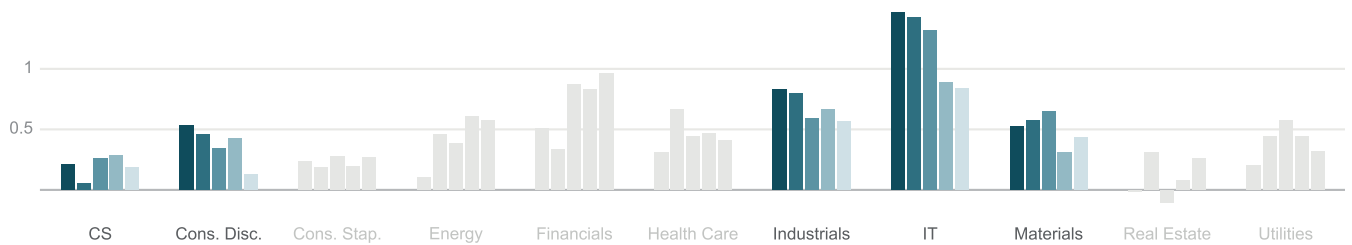
073 ● Adjusted Forward Earnings to Price

Valuation L-S CAGR +7.7%



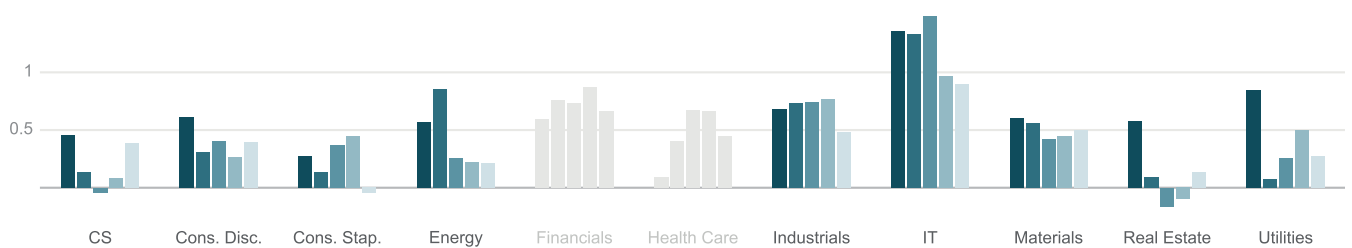
074 ● Gross Profit Margin

Earnings Quality L-S CAGR +7.7%



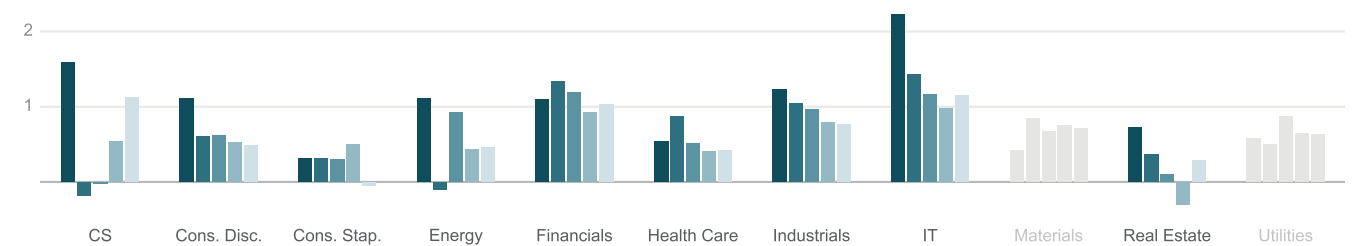
075 ● 60M CAPM Beta

Volatility L-S CAGR +7.6%



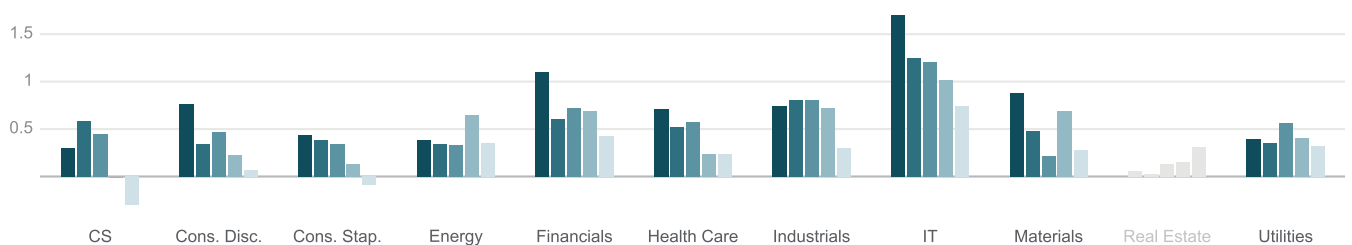
076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +7.6%



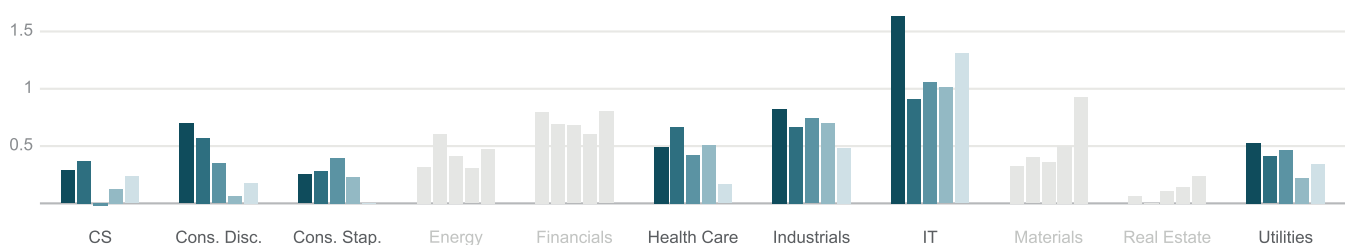
077 ● Adjusted Revision Magnitude

Analyst Expectations L-S CAGR +7.6%



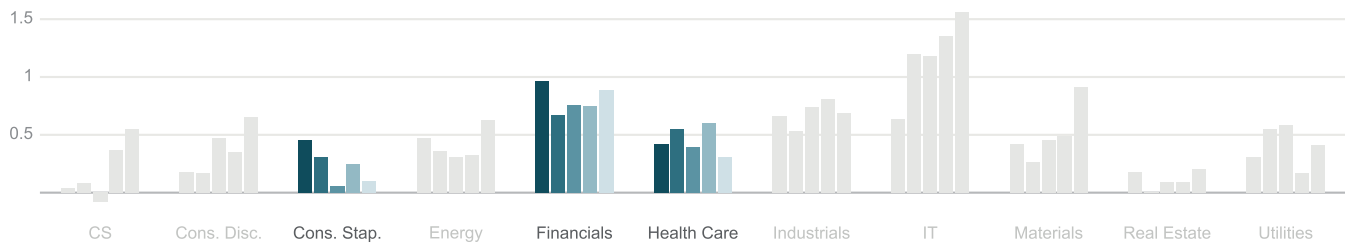
078 ● 1Y Chg in Asset Adjusted EPS

Historical Growth L-S CAGR +7.5%



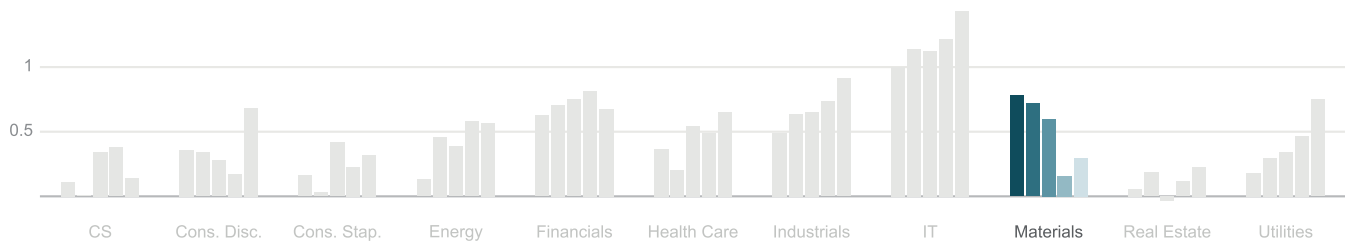
079 ● Net Cash Flow to Price

Valuation L-S CAGR +7.5%



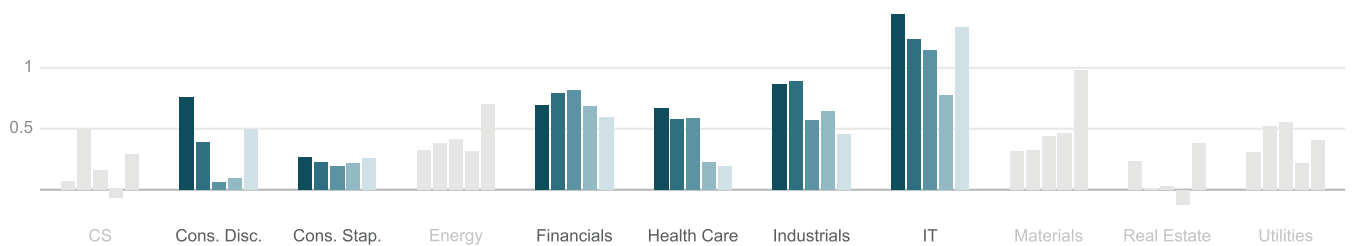
080 ● Sharpe Ratio

Price Momentum L-S CAGR +7.3%



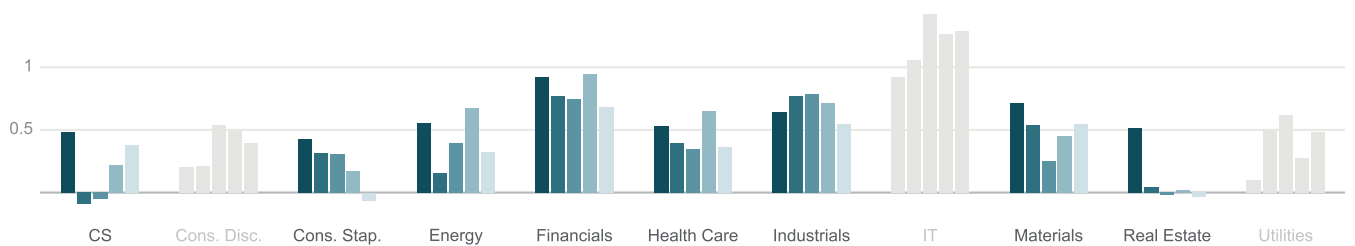
081 ● Return on Equity

Capital Efficiency L-S CAGR +7.2%



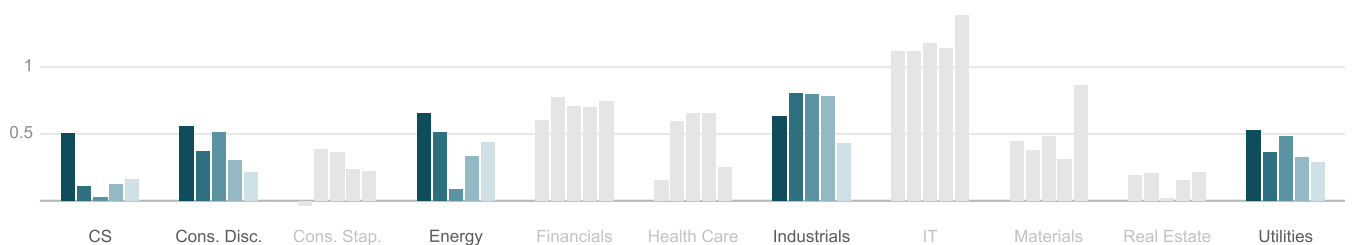
082 ● Operating Cash Flow to Enterprise Value

Valuation L-S CAGR +7.2%



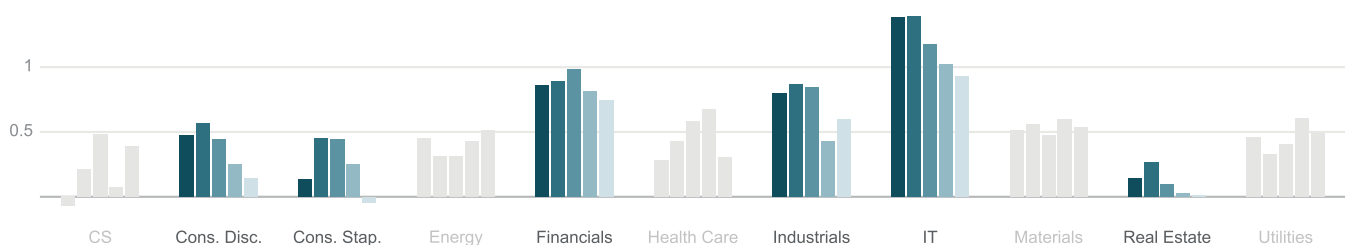
083 ● 1Y Chg in Operating Margin

Historical Growth L-S CAGR +7.2%



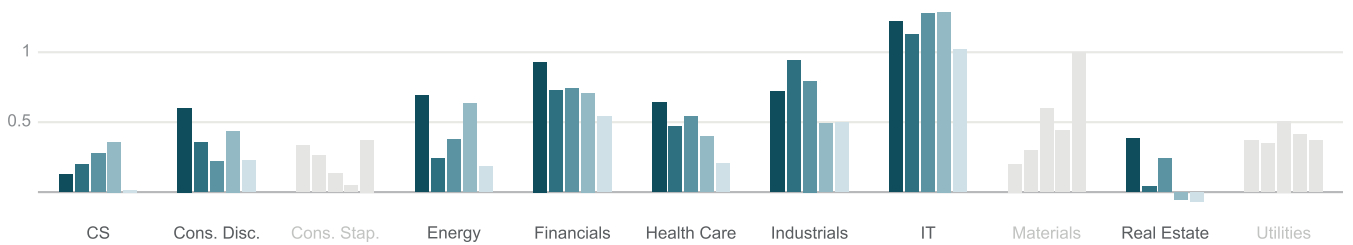
084 ● 1Y Chg in Free Cash Flow per Share

Historical Growth L-S CAGR +7.1%

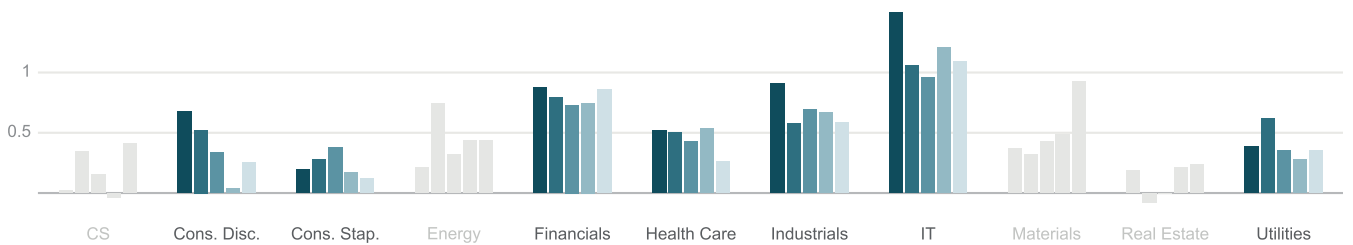


085 Equity Turnover

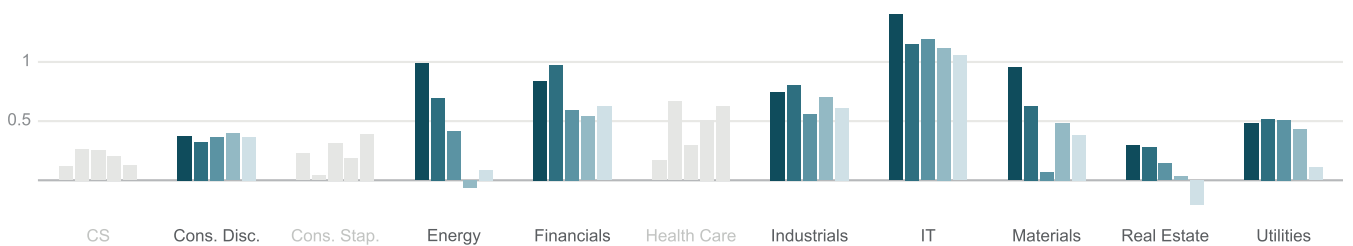
Earnings Quality L-S CAGR +7.1%

**086 1Y Chg in Asset Adjusted Cash Flow**

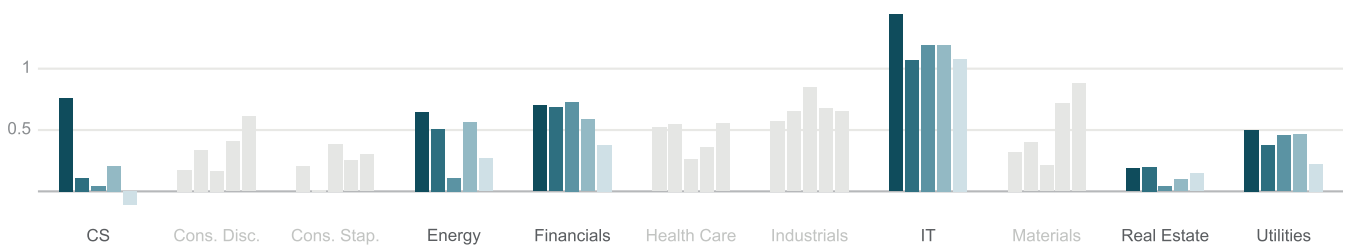
Historical Growth L-S CAGR +7.1%

**087 90 Day Coefficient of Variation**

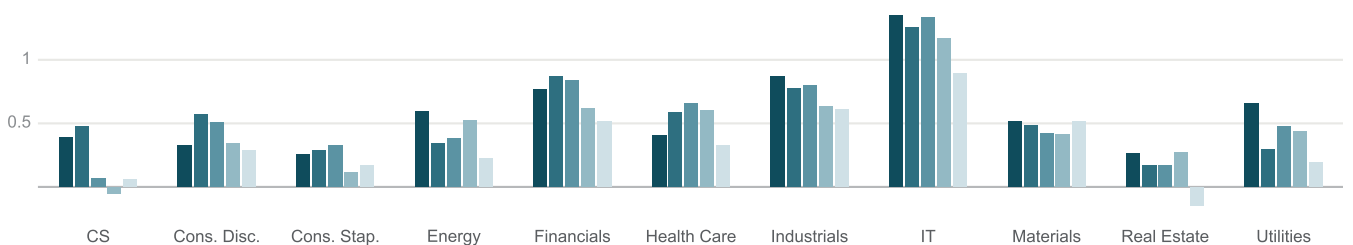
Volatility L-S CAGR +7.1%

**088 Working Capital to Assets Ratio**

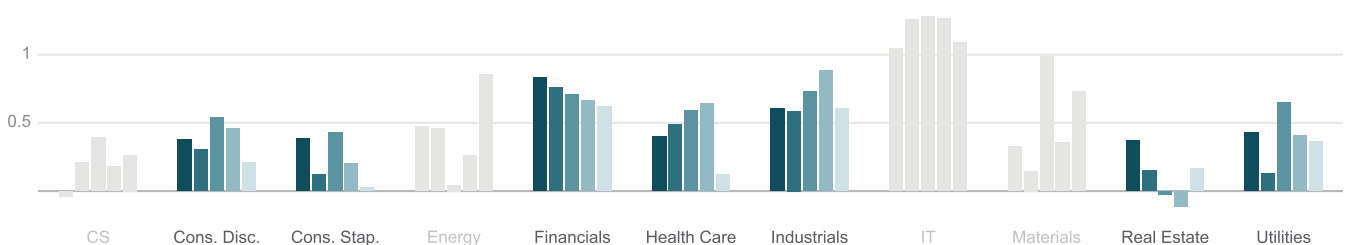
Earnings Quality L-S CAGR +7.0%

**089 Inverse PEG**

Analyst Expectations L-S CAGR +7.0%

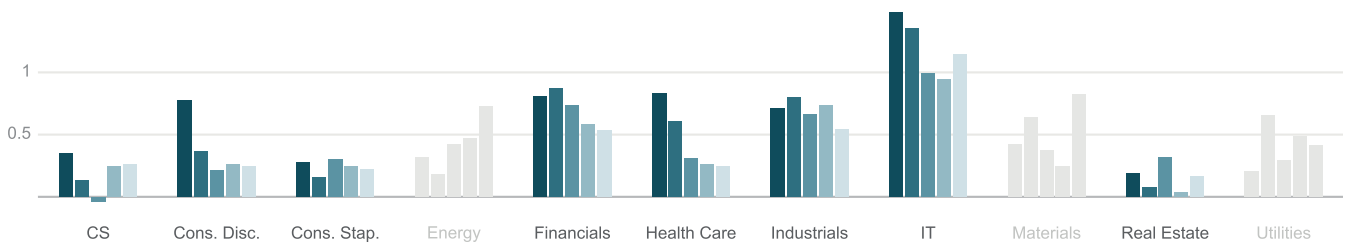
**090 1Y Chg in Sales to Price**

Historical Growth L-S CAGR +7.0%



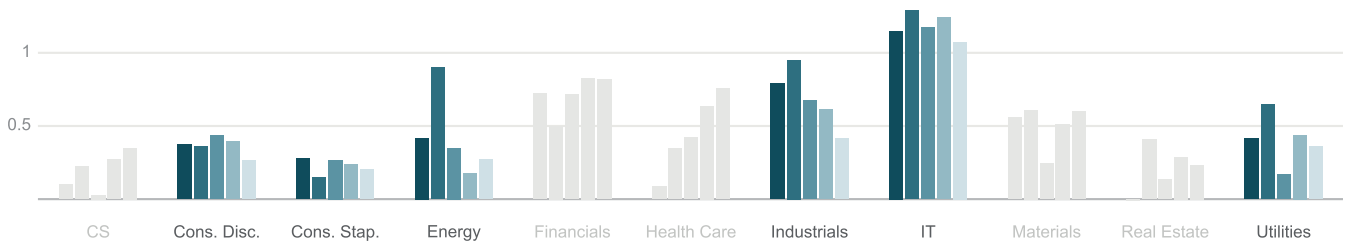
091 ● Normalized ROA

Capital Efficiency L-S CAGR +7.0%



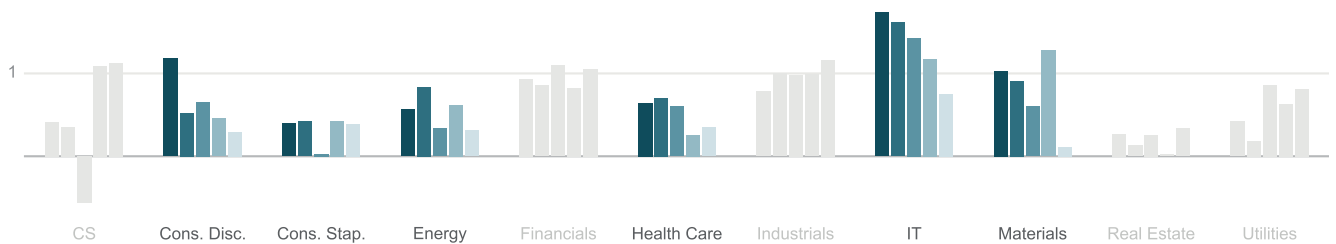
092 ● Capital Expenditure to Total Assets

Capital Efficiency L-S CAGR +6.9%



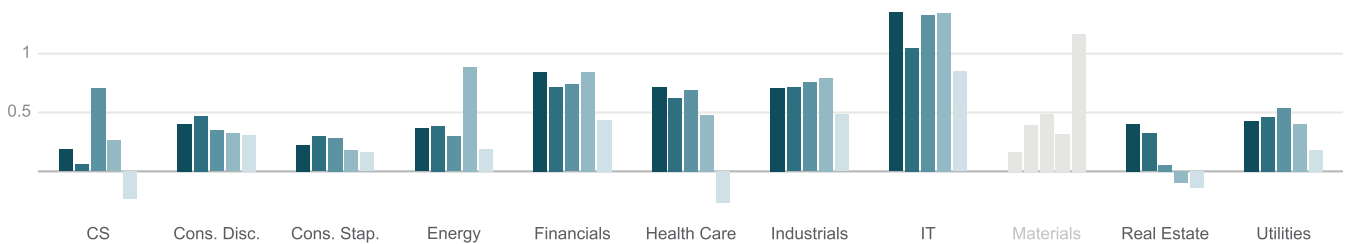
093 ● Consecutive Quarters of Declines in TTM Overhead/Sales

Historical Growth L-S CAGR +6.9%



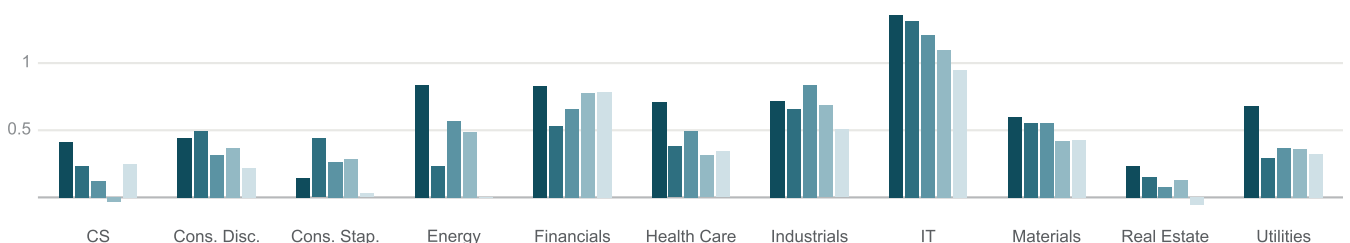
094 ● Sales to Invested Capital

Earnings Quality L-S CAGR +6.9%



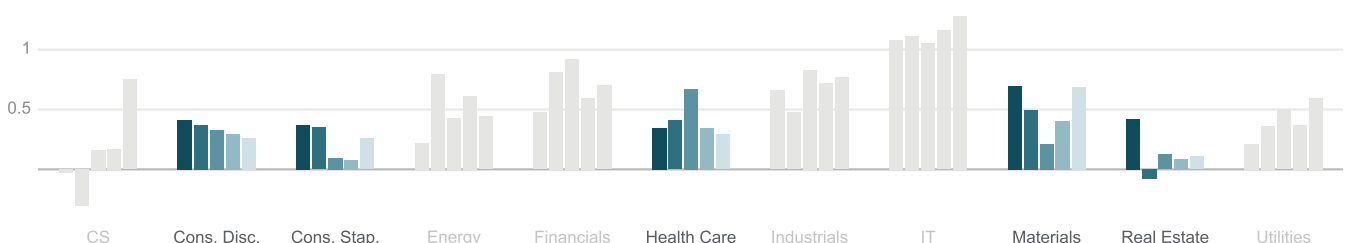
095 ● 26W Price Stat

Price Momentum L-S CAGR +6.9%



096 ● Inventory Turnover

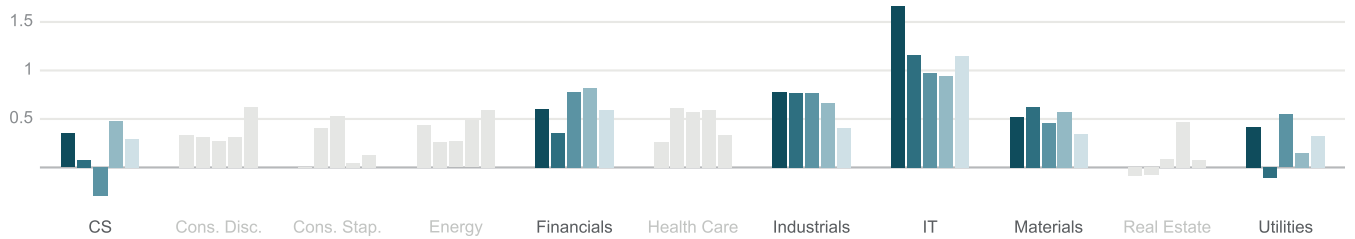
Earnings Quality L-S CAGR +6.8%



097 ● Year over Year Change in SGA to Sales

Earnings Quality

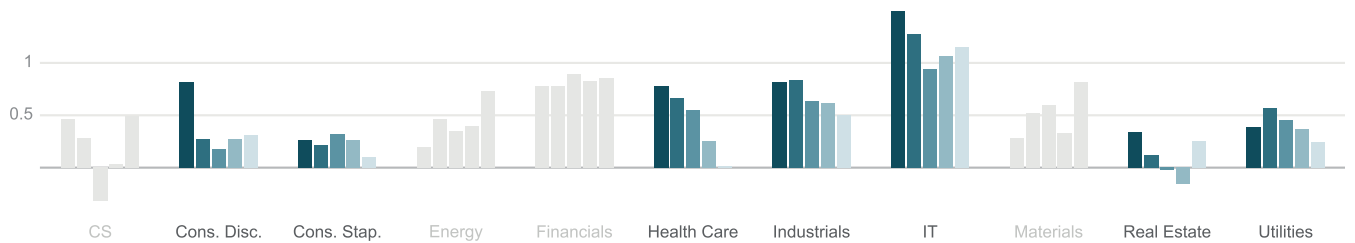
L-S CAGR +6.7%



098 ● EBIT to Assets

Capital Efficiency

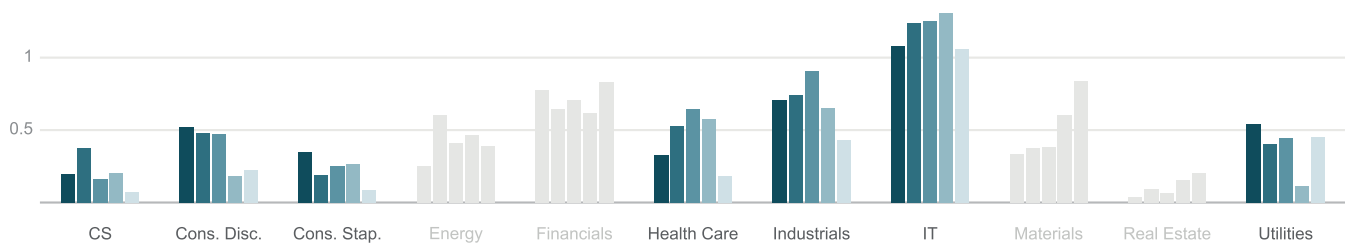
L-S CAGR +6.7%



099 ● 1Y Chg in Earnings to Price

Historical Growth

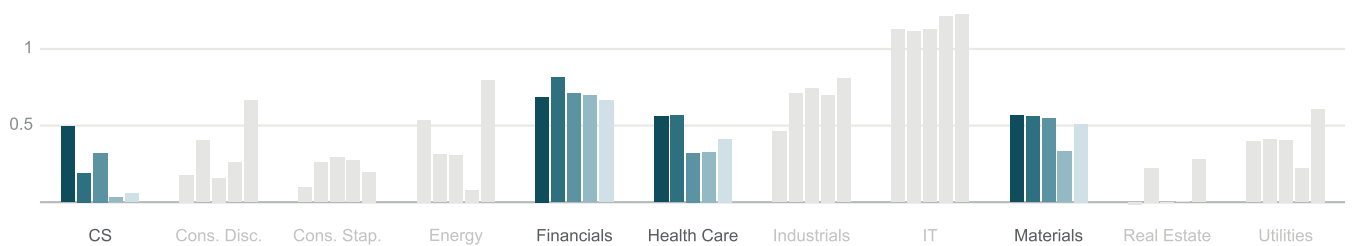
L-S CAGR +6.7%



100 ● Long Term Debt to Equity Ratio

Capital Efficiency

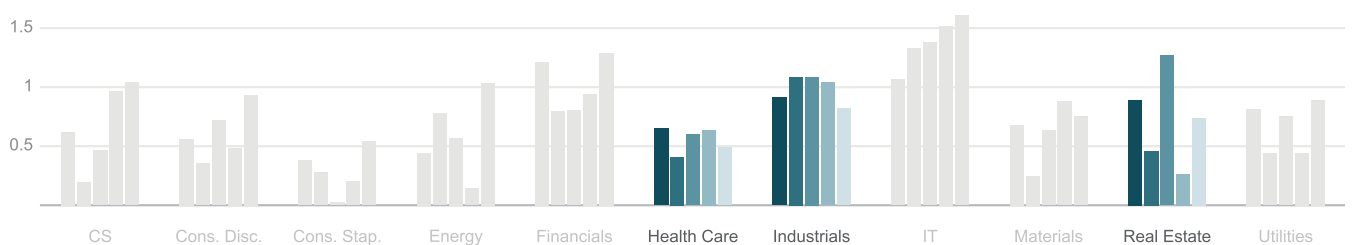
L-S CAGR +6.7%



101 ● Adj. EBITDA to Enterprise Value

Valuation

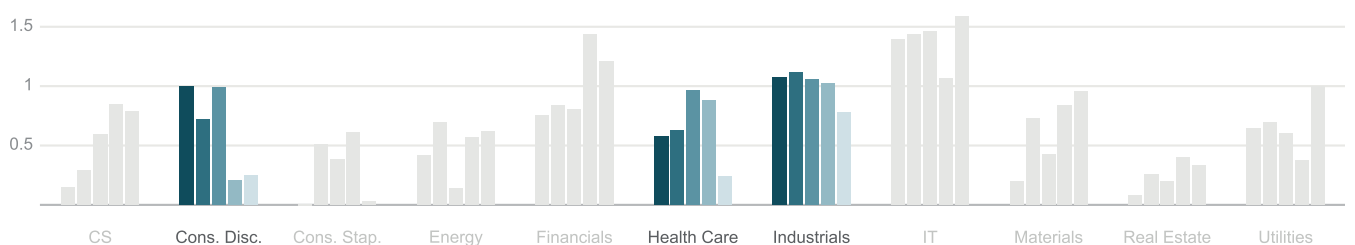
L-S CAGR +6.6%



102 ● Adj 1Y Chg in Earnings per Share

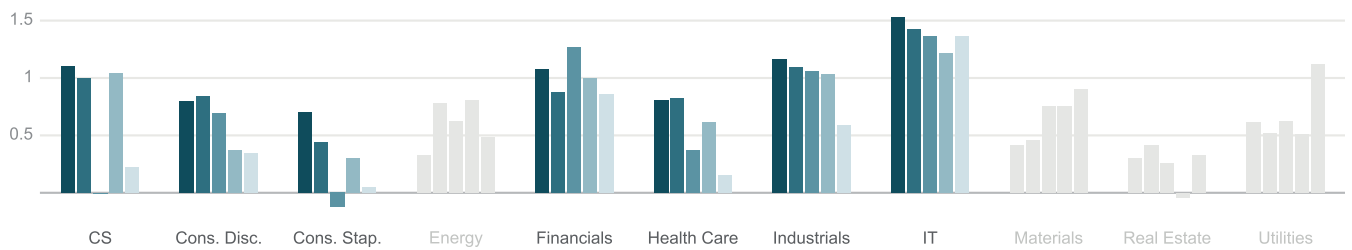
Historical Growth

L-S CAGR +6.6%



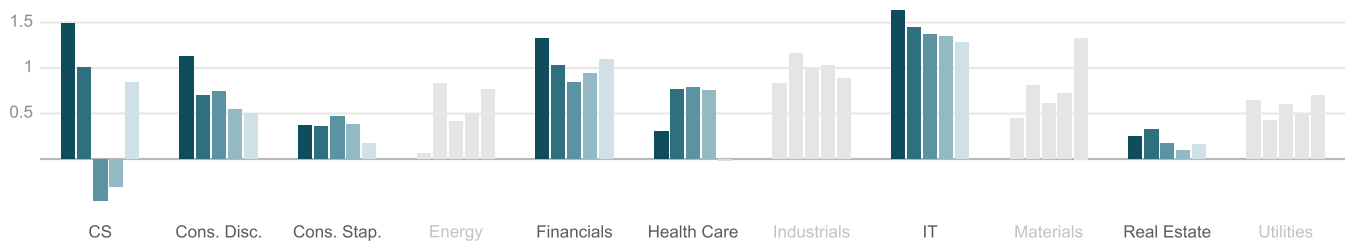
103 ● Short Interest Ratio

Price Momentum L-S CAGR +6.6%



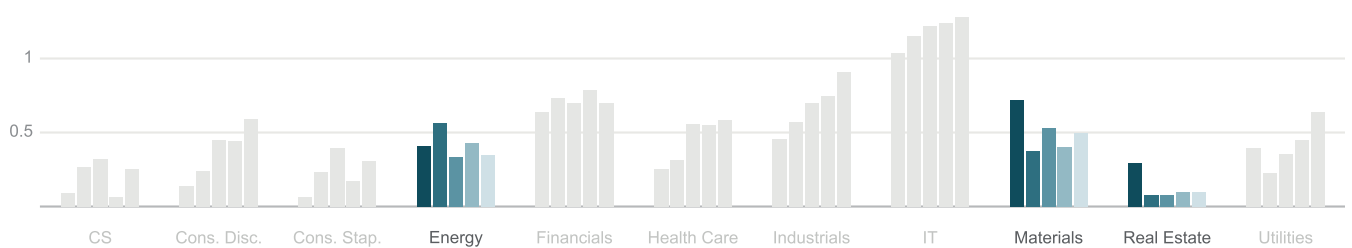
104 ● 12-Quarter Trendline in Trailing 12-Month Cash Flows

Historical Growth L-S CAGR +6.5%



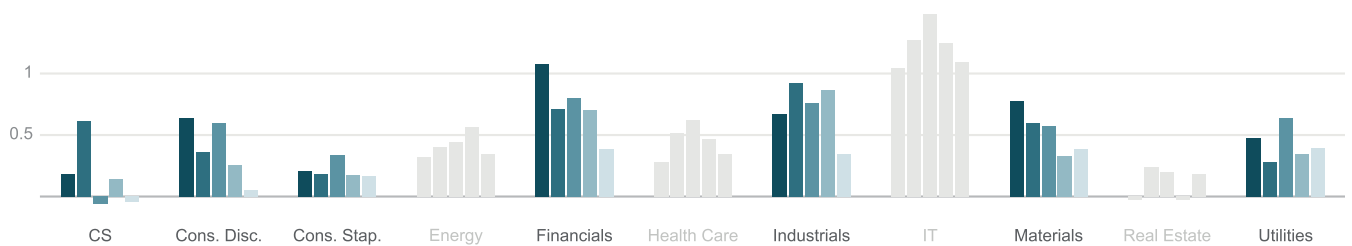
105 ● Analyst Dispersion for FY1 EPS

Analyst Expectations L-S CAGR +6.5%



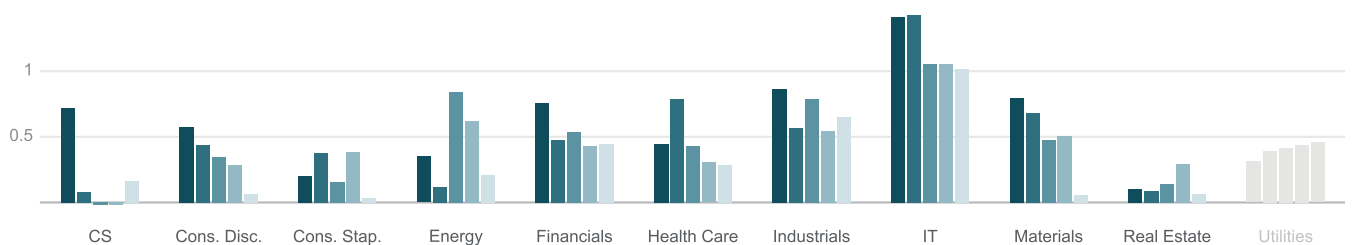
106 ● Earnings Surprise

Analyst Expectations L-S CAGR +6.4%



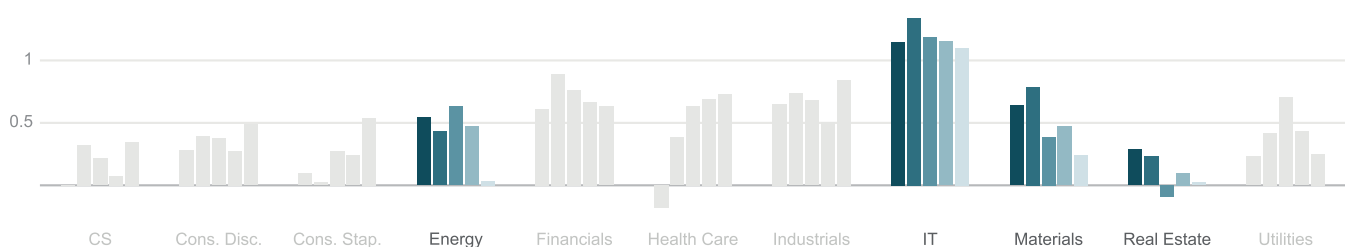
107 ● Operating Cash Flow Ratio

Capital Efficiency L-S CAGR +6.3%



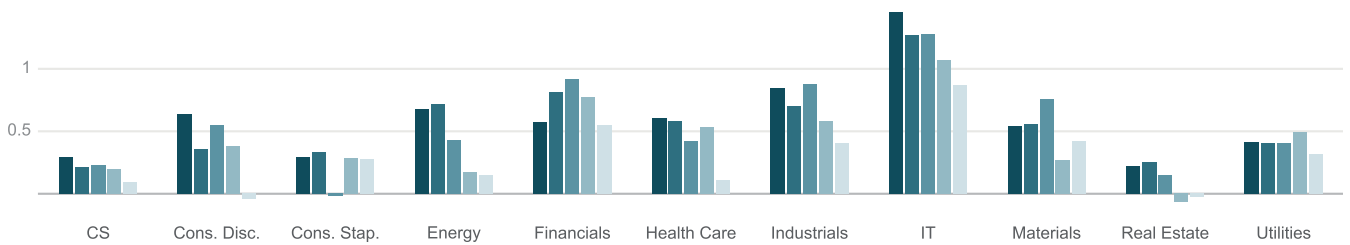
108 ● 1M Realized Price Volatility

Volatility L-S CAGR +6.2%



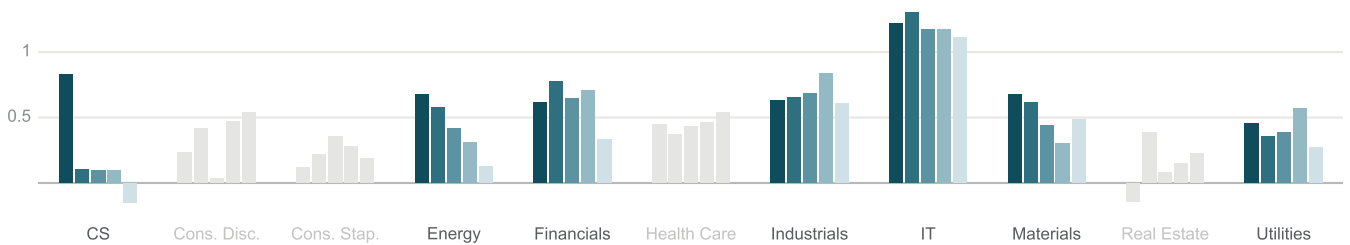
109 ● Volatility adjusted 12M return

Price Momentum L-S CAGR +6.2%



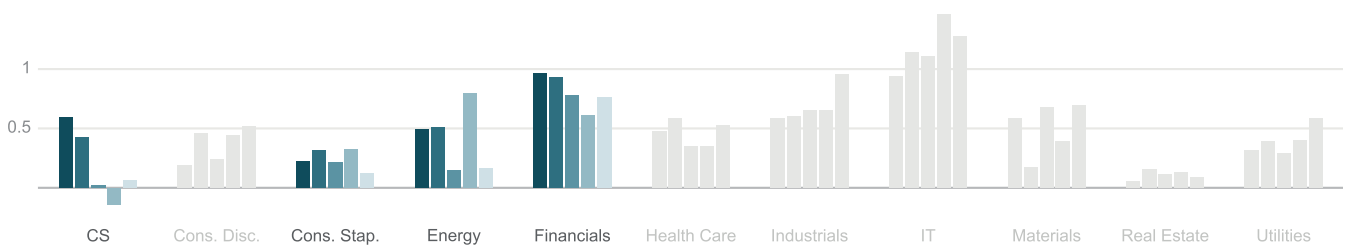
110 ● Current Ratio

Earnings Quality L-S CAGR +6.2%



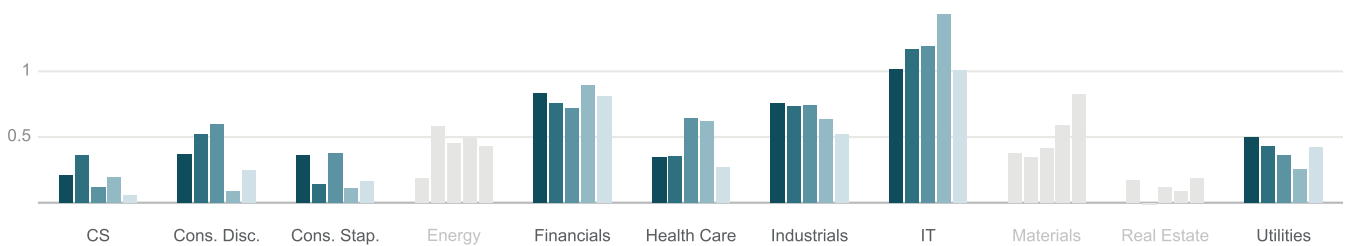
111 ● Cash to EV Ratio

Valuation L-S CAGR +6.2%



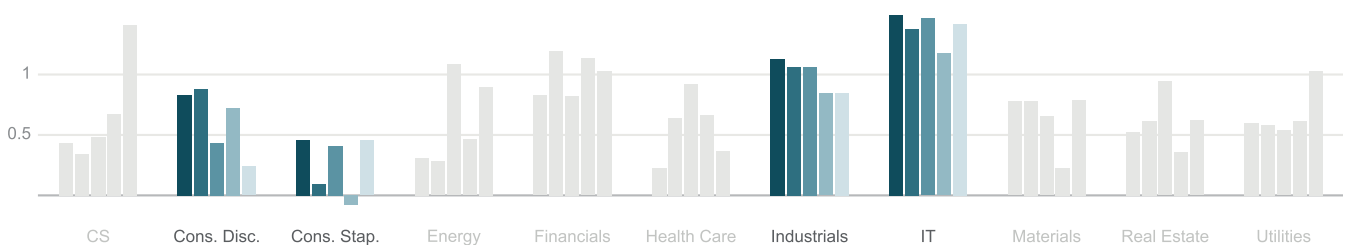
112 ● 1Y Chg in Cash Flow to Price

Historical Growth L-S CAGR +6.1%



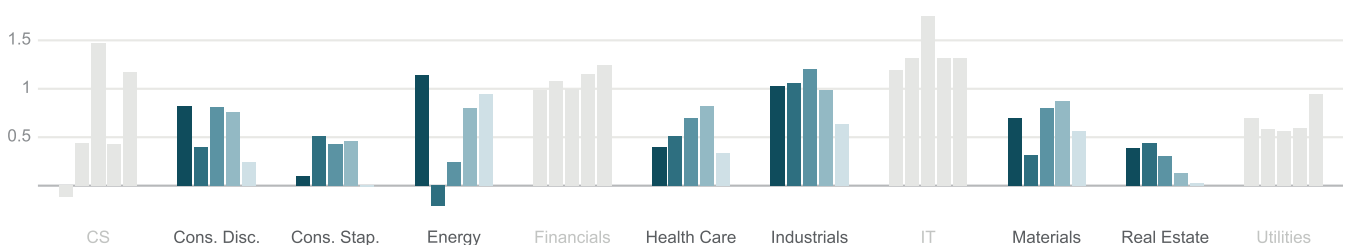
113 ● EBITDA Margin

Earnings Quality L-S CAGR +6.1%



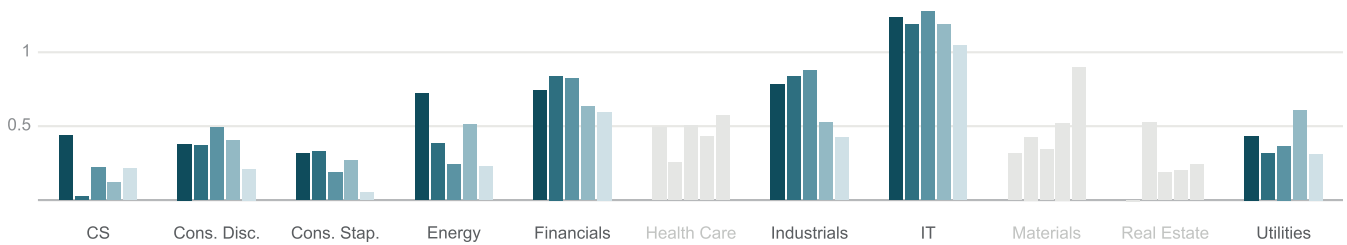
114 ● Adj Free Cash Flow to Price

Valuation L-S CAGR +6.0%



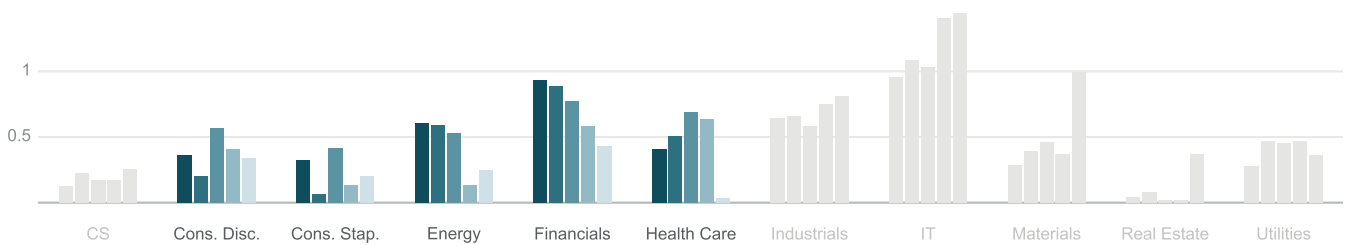
115 ● Capital Expenditure to Sales

Earnings Quality L-S CAGR +6.0%



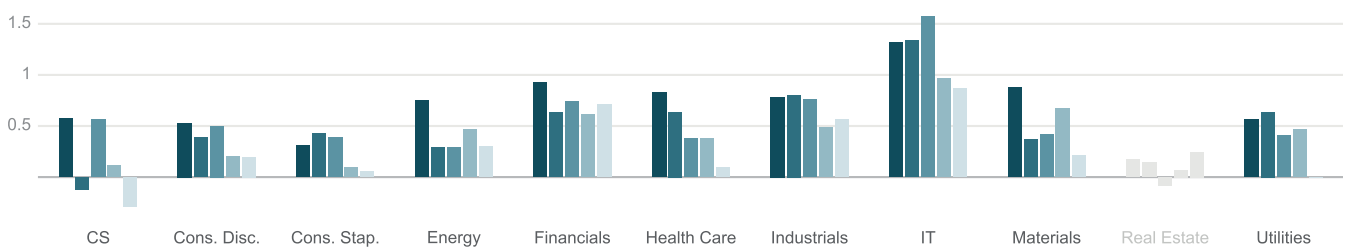
116 ● Trailing 12-Month Overhead/Trailing 12-Month Sales

Historical Growth L-S CAGR +6.0%



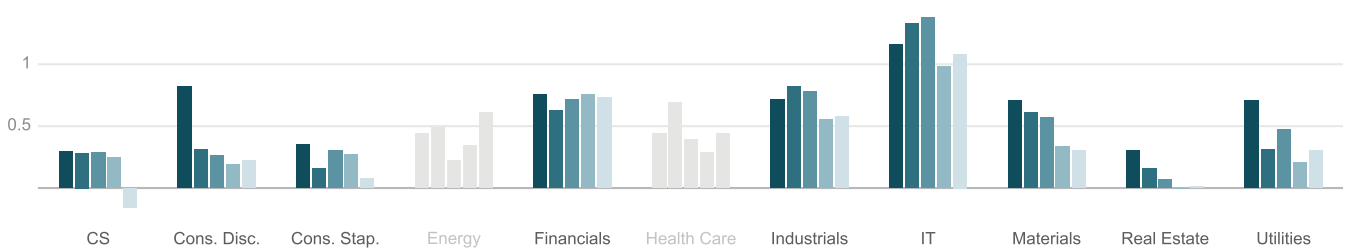
117 ● Adj Number of EPS FY2 Revisions

Analyst Expectations L-S CAGR +6.0%



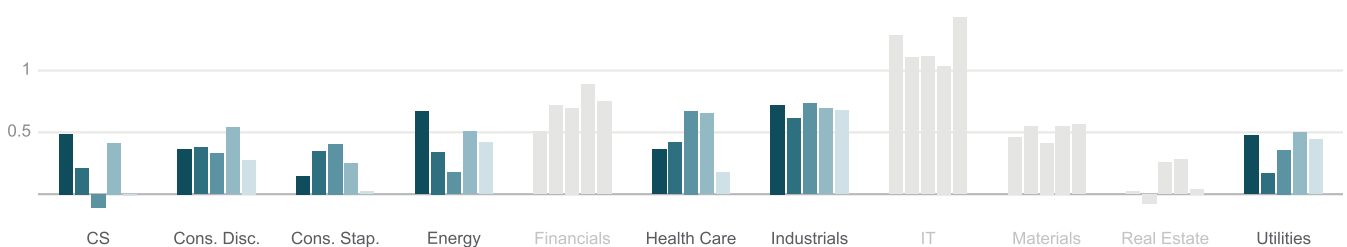
118 ● Working Capital Accruals

Earnings Quality L-S CAGR +6.0%



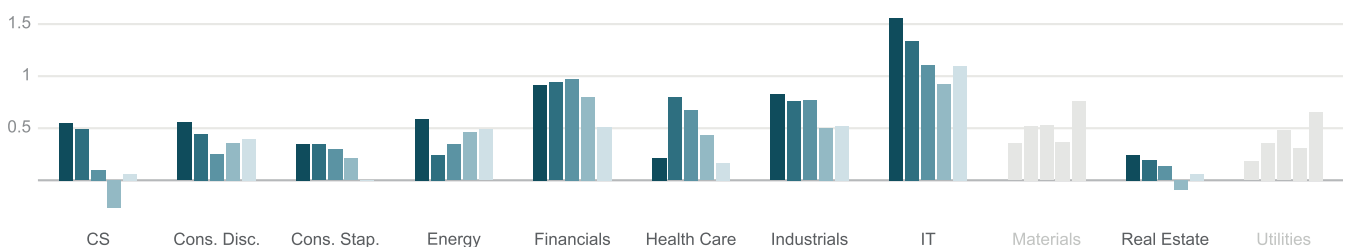
119 ● Asset Growth - 1 Year

Capital Efficiency L-S CAGR +6.0%



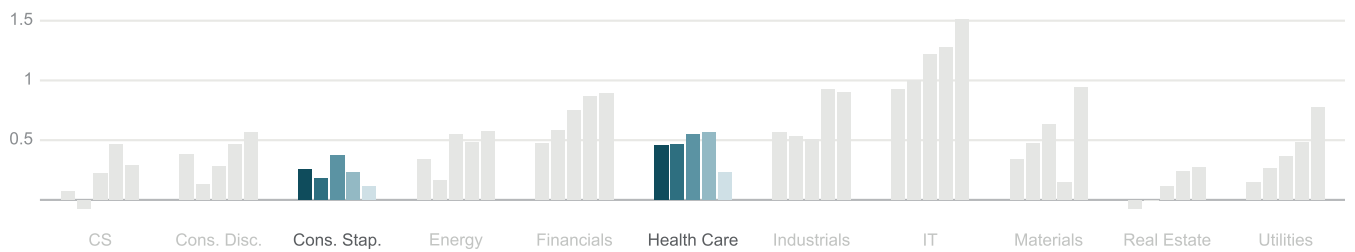
120 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth L-S CAGR +6.0%



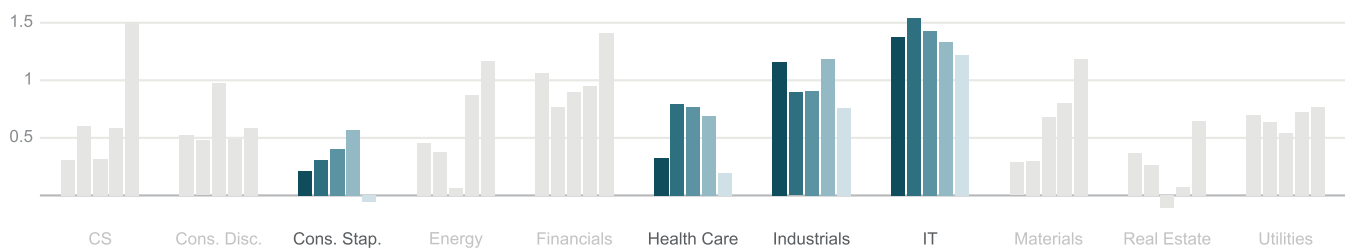
121 Adjusted 6 Month Average Share Turn Over

Price Momentum L-S CAGR +6.0%



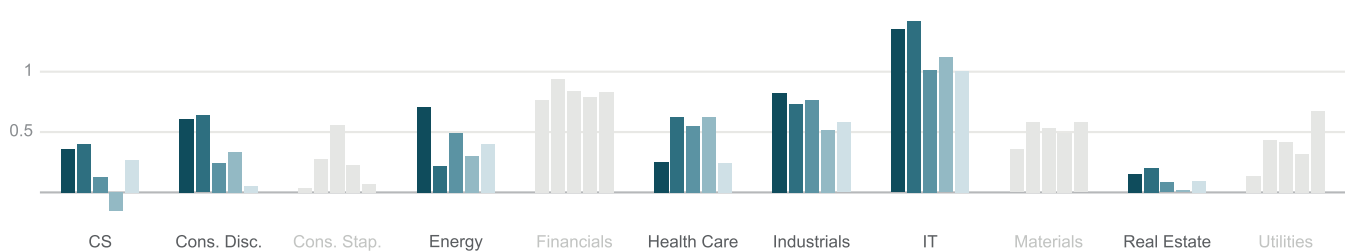
122 Adj Growth Flow to Price

Valuation L-S CAGR +6.0%



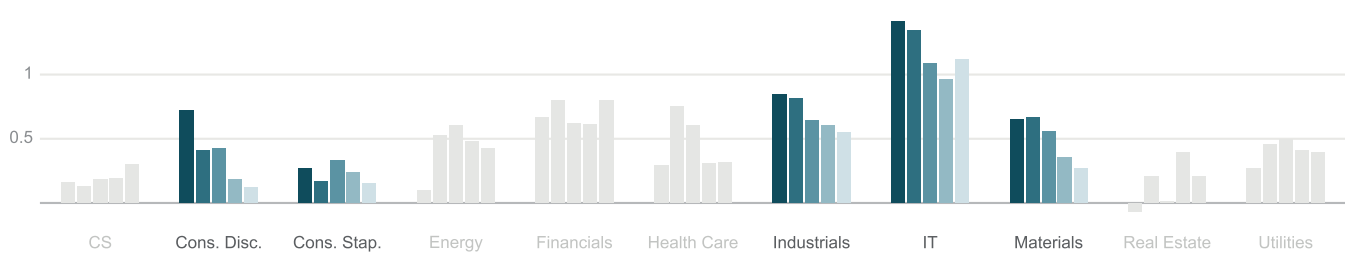
123 1Y Chg in Operating Cash Flow per Share

Historical Growth L-S CAGR +6.0%



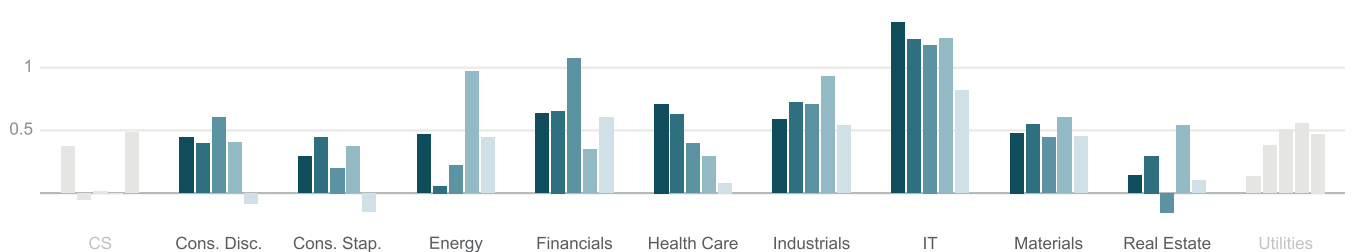
124 Operating Margin

Earnings Quality L-S CAGR +5.9%



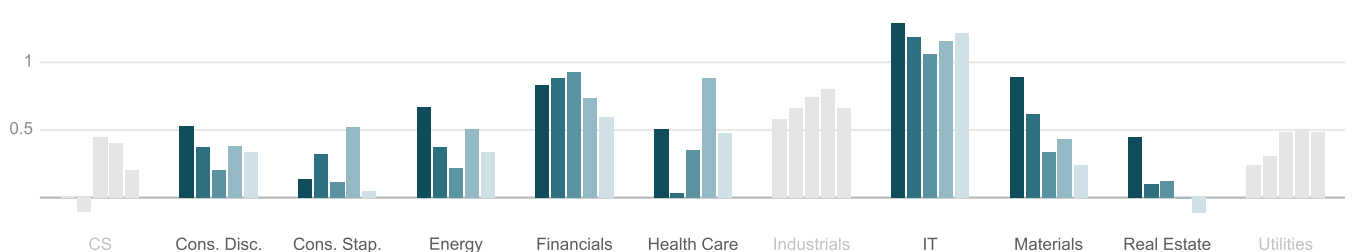
125 Adjusted Interest Coverage Ratio

Capital Efficiency L-S CAGR +5.8%



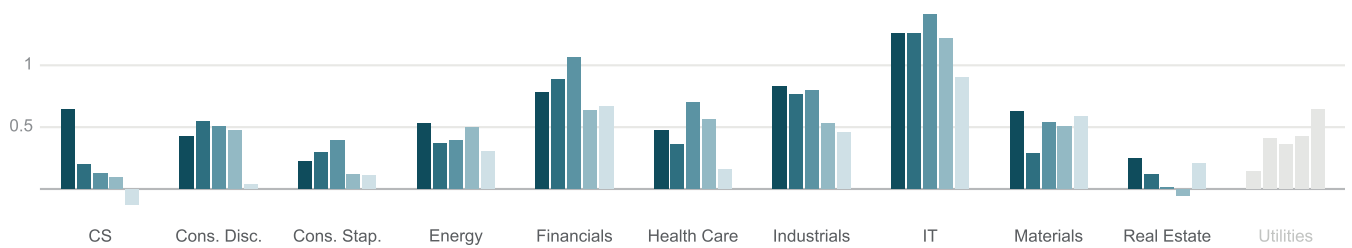
126 Adjusted Accruals

Earnings Quality L-S CAGR +5.8%

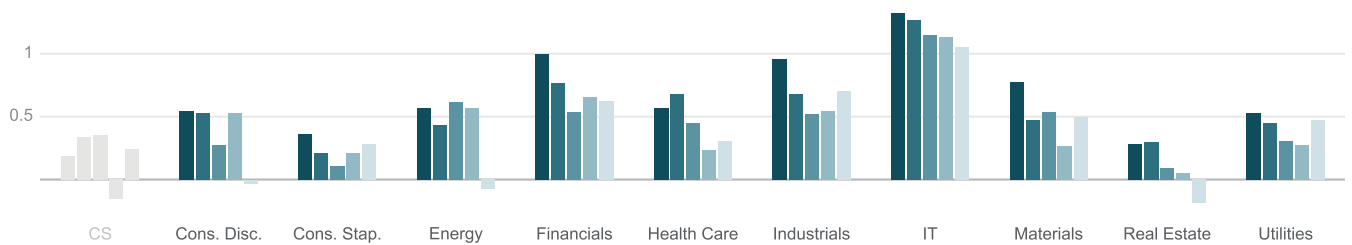


127 3Y Chg in Free Cash Flow to Price

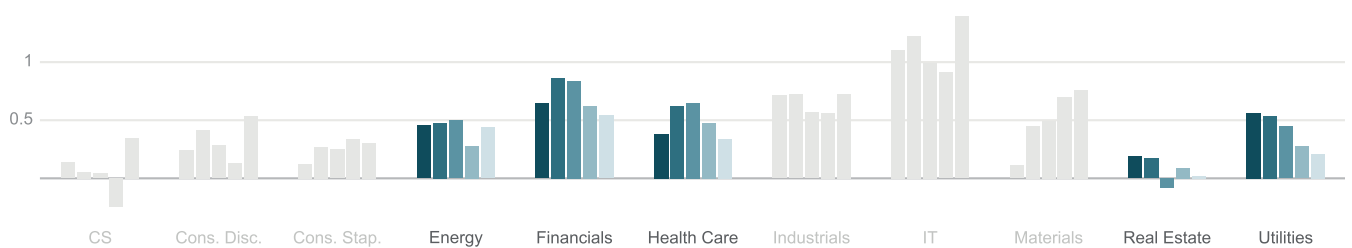
Historical Growth L-S CAGR +5.7%

**128 Ownership Breadth Stability of Institution Ownership by Buyers**

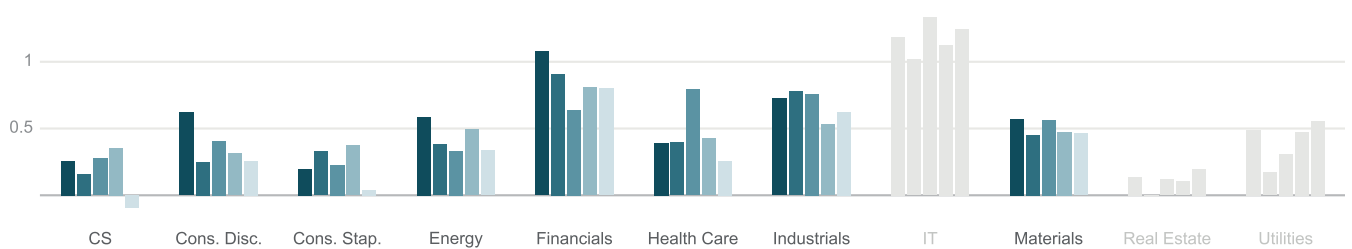
Price Momentum L-S CAGR +5.7%

**129 Dividend per Share Growth**

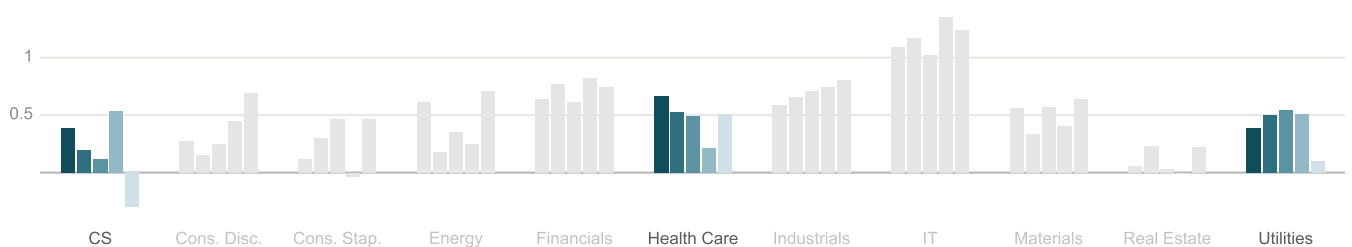
Historical Growth L-S CAGR +5.7%

**130 Accrual Ratio - Balance Sheet**

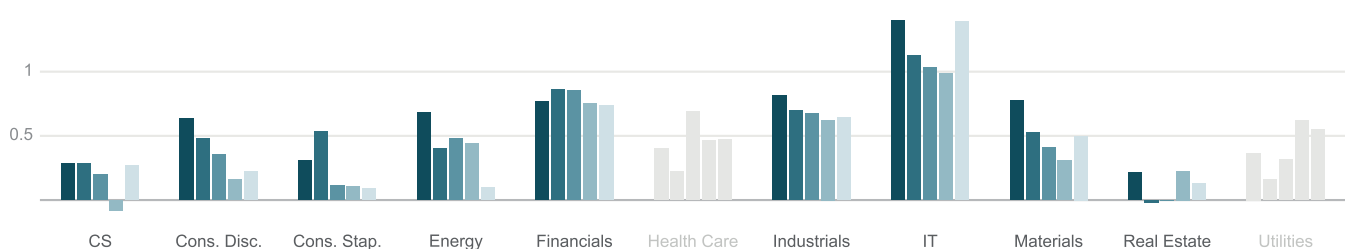
Earnings Quality L-S CAGR +5.6%

**131 Long Term Debt to Assets Ratio**

Capital Efficiency L-S CAGR +5.6%

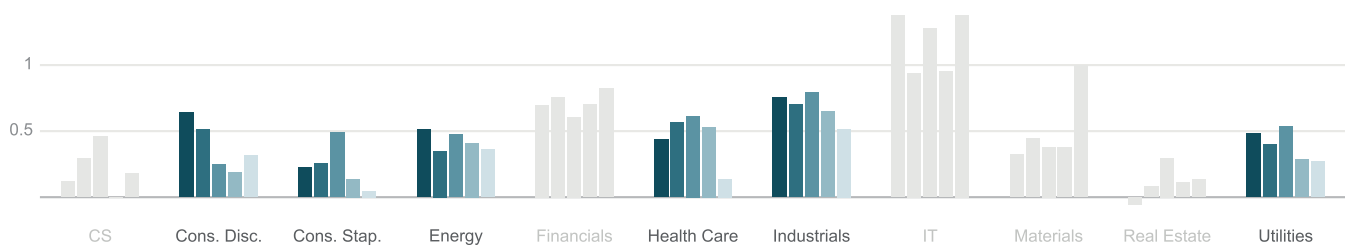
**132 Accrual Ratio - Cash Flows**

Earnings Quality L-S CAGR +5.6%



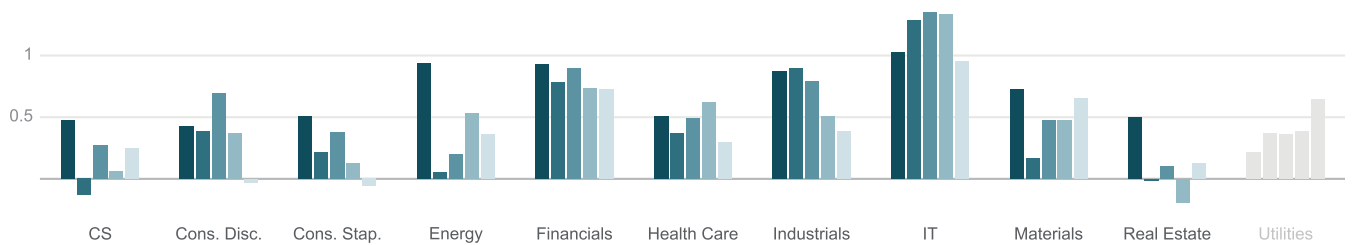
133 ● Year over Year Change of EPS to Sales

Earnings Quality L-S CAGR +5.5%



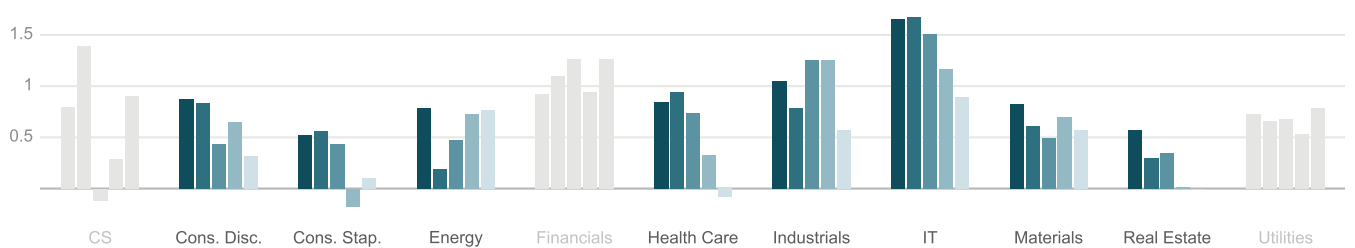
134 ● Free Cash Flow to Enterprise Value

Valuation L-S CAGR +5.5%



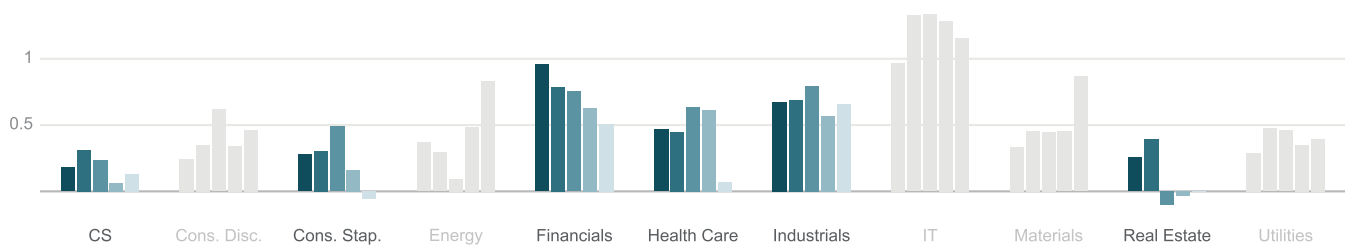
135 ● Operating Cash Flow to Asset

Earnings Quality L-S CAGR +5.5%



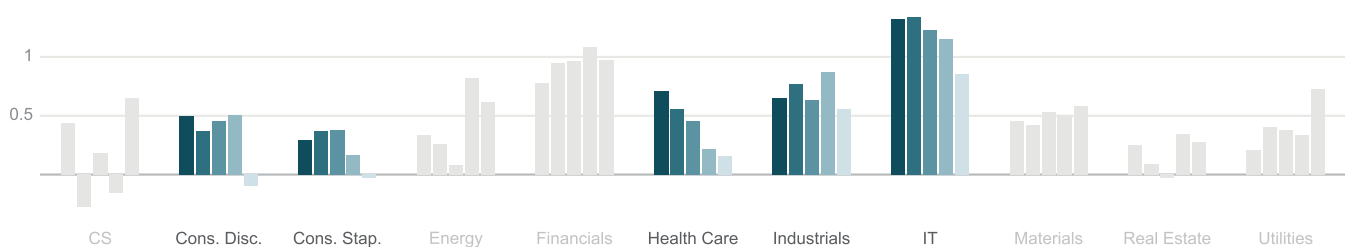
136 ● 3Y Chg in Sales to Price

Historical Growth L-S CAGR +5.5%



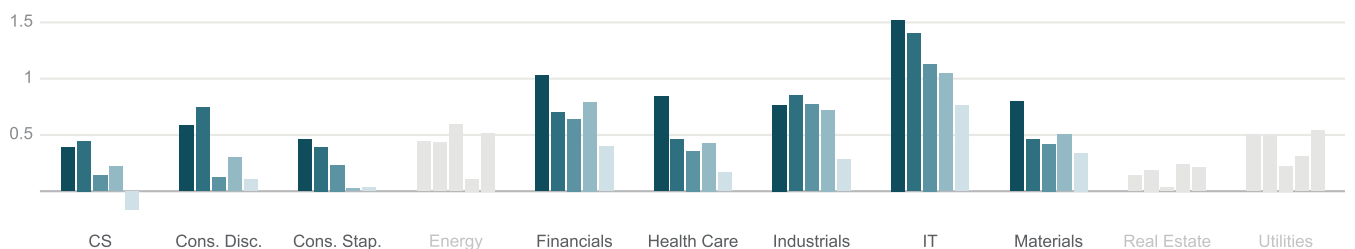
137 ● Interest Coverage Ratio

Capital Efficiency L-S CAGR +5.4%



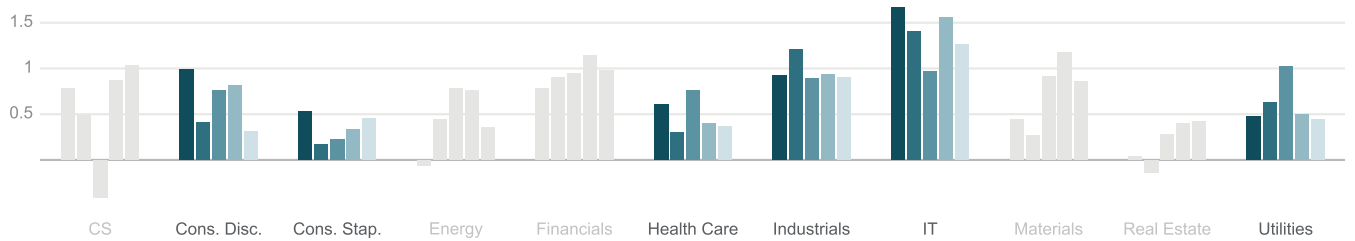
138 ● Number of EPS FY1 Revisions

Analyst Expectations L-S CAGR +5.4%

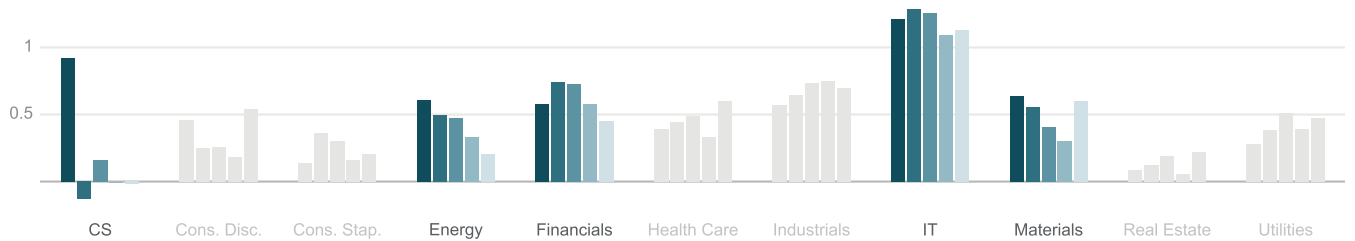


139 ● Num of Quarters of Declines in Receivables and Inventories/TTM Sales

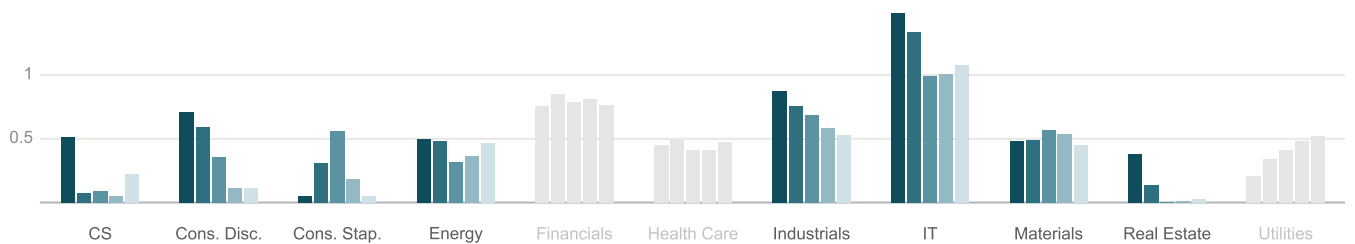
Historical Growth L-S CAGR +5.4%

**140 ● Quick Ratio**

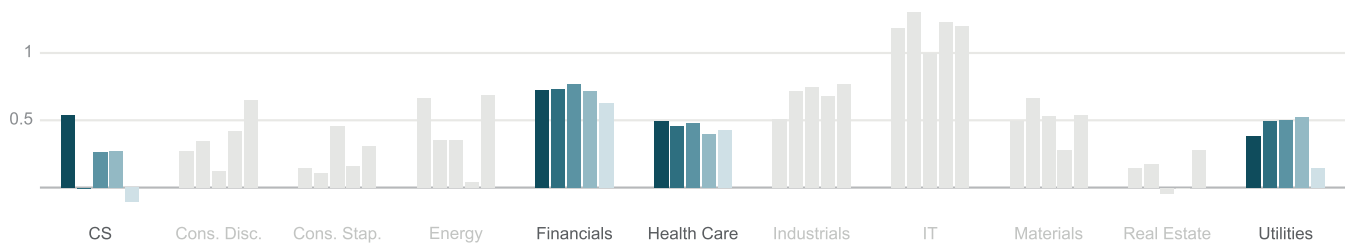
Earnings Quality L-S CAGR +5.3%

**141 ● 1Y Chg in Asset Adjusted Operating Cash Flow**

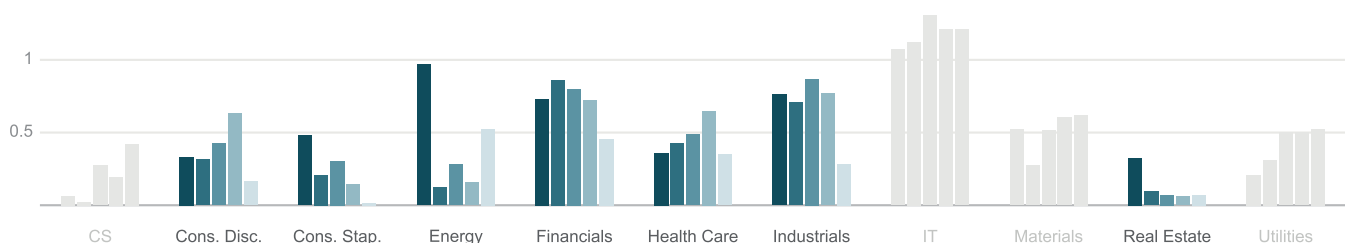
Historical Growth L-S CAGR +5.3%

**142 ● Debt to Assets Ratio**

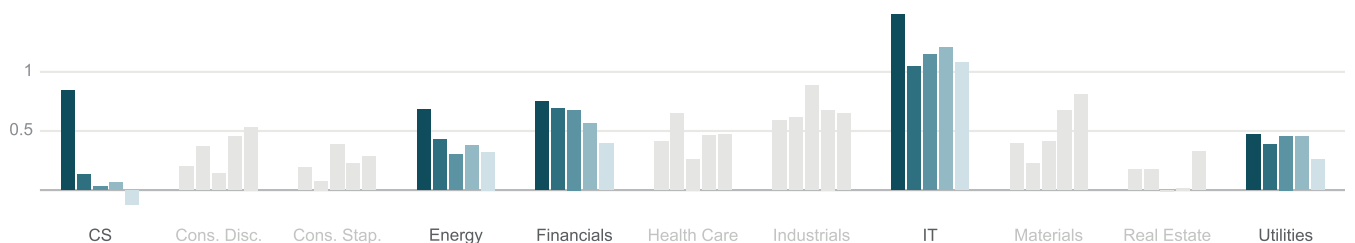
Capital Efficiency L-S CAGR +5.3%

**143 ● Forward Free Cash Flow to Price Ratio**

Valuation L-S CAGR +5.3%

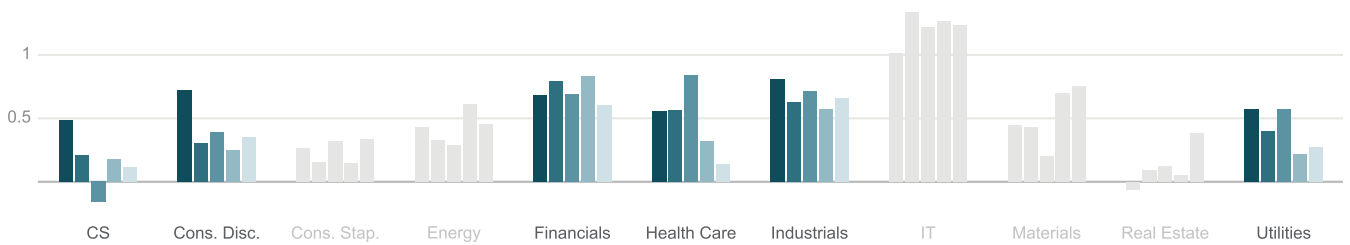
**144 ● Working Capital to Total Assets**

Earnings Quality L-S CAGR +5.2%



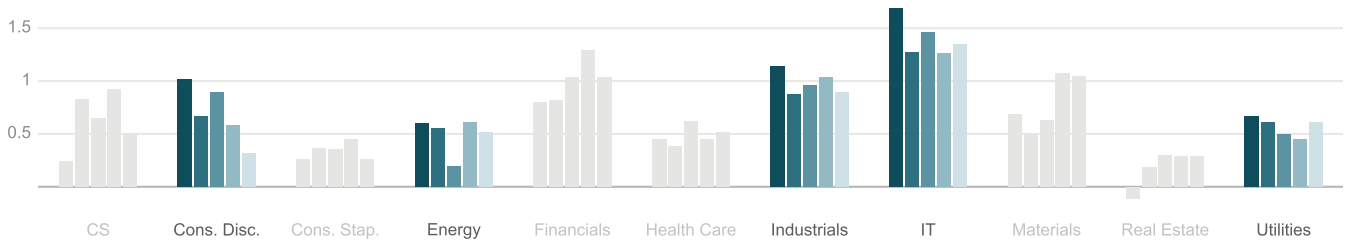
145 ● Net Income Stability

Earnings Quality L-S CAGR +5.2%



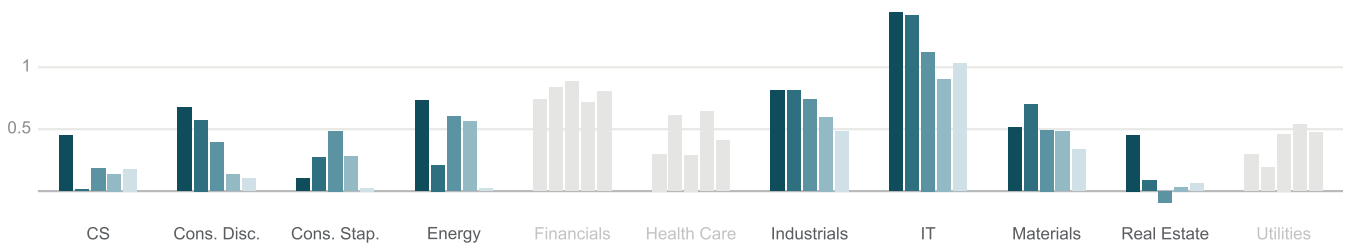
146 ● Return on Equity Stability

Historical Growth L-S CAGR +5.2%



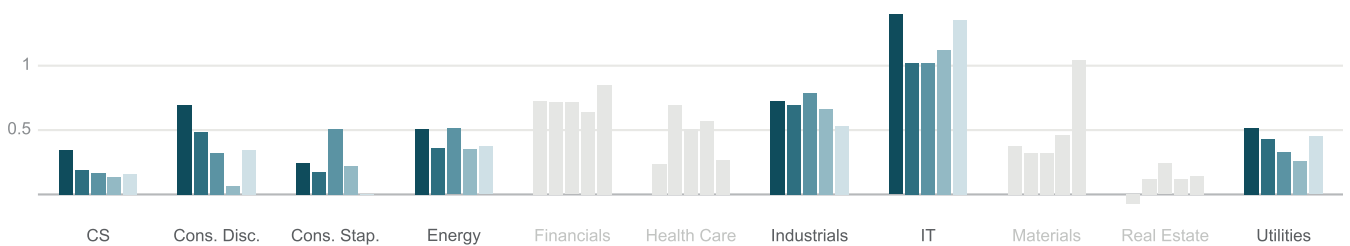
147 ● 1Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +5.2%



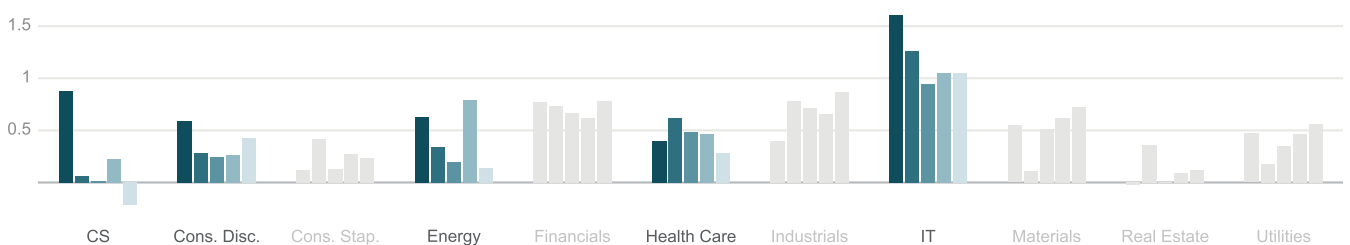
148 ● Change EPS to Sales

Earnings Quality L-S CAGR +5.2%



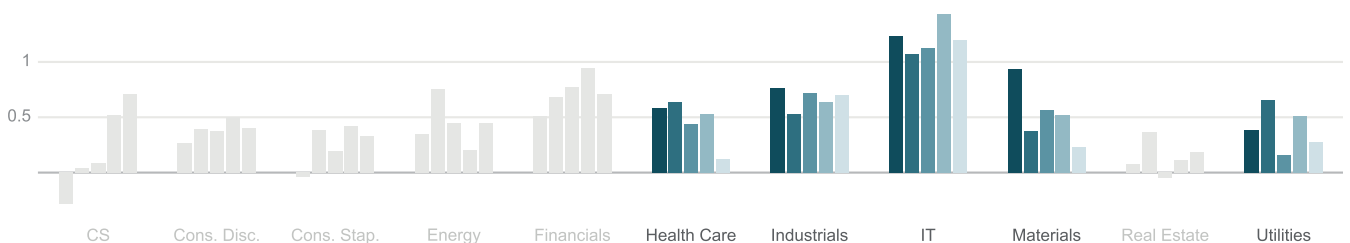
149 ● Cash to Total Assets

Valuation L-S CAGR +5.1%



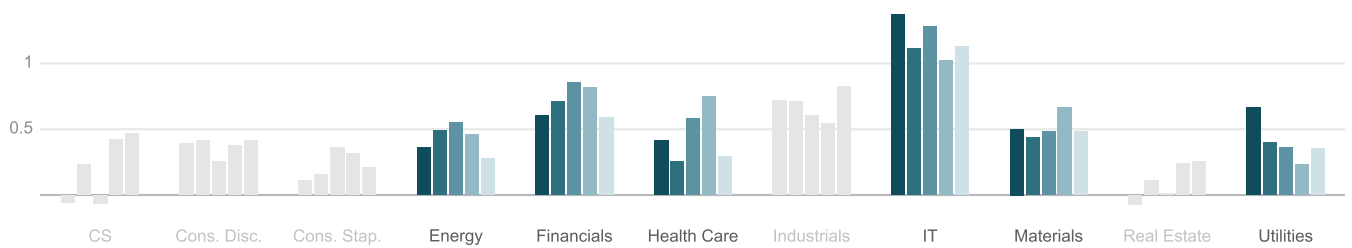
150 ● 6M Momentum in Trailing 12M Sales

Historical Growth L-S CAGR +5.1%

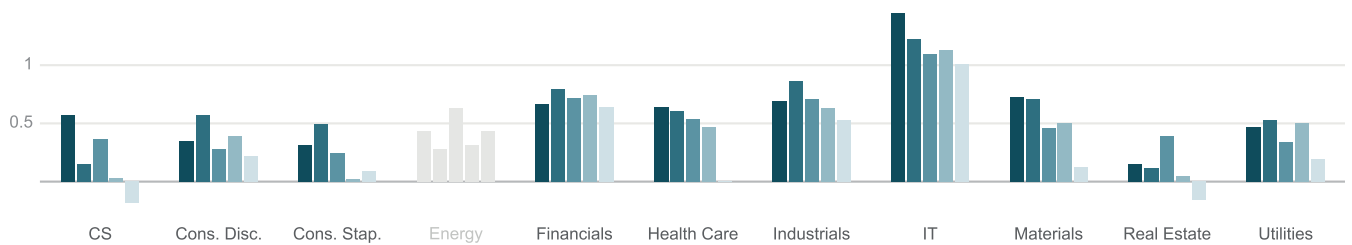


151 ● **Change in Number of Holders**

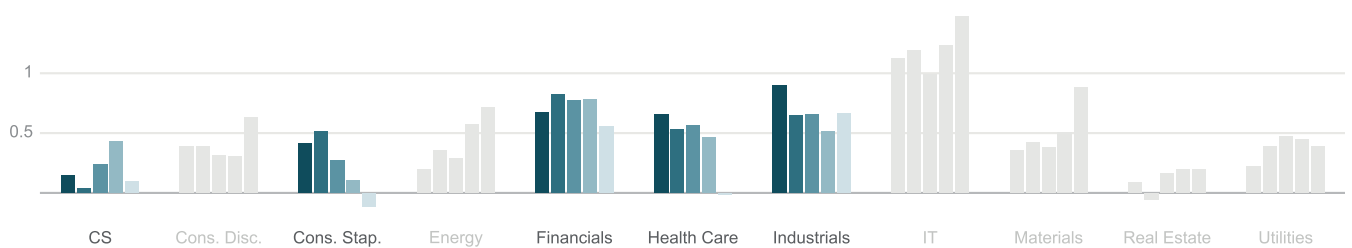
Price Momentum L-S CAGR +5.1%

152 ● **1Y Chg in Amihud**

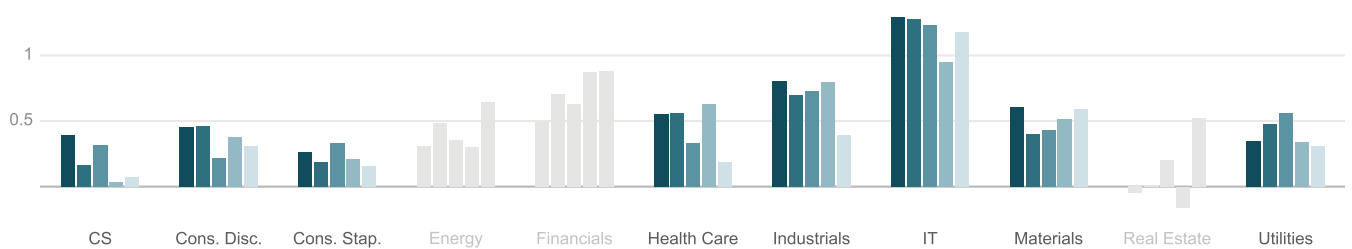
Volatility L-S CAGR +5.0%

153 ● **24M Residual Return Variance**

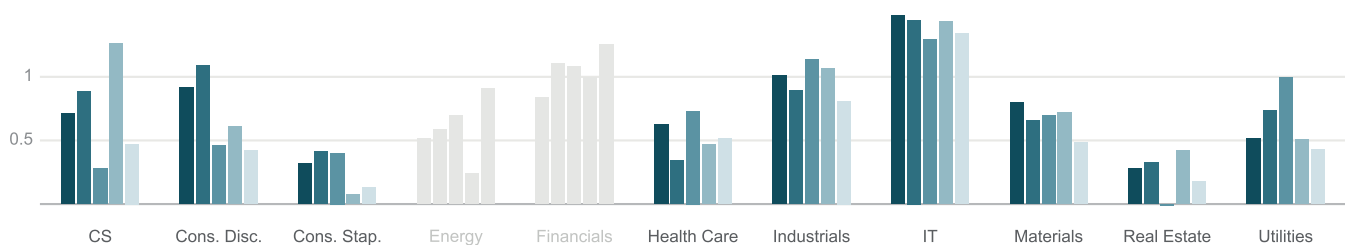
Price Momentum L-S CAGR +5.0%

154 ● **Closing Price to 52 Week High**

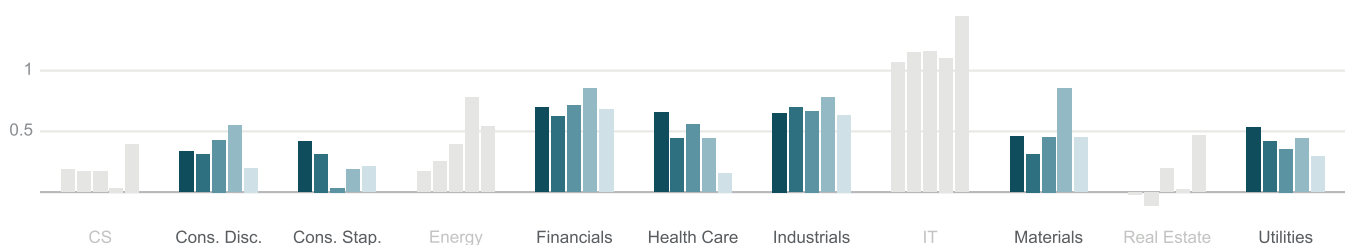
Price Momentum L-S CAGR +5.0%

155 ● **Free Cash Flow Stability**

Earnings Quality L-S CAGR +5.0%

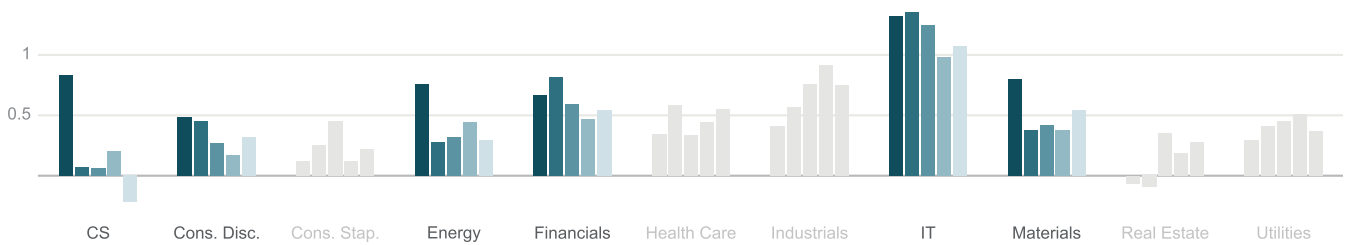
156 ● **30 Day Coefficient of Variation of Volume to Price**

Price Momentum L-S CAGR +5.0%

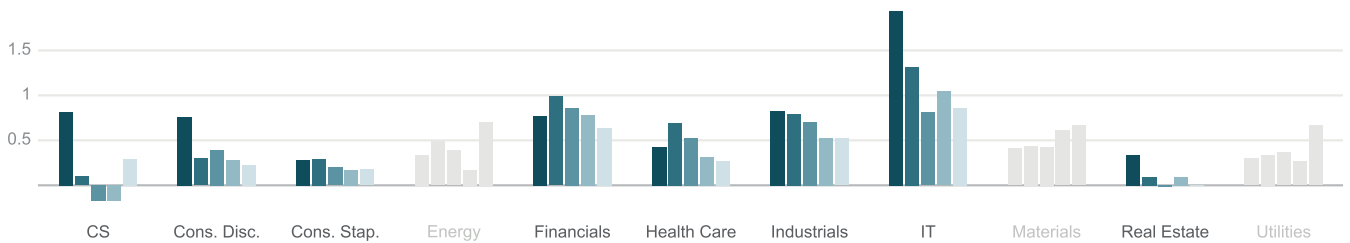


157 ● **Cash Ratio**

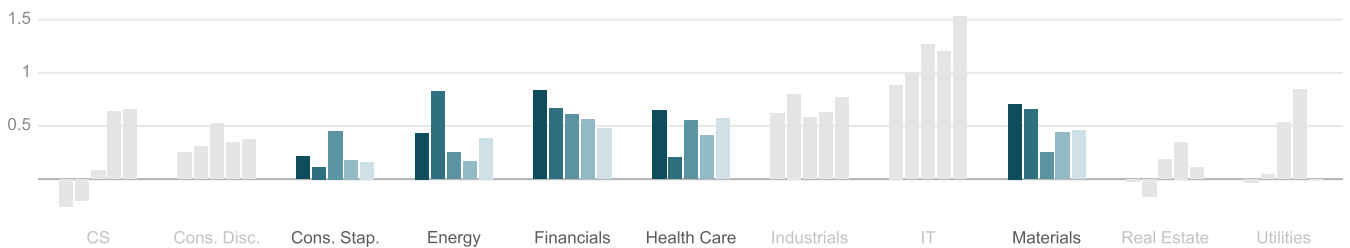
Earnings Quality L-S CAGR +5.0%

158 ● **3Y Chg in Asset Adjusted Operating Cash Flow**

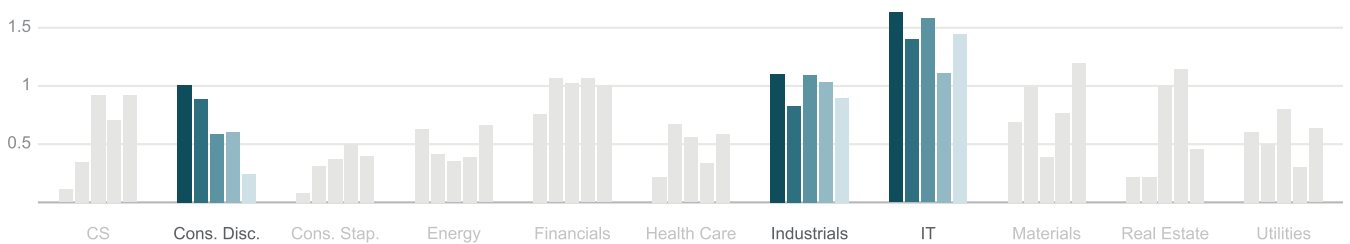
Historical Growth L-S CAGR +4.9%

159 ● **SG&A to Sales**

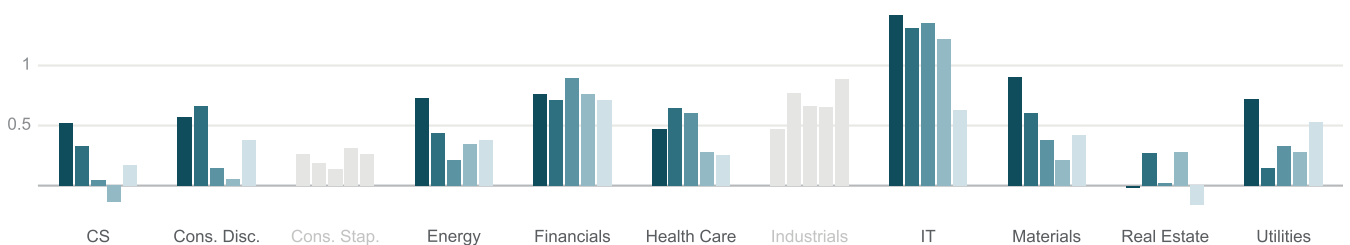
Earnings Quality L-S CAGR +4.9%

160 ● **Return on Assets Stability**

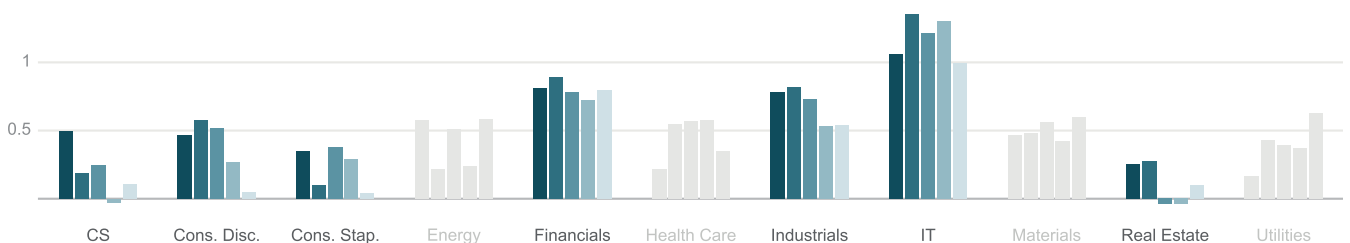
Historical Growth L-S CAGR +4.9%

161 ● **Receivables Turnover Ratio**

Capital Efficiency L-S CAGR +4.9%

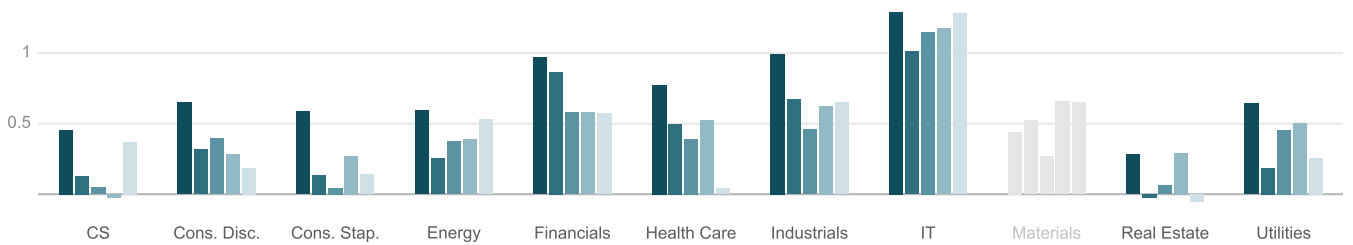
162 ● **1Y Chg in Operating Cash Flow to Price**

Historical Growth L-S CAGR +4.8%



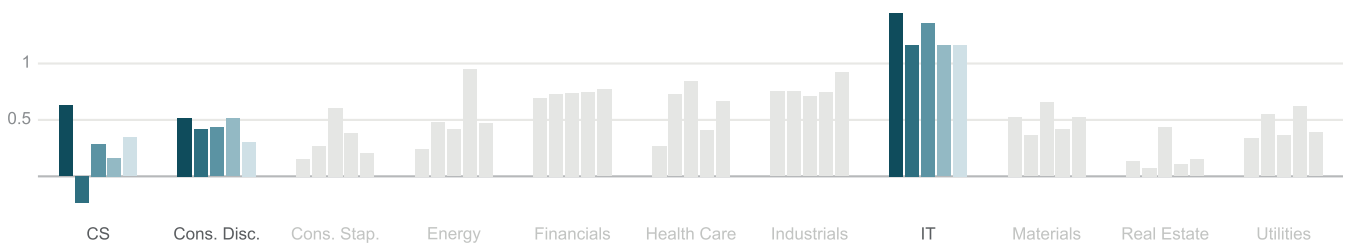
163 ● 1Y Chg in Shares Outstanding

Capital Efficiency L-S CAGR +4.8%



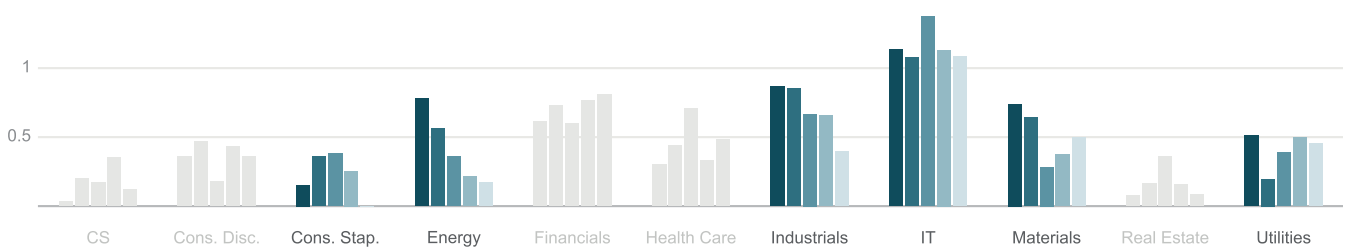
164 ● 1Y Chg in Stock Buybacks

Capital Efficiency L-S CAGR +4.8%



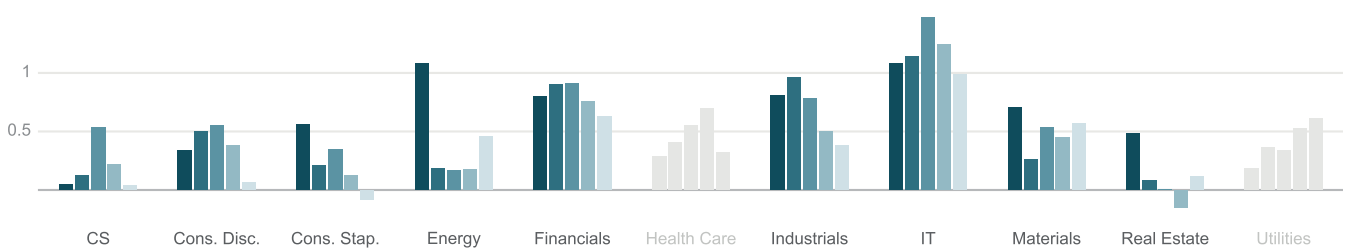
165 ● Depreciation to Capex Ratio

Earnings Quality L-S CAGR +4.8%



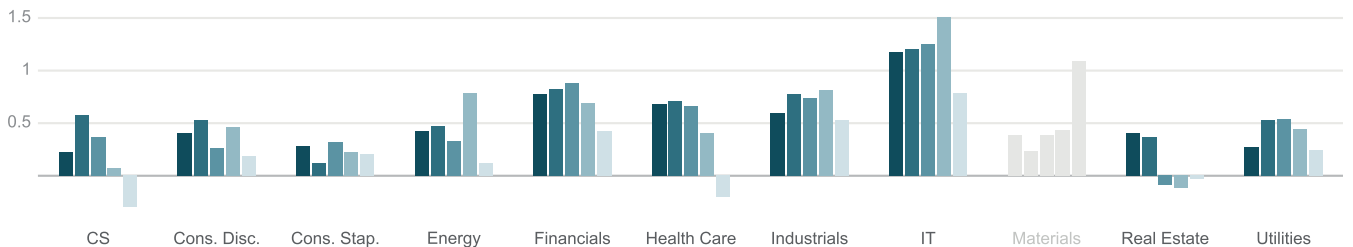
166 ● Free Cash Flow to Price

Valuation L-S CAGR +4.7%



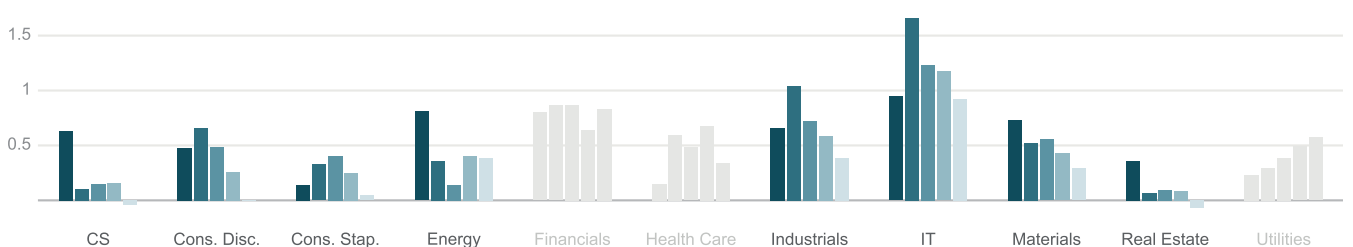
167 ● Asset Turnover

Earnings Quality L-S CAGR +4.7%



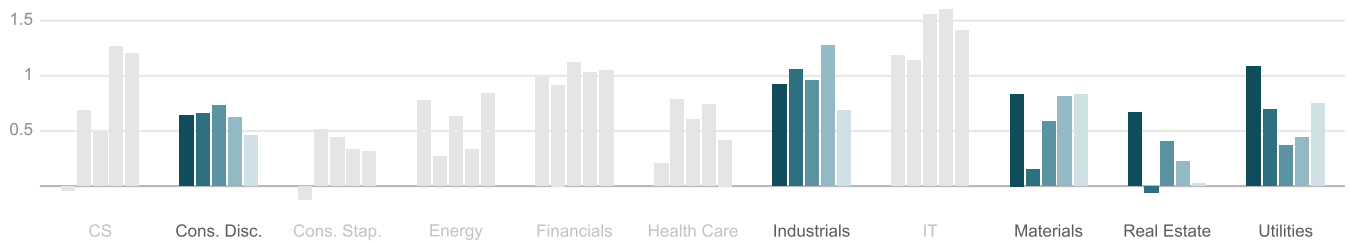
168 ● 1Y Chg in Free Cash Flow to Price

Historical Growth L-S CAGR +4.6%

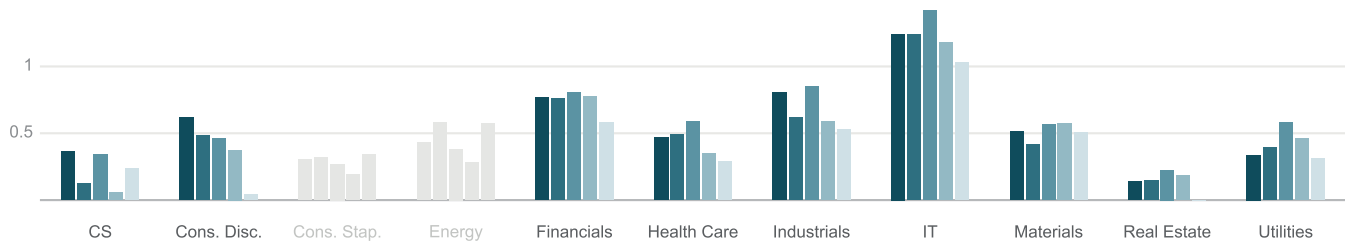


169 ● **Adj Forward Free Cash Flow to Price Ratio**

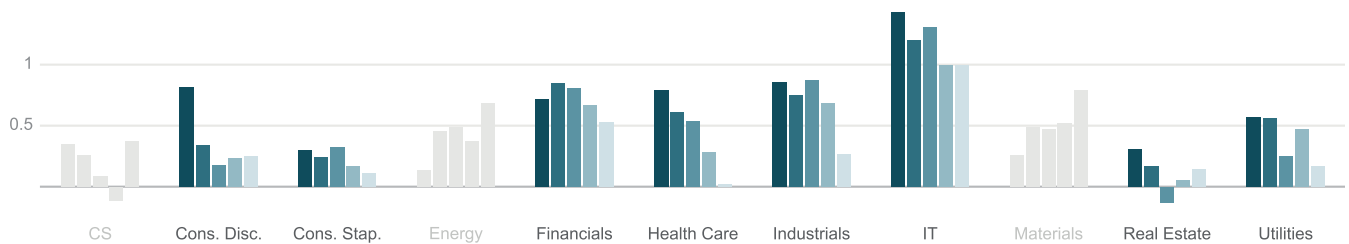
Valuation L-S CAGR +4.6%

170 ● **EPS Stability**

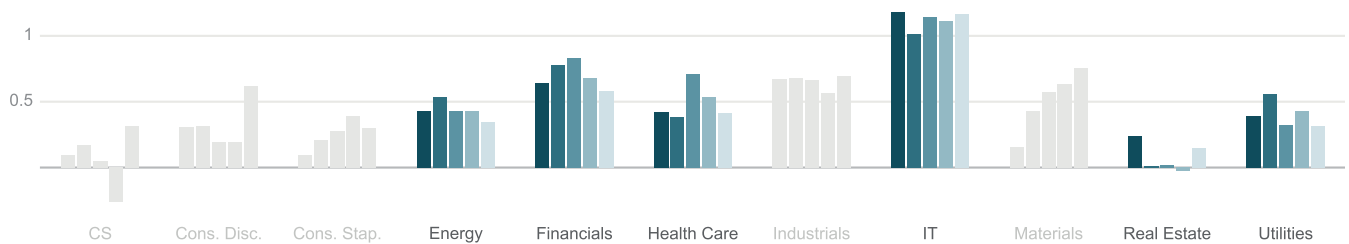
Historical Growth L-S CAGR +4.6%

171 ● **Cash Flow to Invested Capital**

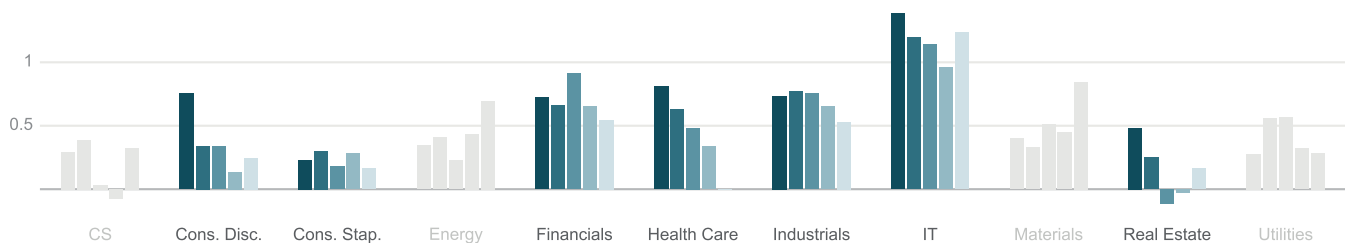
Earnings Quality L-S CAGR +4.6%

172 ● **Dividend Growth**

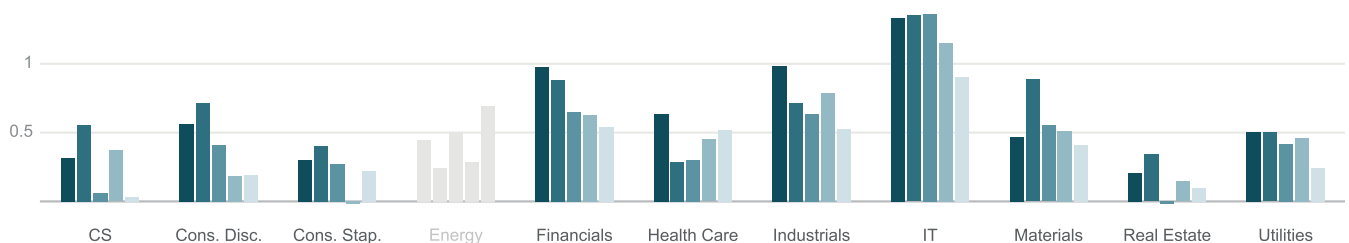
Historical Growth L-S CAGR +4.5%

173 ● **Return on Invested Capital**

Capital Efficiency L-S CAGR +4.5%

174 ● **Standardized Unexpected Earnings**

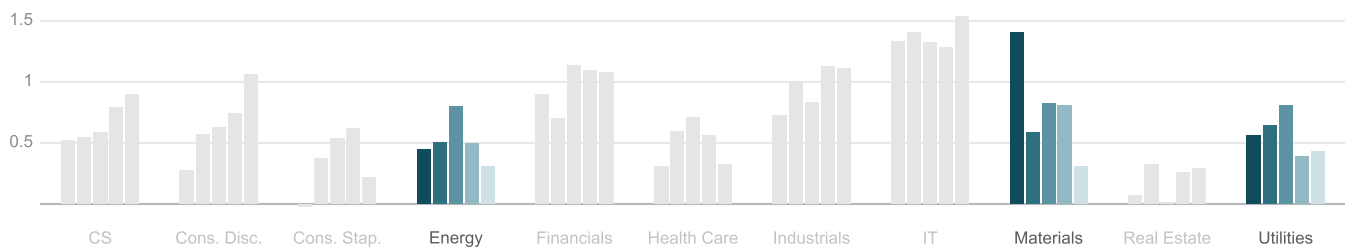
Analyst Expectations L-S CAGR +4.4%



175 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**

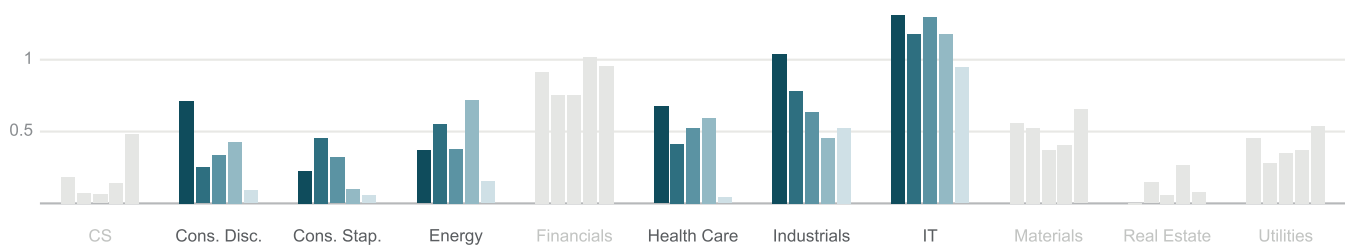
Historical Growth

L-S CAGR +4.3%

176 ● **Cash Burn Rate**

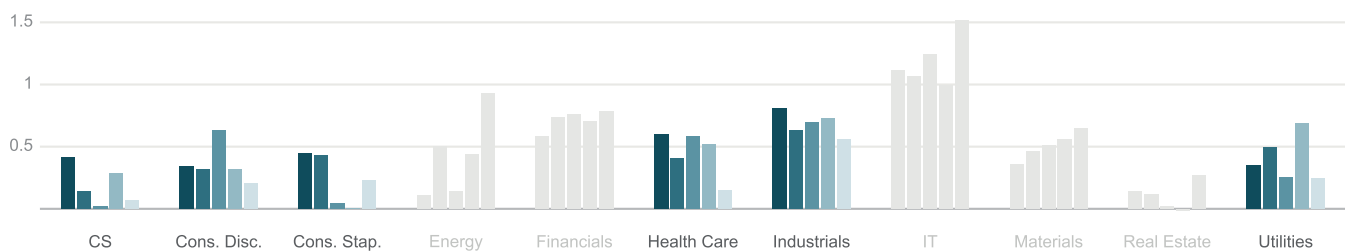
Earnings Quality

L-S CAGR +4.2%

177 ● **20 Day Coefficient of Variation of Volume to Price**

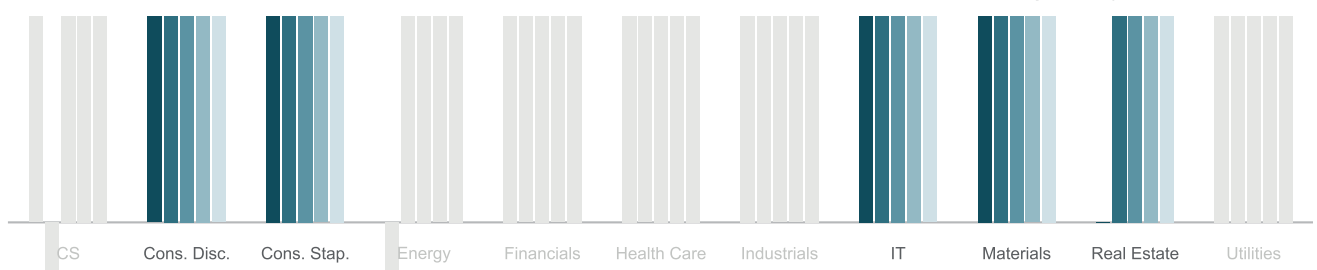
Price Momentum

L-S CAGR +4.2%

178 ● **Inventory to Assets Ratio**

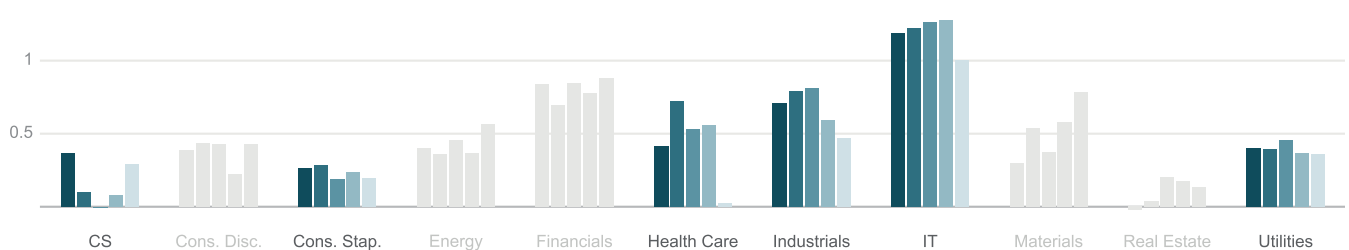
Earnings Quality

L-S CAGR +4.1%

179 ● **3Y Chg in Cash Flow to Price**

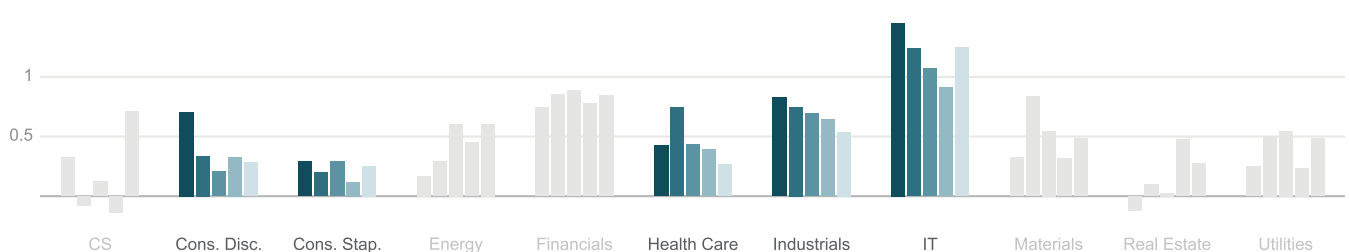
Historical Growth

L-S CAGR +4.1%

180 ● **EBIT Margin**

Earnings Quality

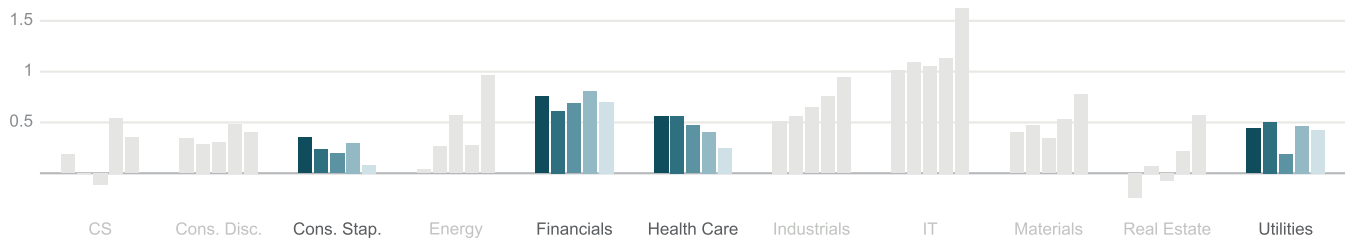
L-S CAGR +4.1%



181 ● 60 Day Coefficient of Variation of Volume to Price

Price Momentum

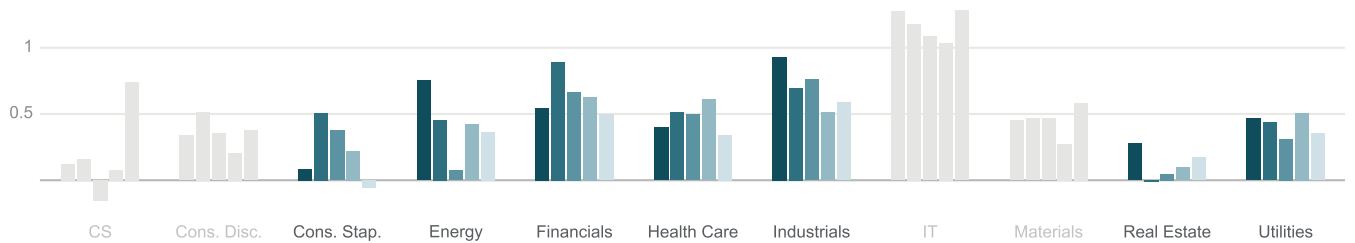
L-S CAGR +4.0%



182 ● Degree of Financial Leverage

Capital Efficiency

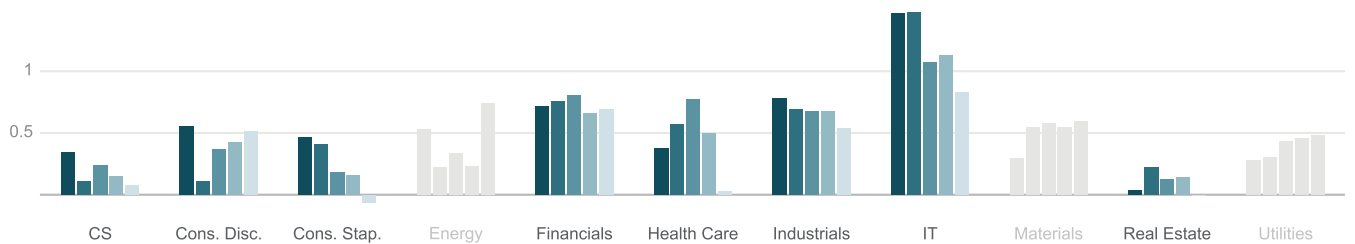
L-S CAGR +4.0%



183 ● 3Y Chg in Sales Turnover

Historical Growth

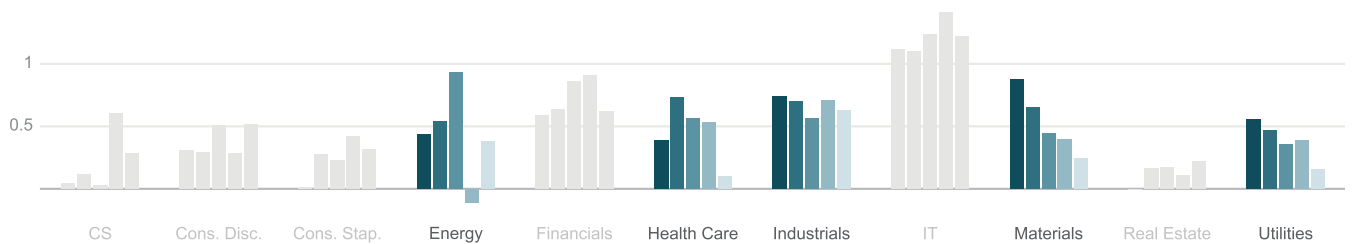
L-S CAGR +4.0%



184 ● 3M Momentum in Trailing 12M Sales

Historical Growth

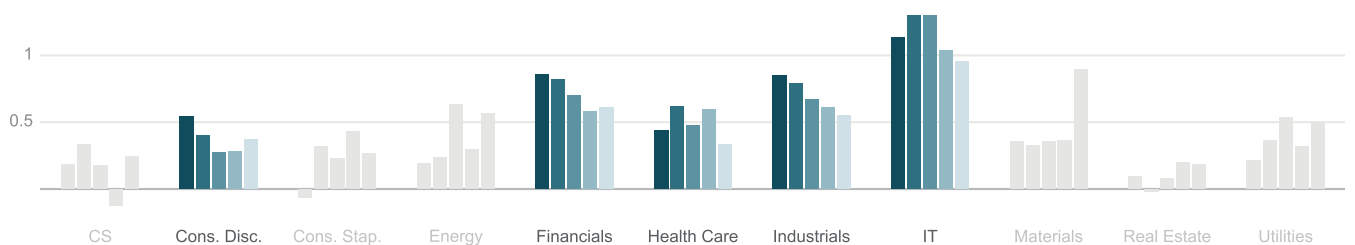
L-S CAGR +4.0%



185 ● 3Y Average Annual Earnings Growth

Historical Growth

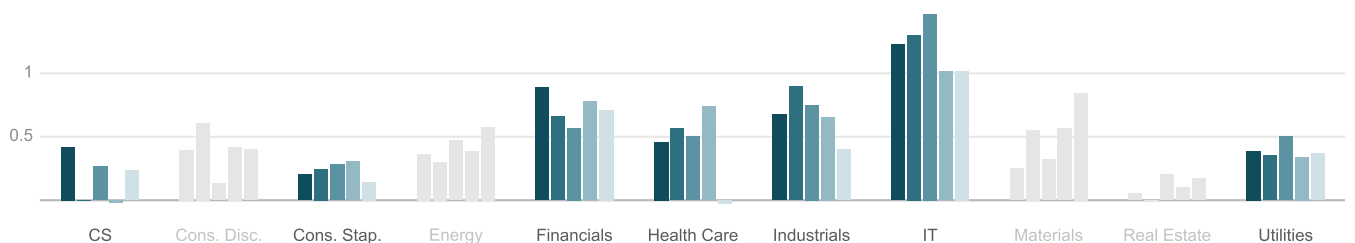
L-S CAGR +4.0%



186 ● 3Y Chg in Earnings to Price

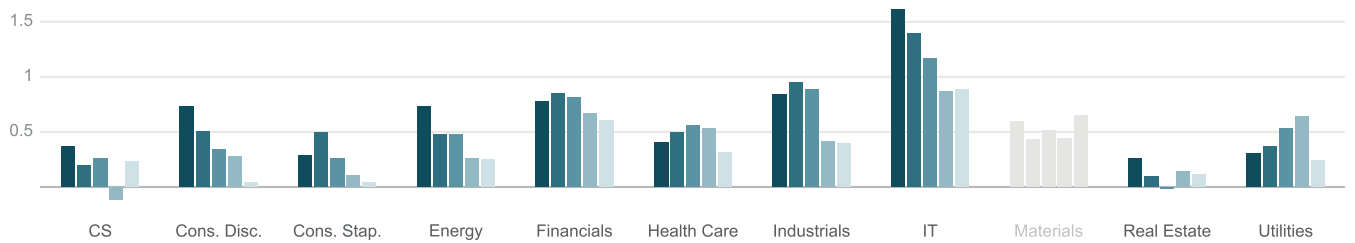
Historical Growth

L-S CAGR +3.9%

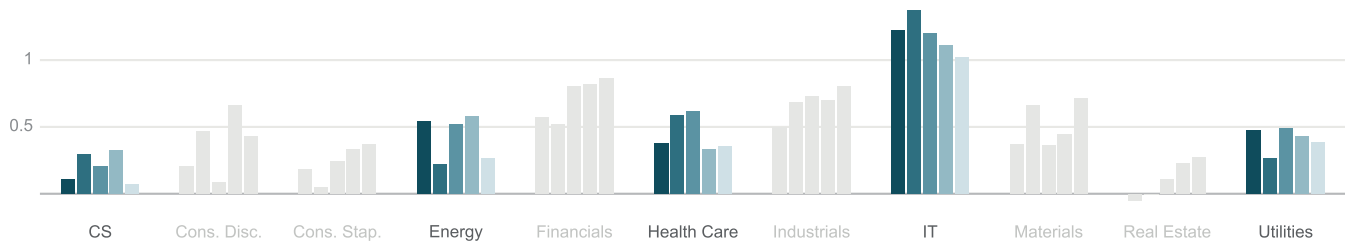


187 Free Cash Flow to Sales

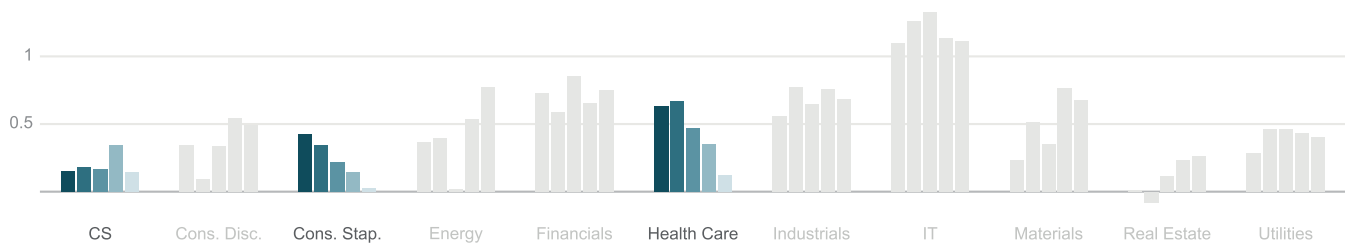
Earnings Quality L-S CAGR +3.9%

**188 Change in Number of Buyers**

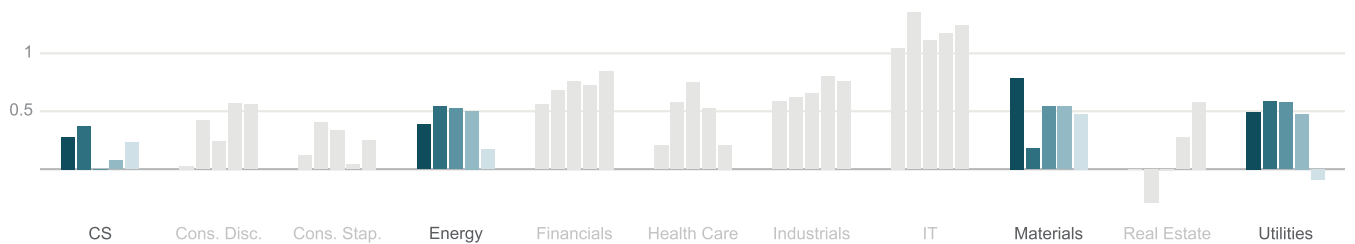
Price Momentum L-S CAGR +3.9%

**189 130 Day Minimum Return**

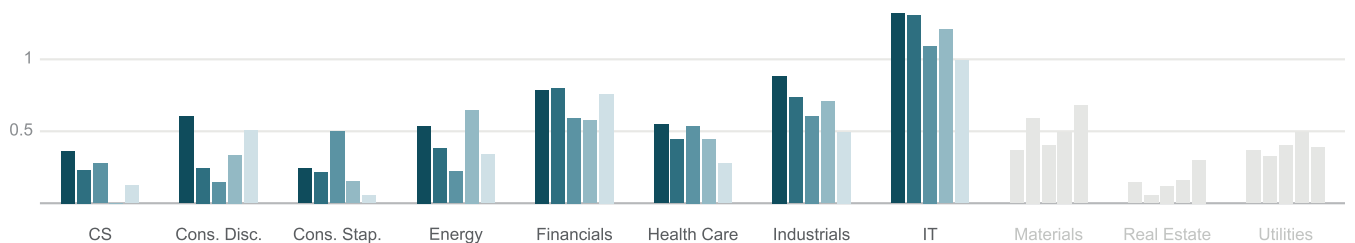
Volatility L-S CAGR +3.8%

**190 10 Day MACD Trend**

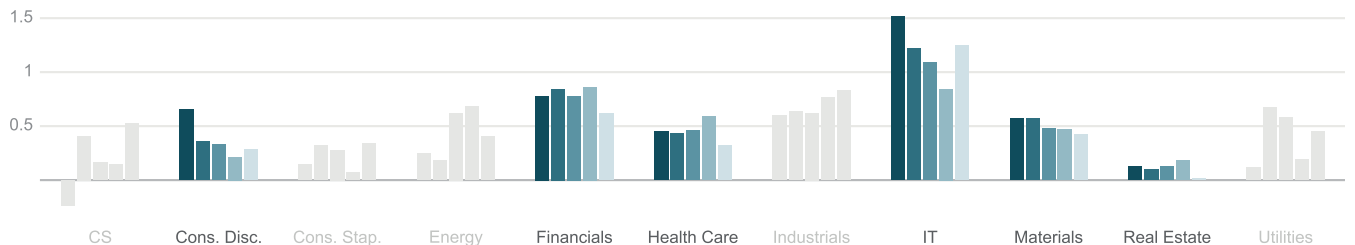
Price Momentum L-S CAGR +3.8%

**191 ROA 60 Month Slope**

Earnings Quality L-S CAGR +3.8%

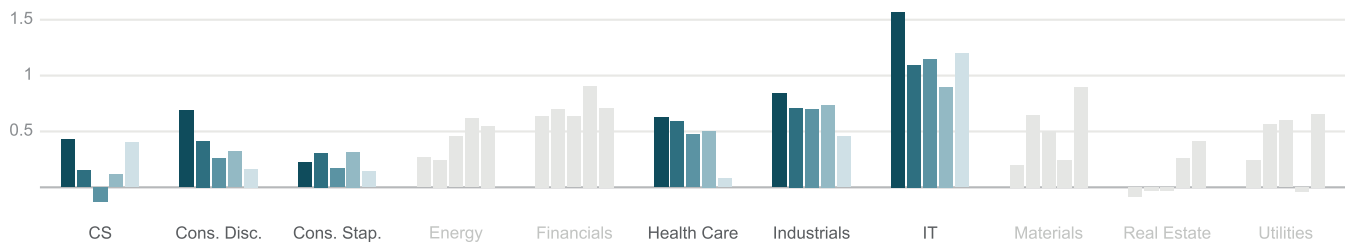
**192 Cash Conversion Cycle**

Earnings Quality L-S CAGR +3.8%



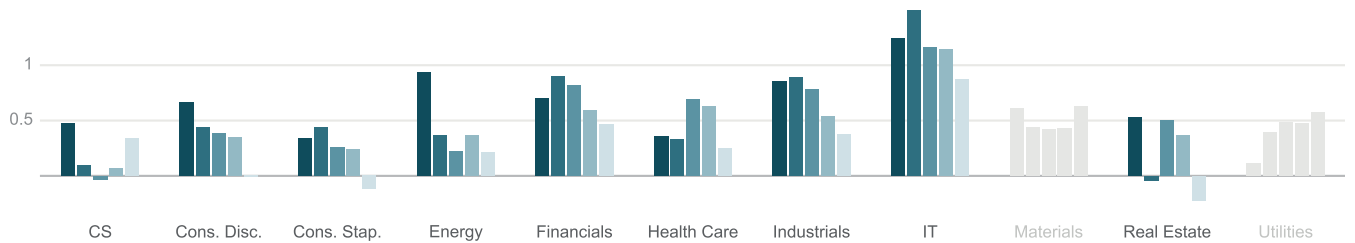
193 ● Net Profit Margin

Earnings Quality L-S CAGR +3.8%



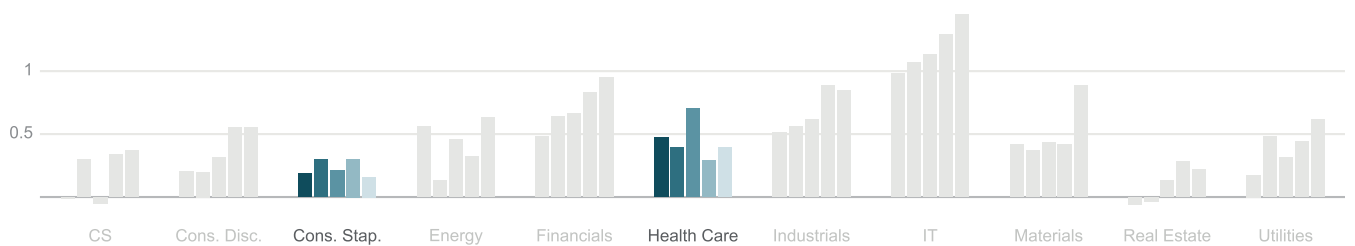
194 ● Capital Acquisition Ratio

Capital Efficiency L-S CAGR +3.8%



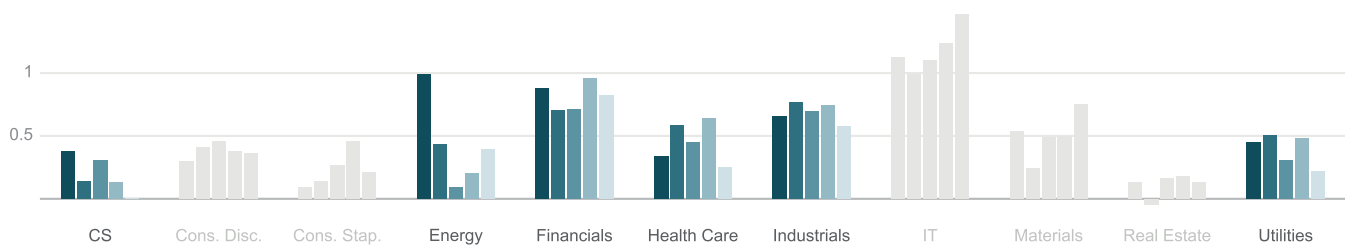
195 ● Share Turnover

Price Momentum L-S CAGR +3.7%



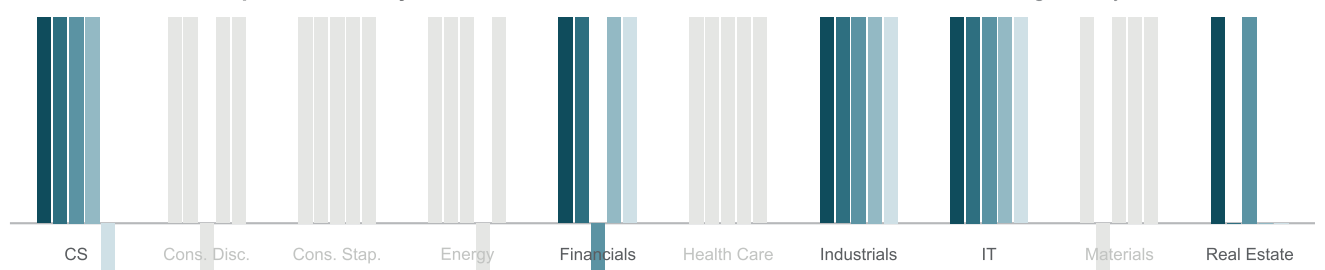
196 ● Sales to Gross Profit Margin Growth

Earnings Quality L-S CAGR +3.7%



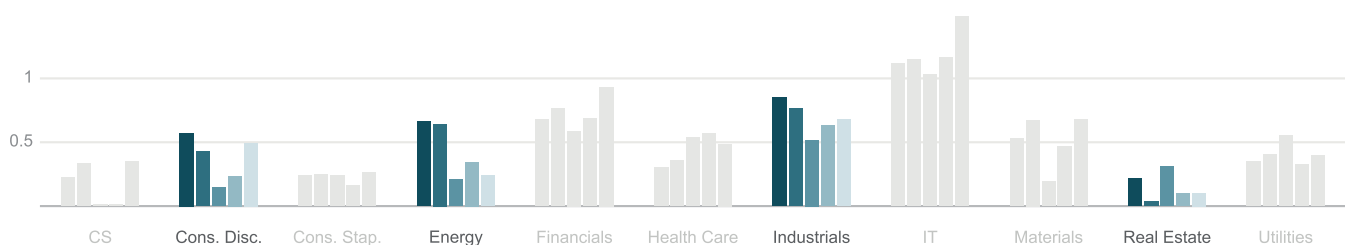
197 ● Research & Development Intensity

Earnings Quality L-S CAGR +3.7%



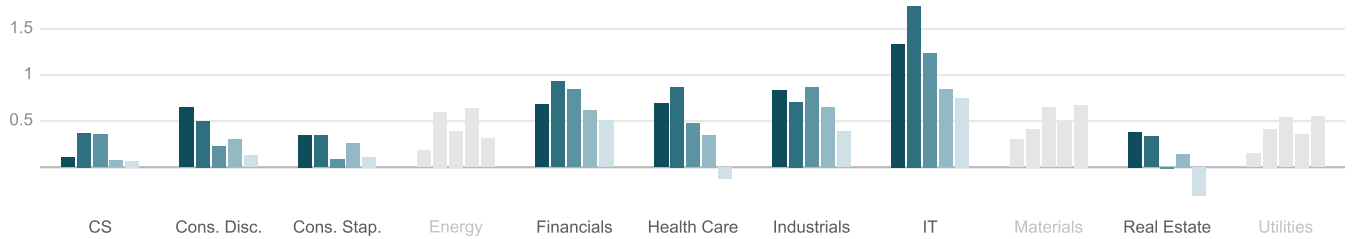
198 ● Asset Adjusted Capital Investments

Earnings Quality L-S CAGR +3.7%

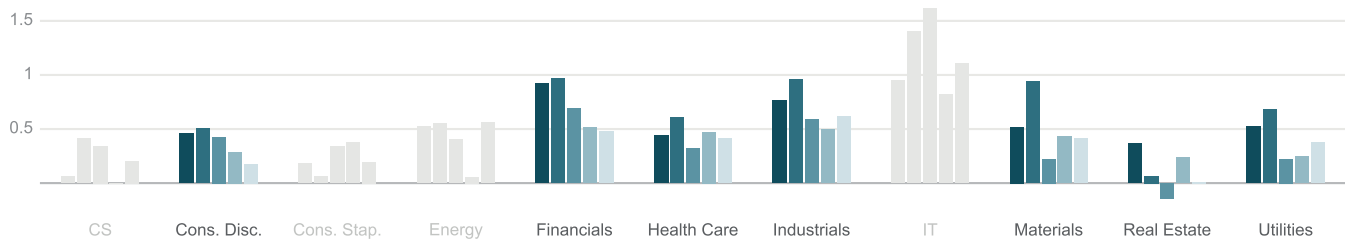


199 ● Cash Flow Return on Invested Capital

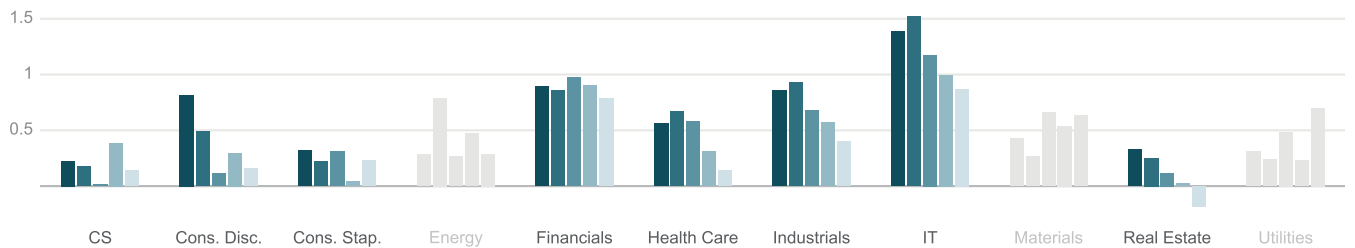
Capital Efficiency L-S CAGR +3.6%

**200 ● Institution Ownership Level**

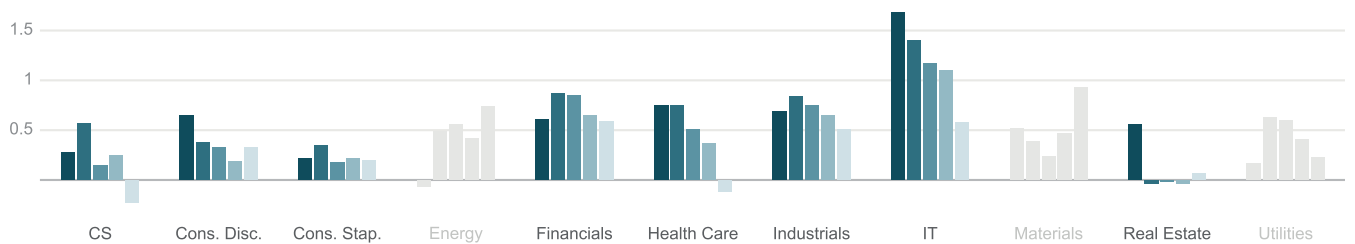
Price Momentum L-S CAGR +3.6%

**201 ● Operating Cash Flow to Equity**

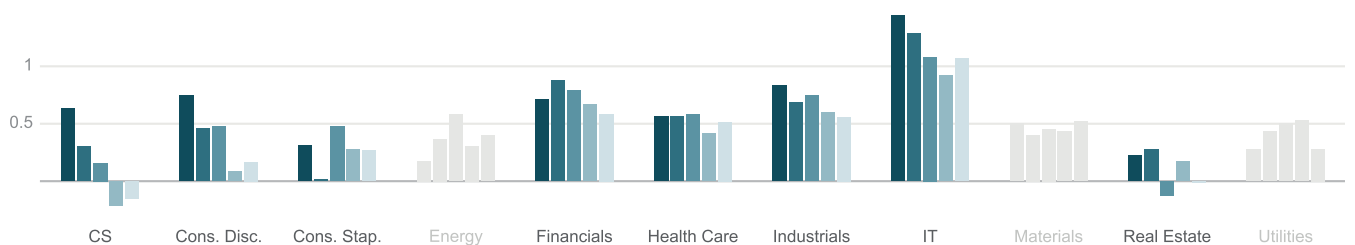
Capital Efficiency L-S CAGR +3.6%

**202 ● Gross Profit to Assets**

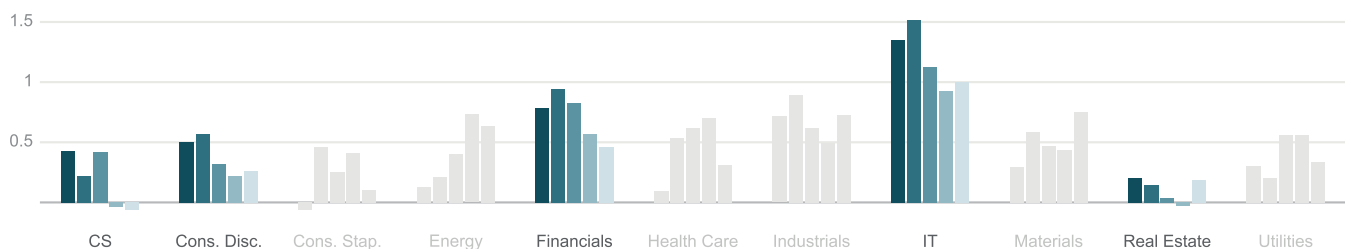
Capital Efficiency L-S CAGR +3.6%

**203 ● Payout Ratio**

Capital Efficiency L-S CAGR +3.6%

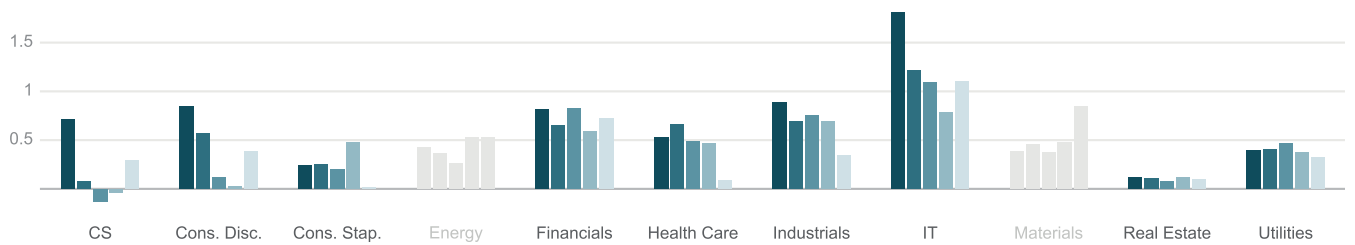
**204 ● 3Y Average Annual Sales Growth**

Historical Growth L-S CAGR +3.5%



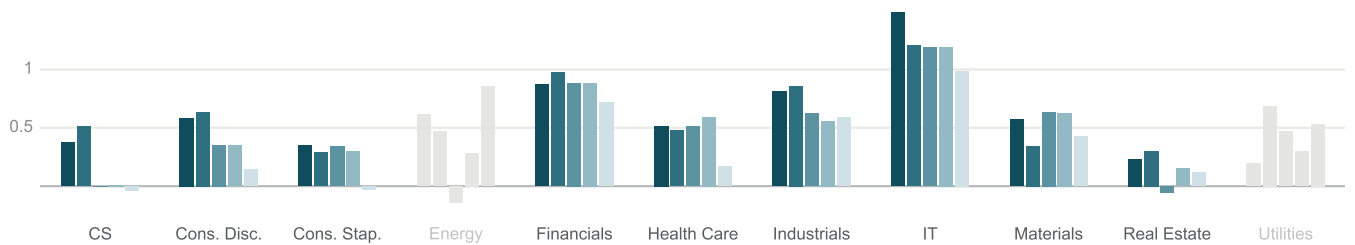
205 ● 3Y Chg in Asset Adjusted EPS

Historical Growth L-S CAGR +3.5%



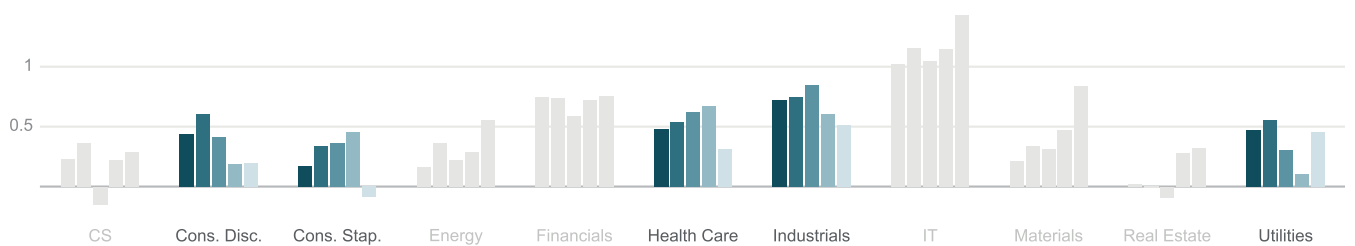
206 ● 3Y Chg in Free Cash Flow per Share

Historical Growth L-S CAGR +3.5%



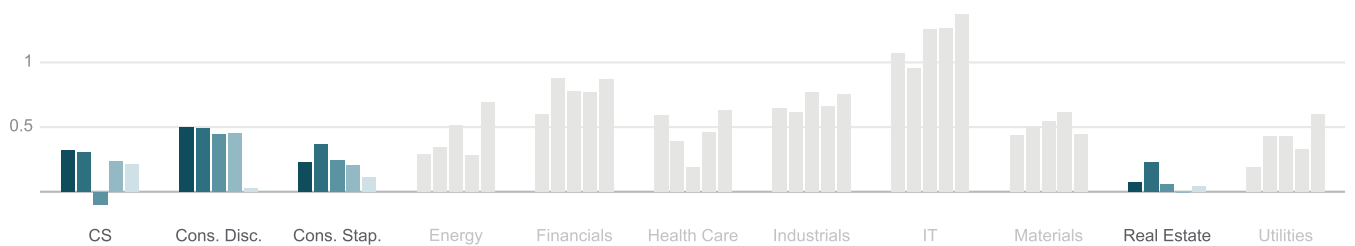
207 ● 1Y Chg in Earnings per Share

Historical Growth L-S CAGR +3.5%



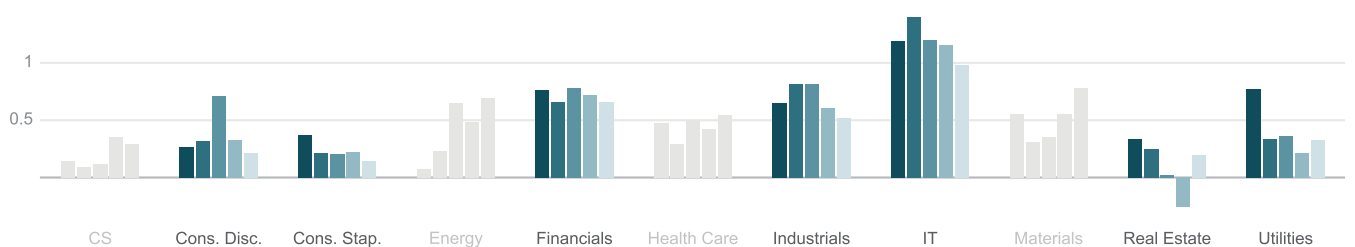
208 ● Unexpected Receivables Change

Earnings Quality L-S CAGR +3.5%



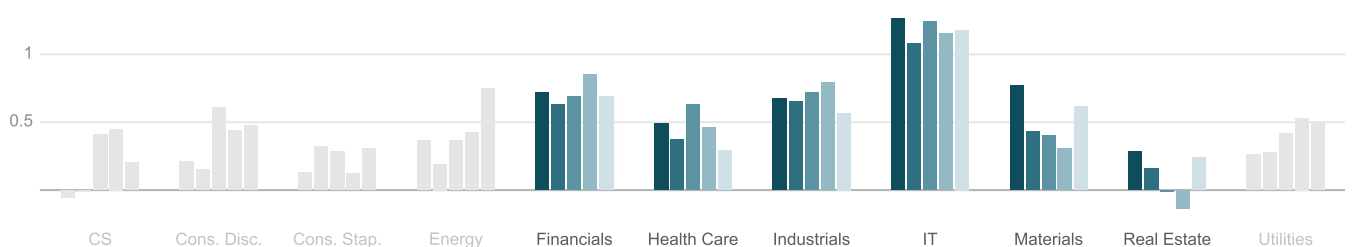
209 ● 6M Chg in 36M CAPM Alpha

Price Momentum L-S CAGR +3.5%



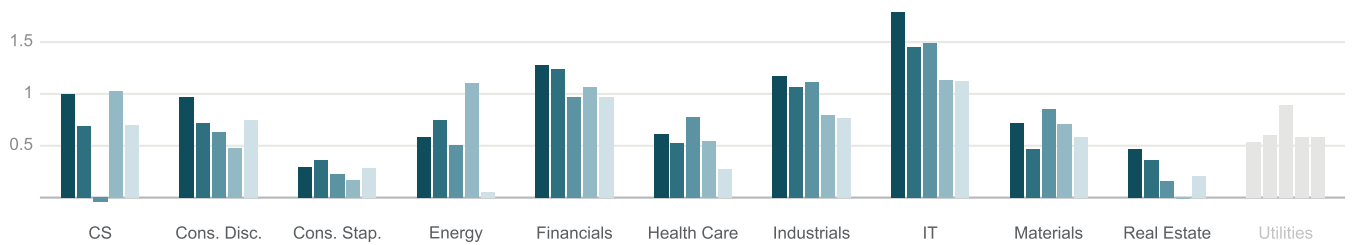
210 ● 1Y Chg in Sales to Earnings

Earnings Quality L-S CAGR +3.5%

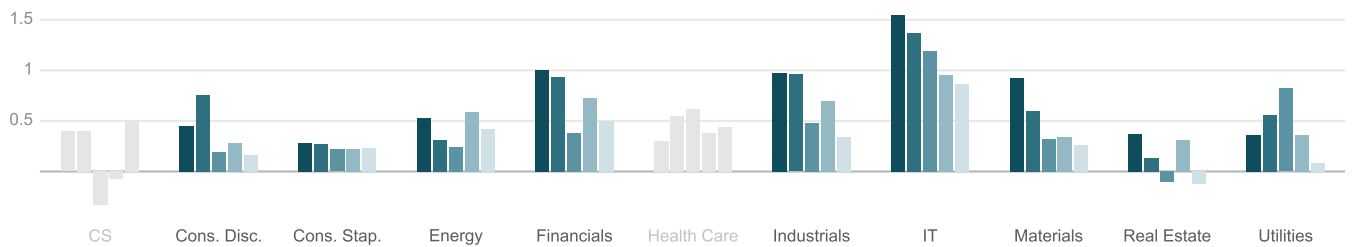


211 Operating Cash Flow Stability

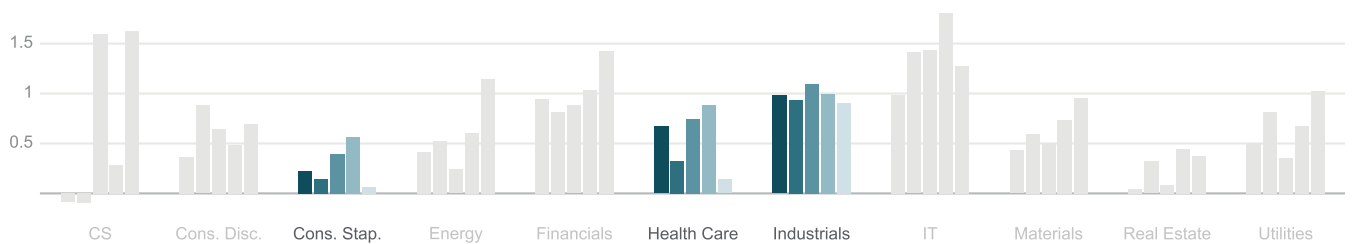
Earnings Quality L-S CAGR +3.4%

**212 6 Month Average Share Turn Over**

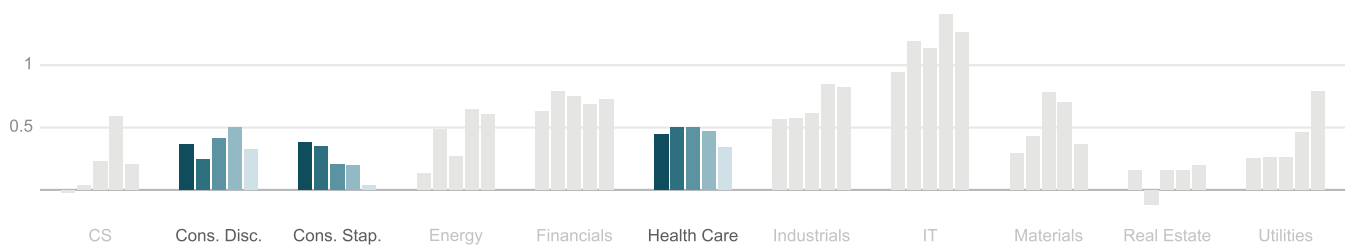
Price Momentum L-S CAGR +3.4%

**213 EBIT to Price**

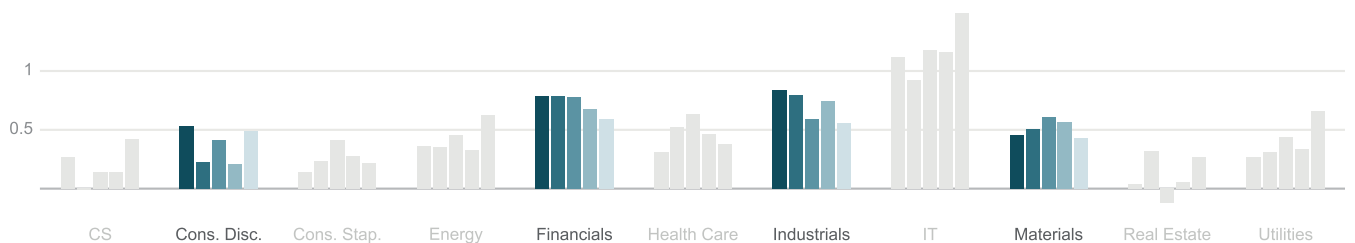
Valuation L-S CAGR +3.4%

**214 Ownership Breadth Stability of Institution Ownership by Holders**

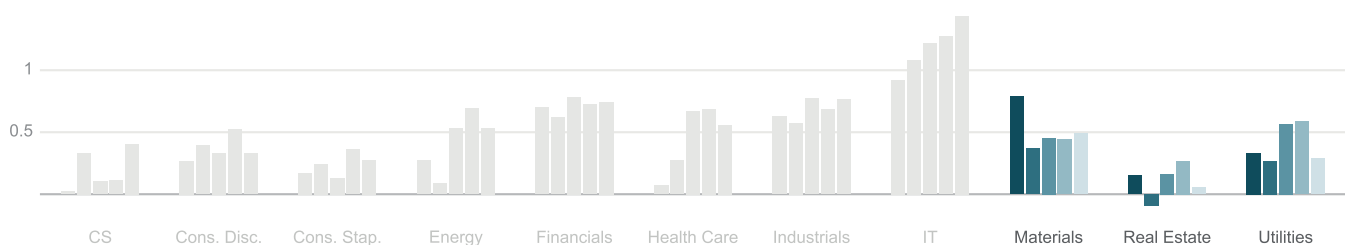
Price Momentum L-S CAGR +3.3%

**215 Year over Year Change of Total Debt**

Capital Efficiency L-S CAGR +3.3%

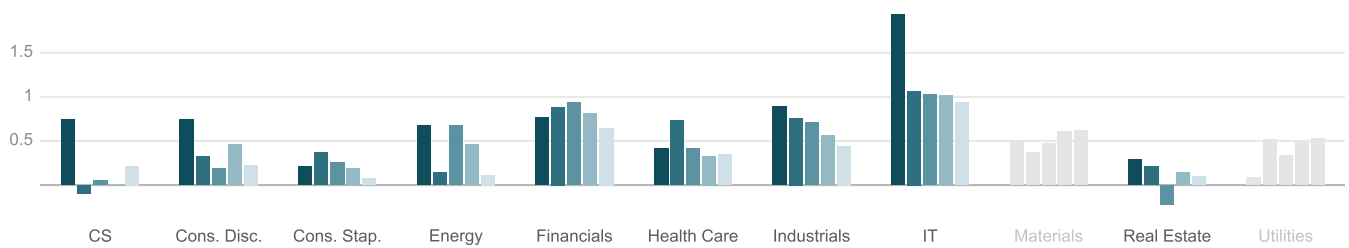
**216 Foreign Institution Ownership**

Price Momentum L-S CAGR +3.3%



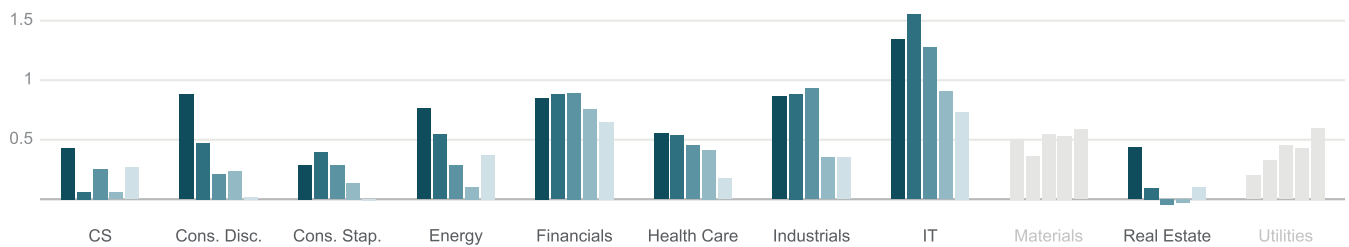
217 ● 3Y Chg in Asset Adjusted Free Cash Flow

Historical Growth L-S CAGR +3.2%



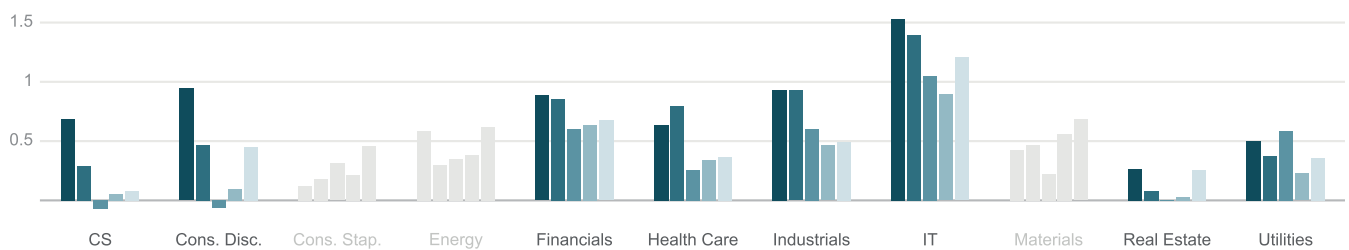
218 ● Free Cash Flow to Common Equity

Valuation L-S CAGR +3.2%



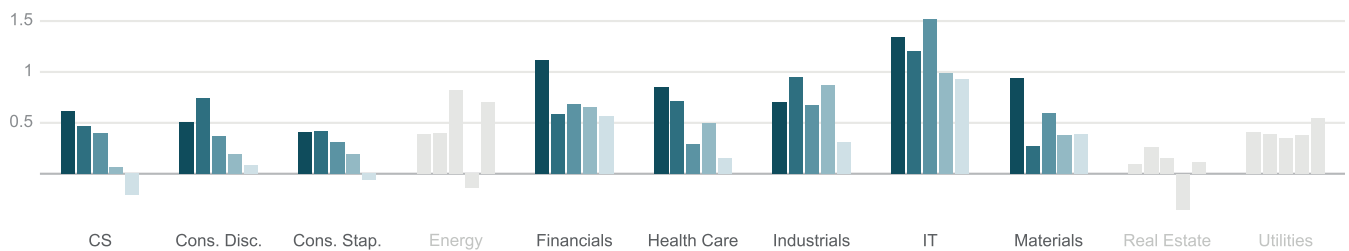
219 ● Sustainable Growth Rate

Historical Growth L-S CAGR +3.1%



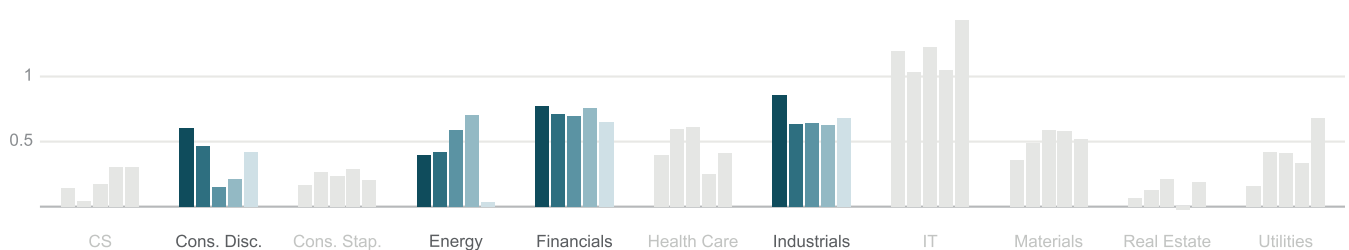
220 ● Adjusted Number of EPS FY1 Revisions

Analyst Expectations L-S CAGR +3.1%



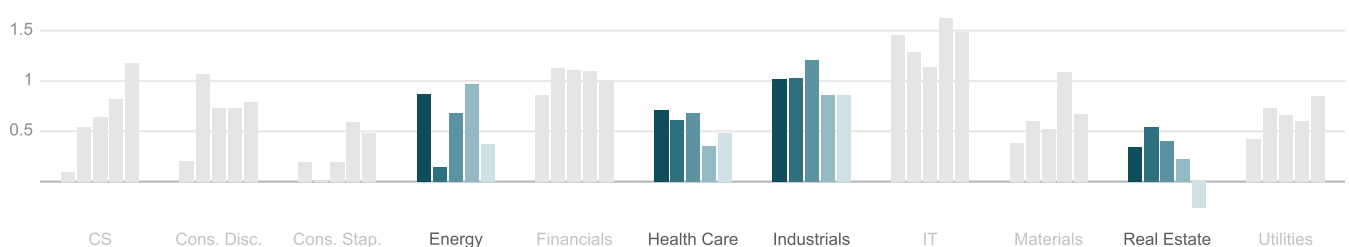
221 ● 1Y Chg in Debt to Assets

Volatility L-S CAGR +3.0%



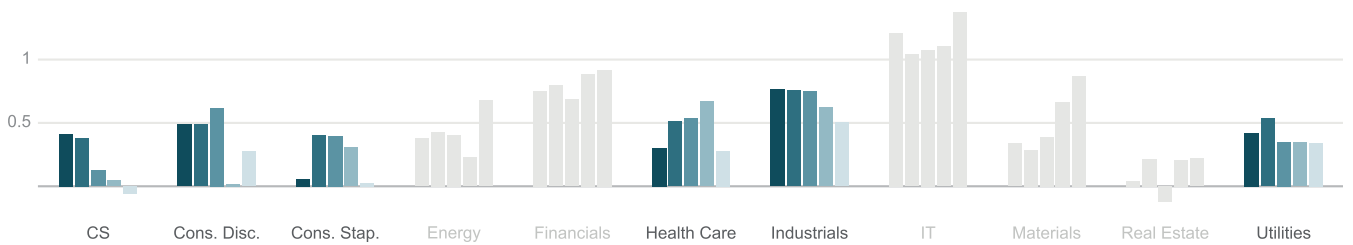
222 ● Standardized 36M Residual

Price Momentum L-S CAGR +2.9%



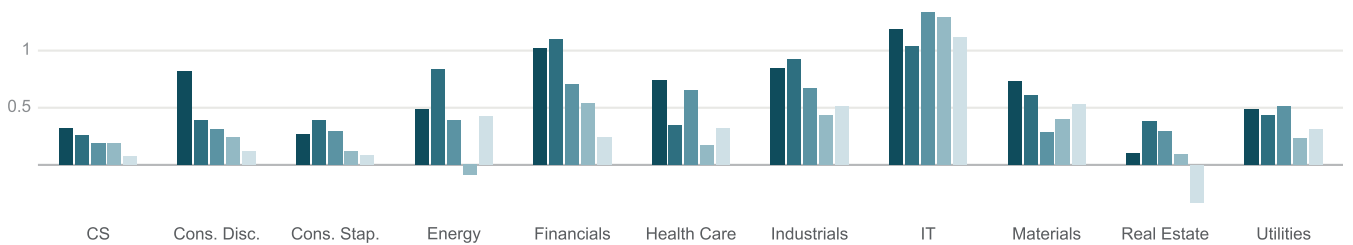
223 ● 1Y Chg in Cash Flow per Share

Historical Growth L-S CAGR +2.8%



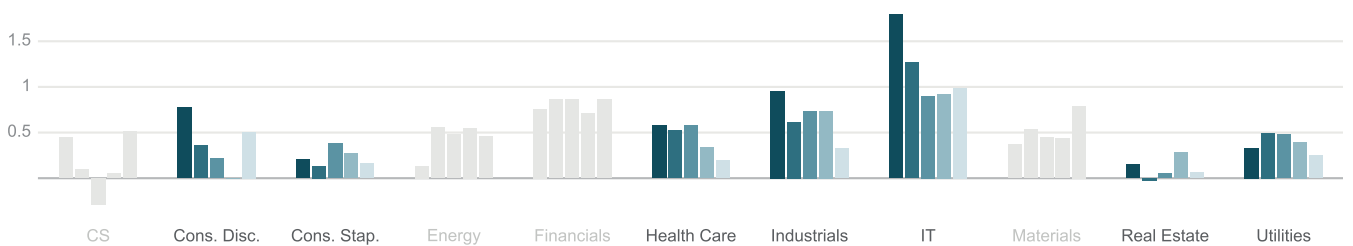
224 ● Target Price Gap to 6M EMA

Analyst Expectations L-S CAGR +2.8%



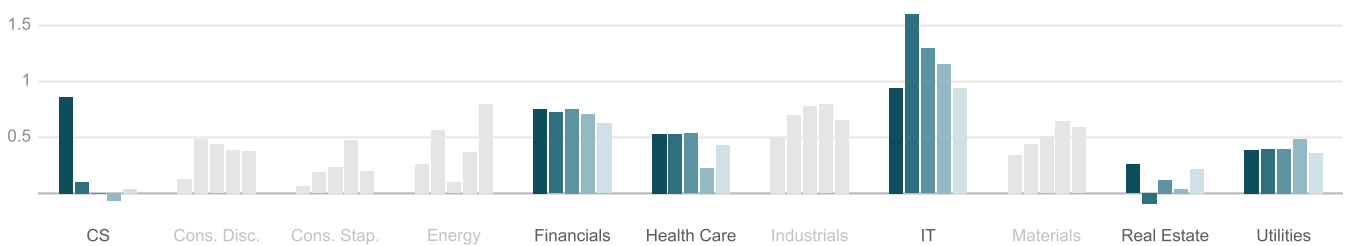
225 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth L-S CAGR +2.8%



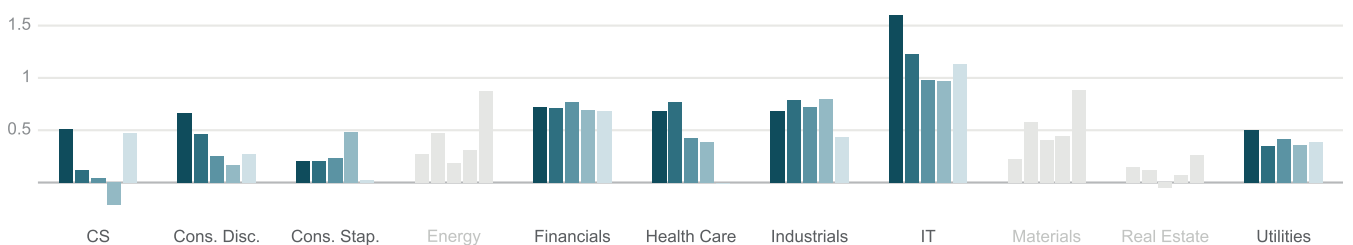
226 ● Net Current Assets to Price Ratio

Valuation L-S CAGR +2.7%



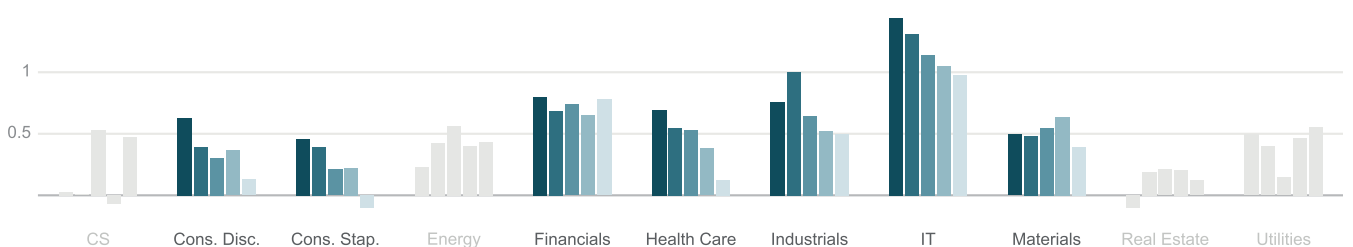
227 ● Pretax Income to Net Operating Assets

Earnings Quality L-S CAGR +2.7%



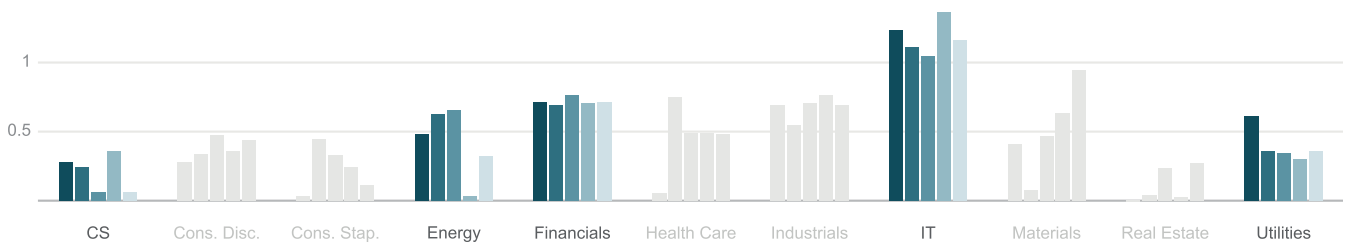
228 ● EBIT Estimate Dispersion

Analyst Expectations L-S CAGR +2.7%



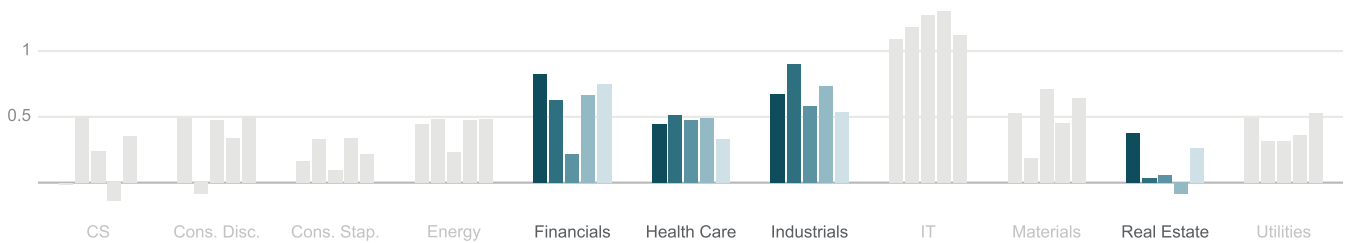
229 ● 1Y Chg in EPS to Operating Cash Flow

Earnings Quality L-S CAGR +2.6%



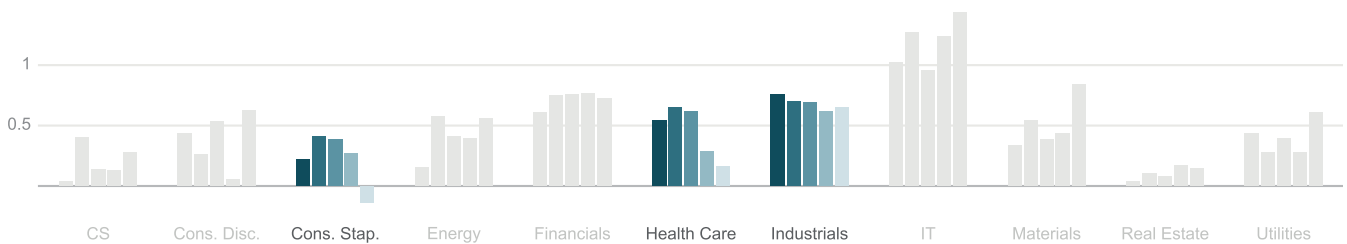
230 ● Working Capital Turnover Ratio

Capital Efficiency L-S CAGR +2.6%



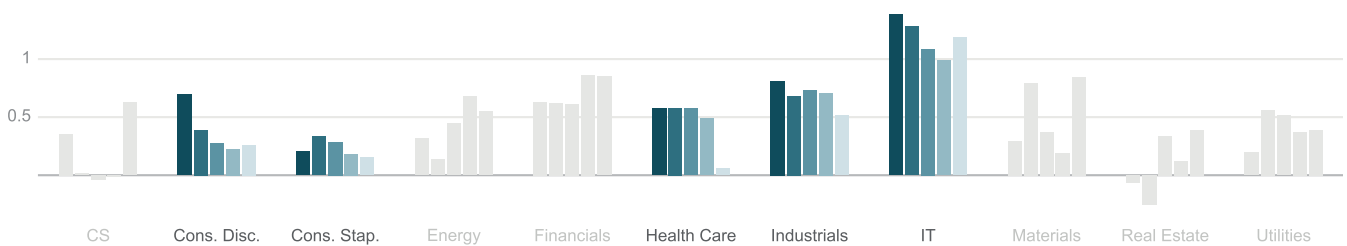
231 ● Change of Sales to Change of EPS

Earnings Quality L-S CAGR +2.5%



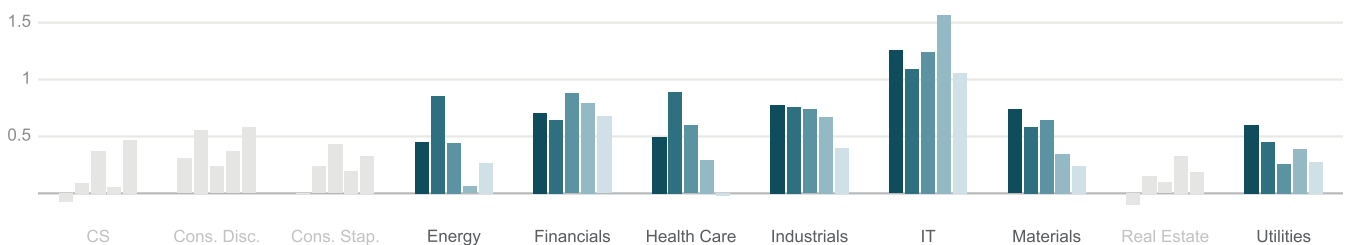
232 ● Pretax Profit Margin

Capital Efficiency L-S CAGR +2.5%



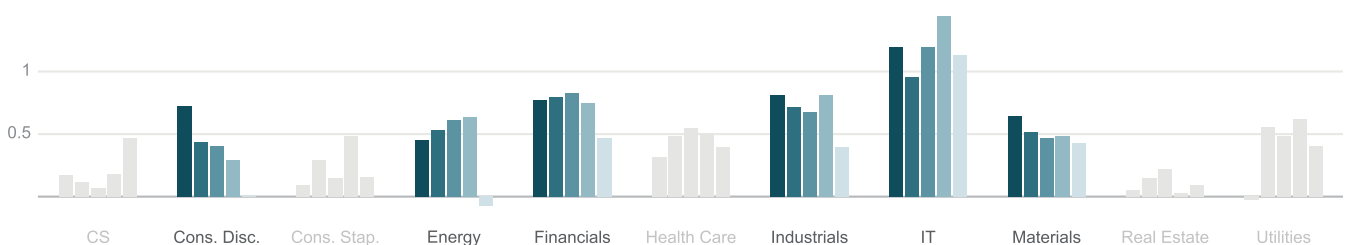
233 ● Sales Acceleration

Historical Growth L-S CAGR +2.4%



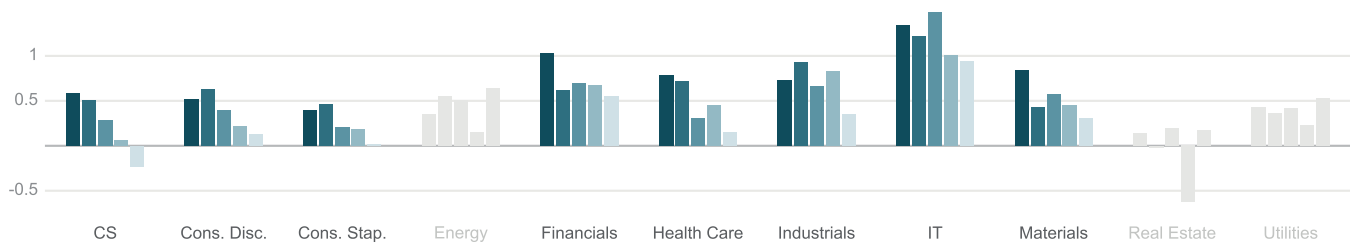
234 ● Risk Adjusted Relative Strength

Price Momentum L-S CAGR +2.4%



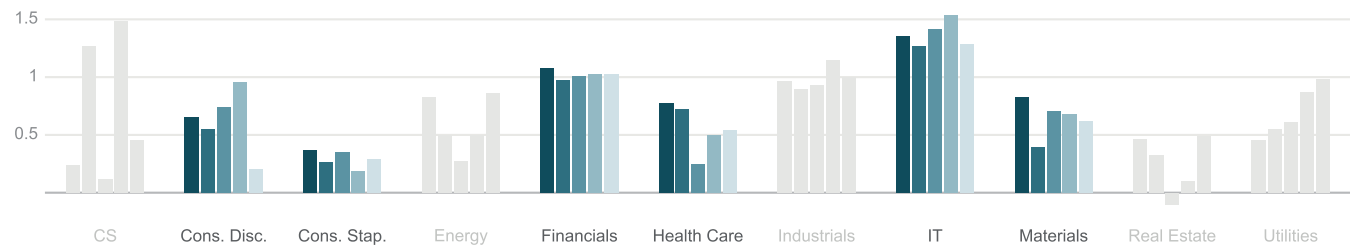
235 ● Analyst Earnings Estimate Diffusion

Analyst Expectations L-S CAGR +2.4%



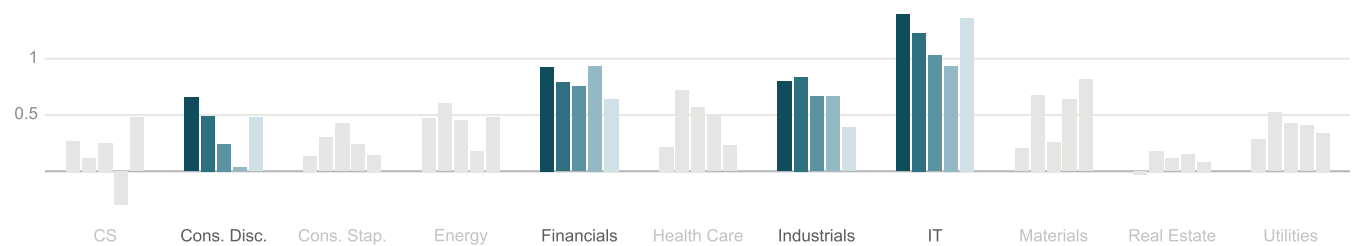
236 ● Change of Ownership Level - Hedge Fund

Price Momentum L-S CAGR +2.4%



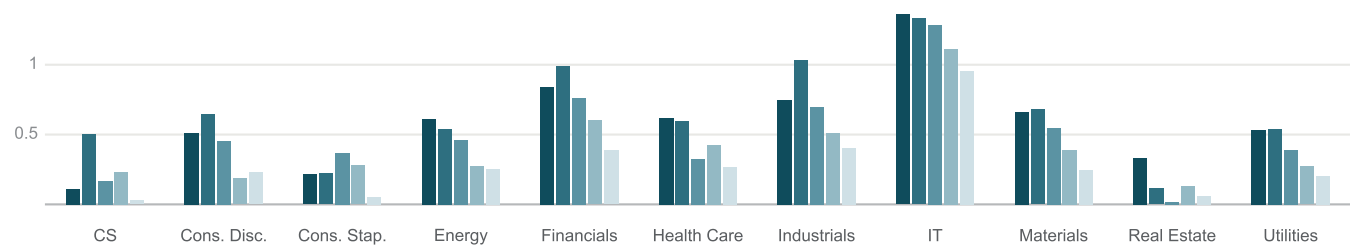
237 ● 3Y Chg in Cash Flow per Share

Historical Growth L-S CAGR +2.3%



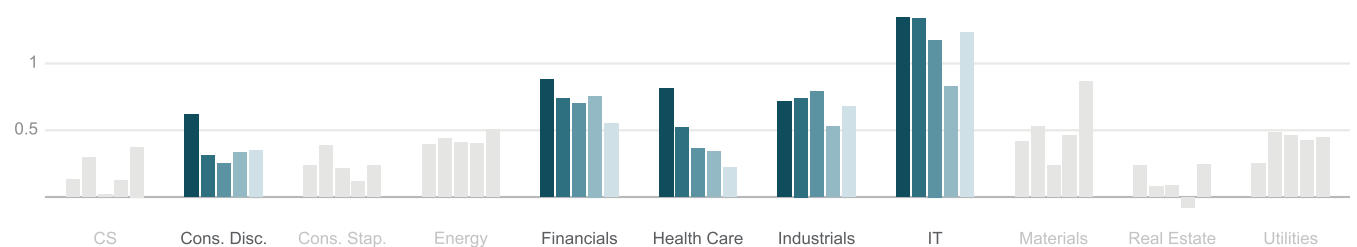
238 ● Investment Duration

Price Momentum L-S CAGR +2.2%



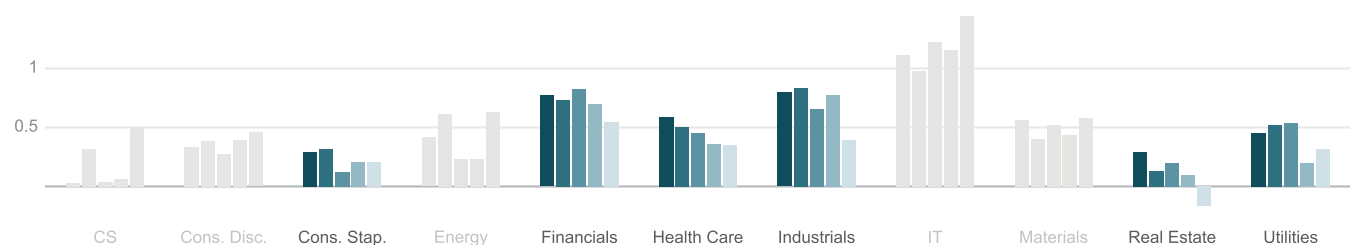
239 ● Normalized ROE

Capital Efficiency L-S CAGR +2.1%



240 ● Cash to Sales

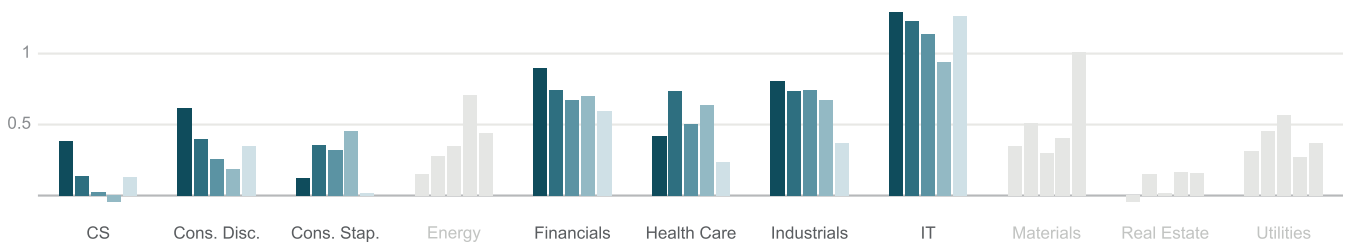
Earnings Quality L-S CAGR +2.1%



241 ● 3Y Chg in Earnings per Share

Historical Growth

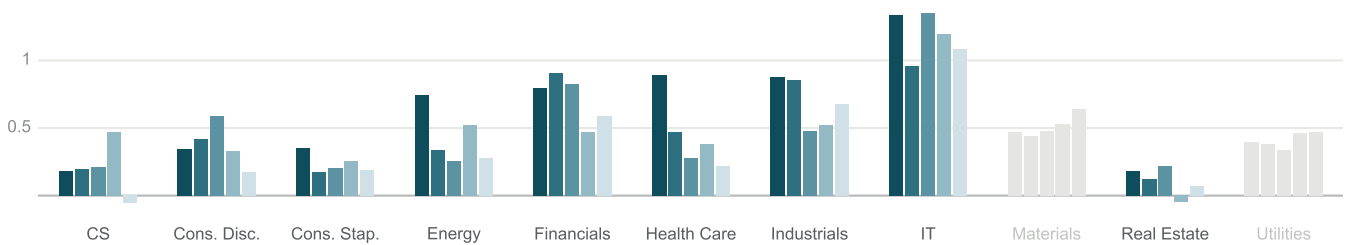
L-S CAGR +2.0%



242 ● Institution Ownership Turnover

Price Momentum

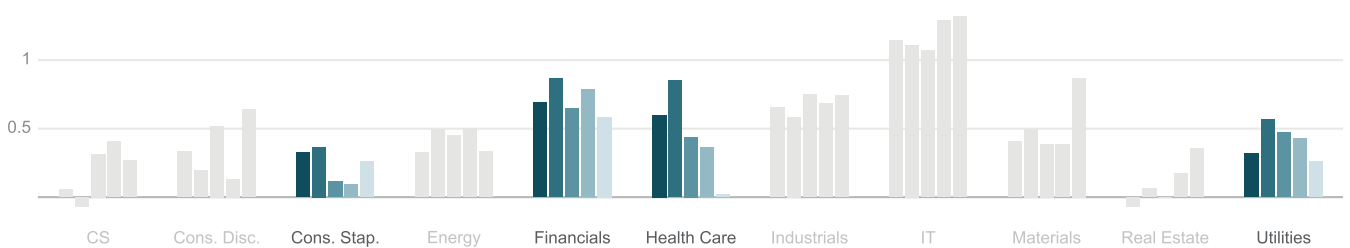
L-S CAGR +2.0%



243 ● Max Daily Return in the Past 6 Months

Price Momentum

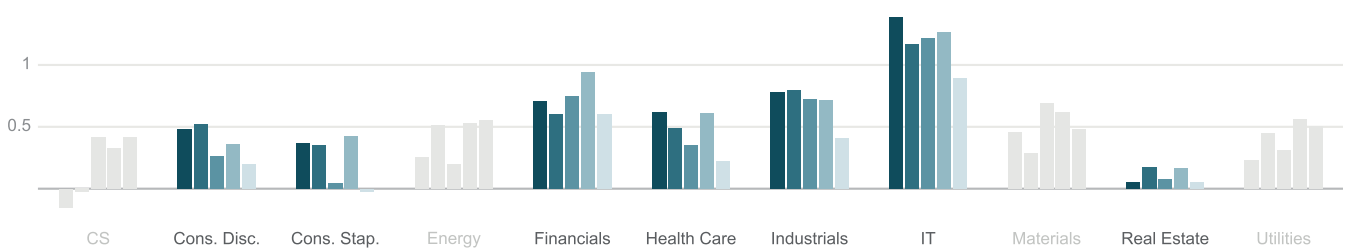
L-S CAGR +2.0%



244 ● Revenue Estimate Dispersion

Analyst Expectations

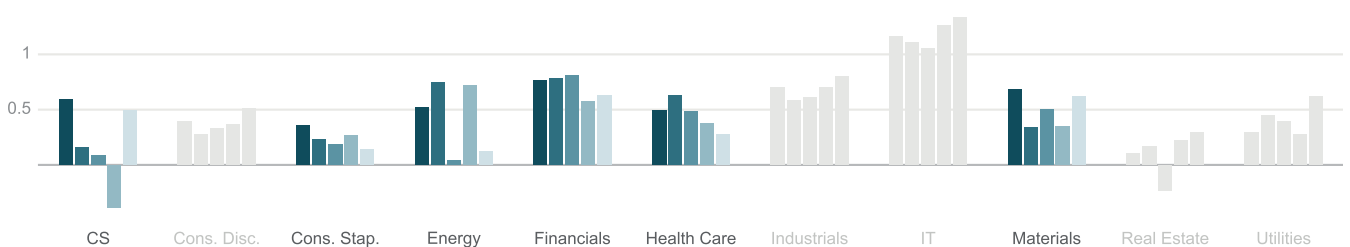
L-S CAGR +1.9%



245 ● Retained Earnings to Total Assets

Earnings Quality

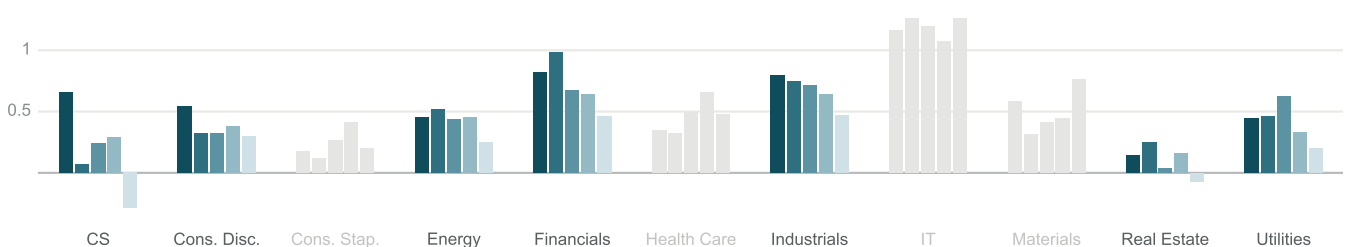
L-S CAGR +1.9%



246 ● Buy-to-Sell Recommendation Ratio Less 3M SMA

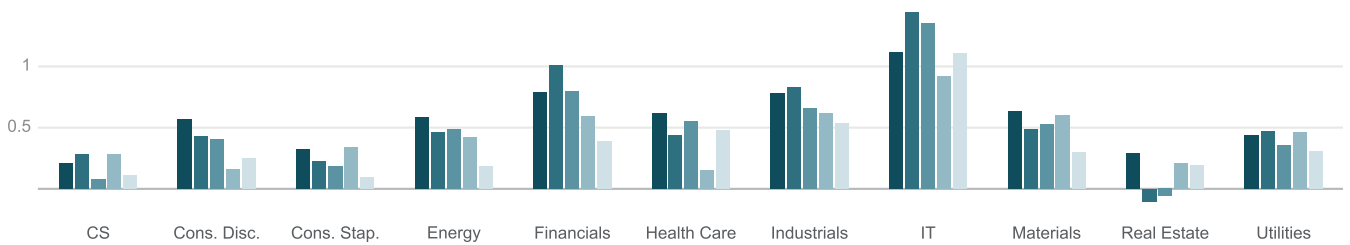
Analyst Expectations

L-S CAGR +1.9%

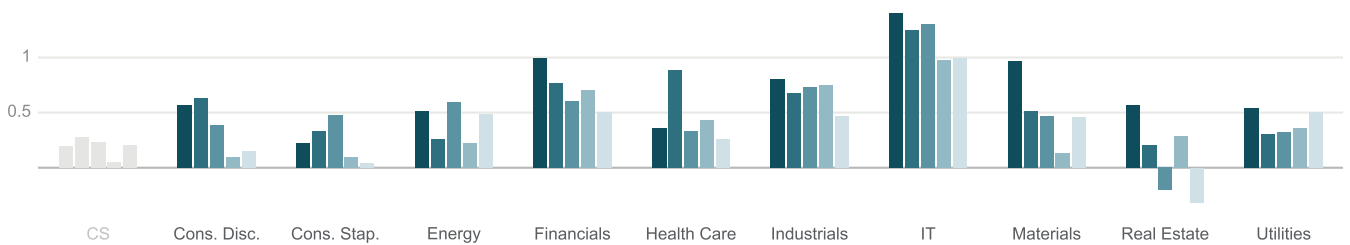


247 ● **Institution Ownership Concentration**

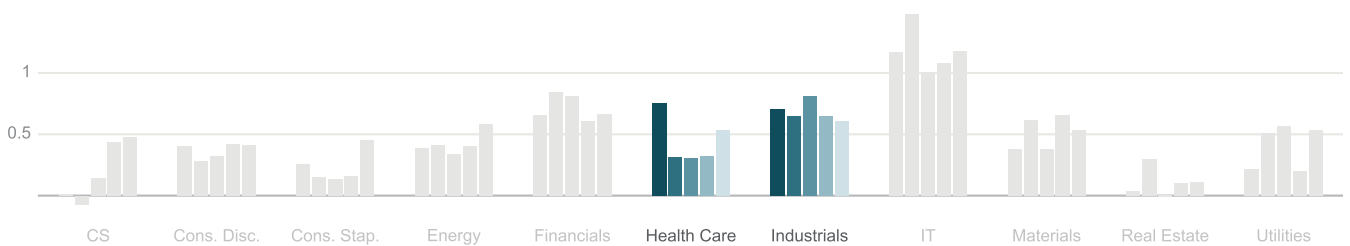
Price Momentum L-S CAGR +1.8%

248 ● **5 Day Price Reversal**

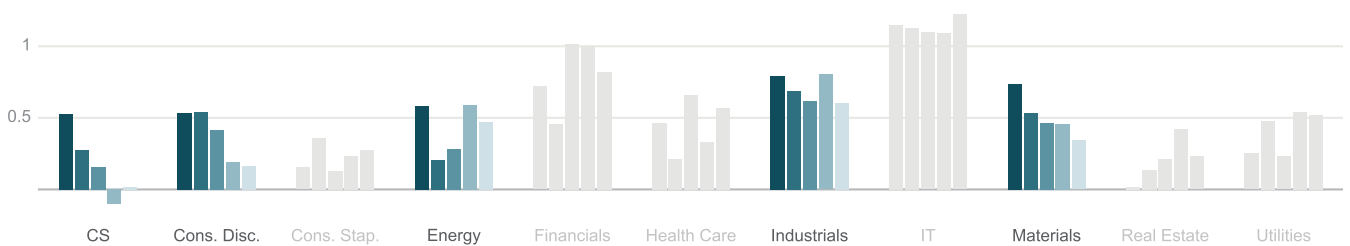
Price Momentum L-S CAGR +1.7%

249 ● **6M Chg in 18M CAPM Alpha**

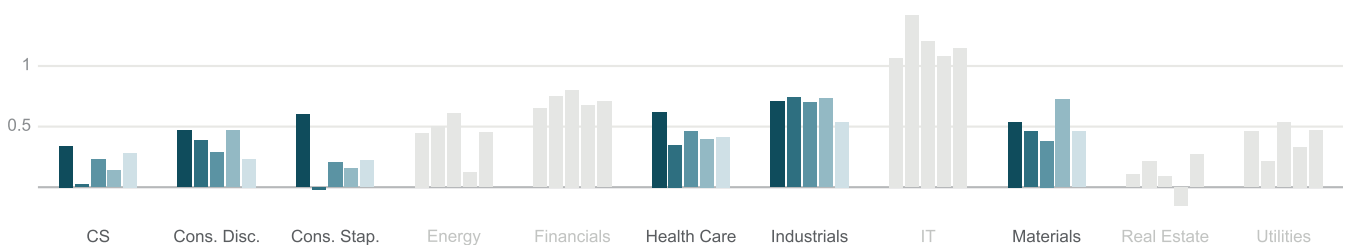
Price Momentum L-S CAGR +1.6%

250 ● **Year over Year Change of Inventory to Assets**

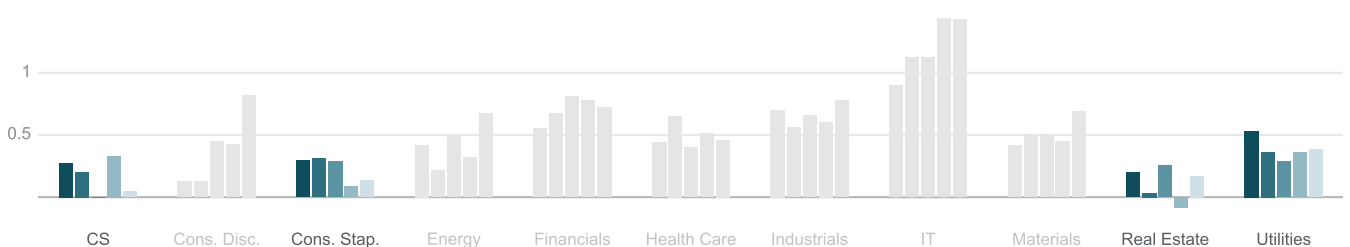
Earnings Quality L-S CAGR +1.6%

251 ● **6M Chg in 12M CAPM Alpha**

Price Momentum L-S CAGR +1.4%

252 ● **ROE 20 Qtr Standard Deviation**

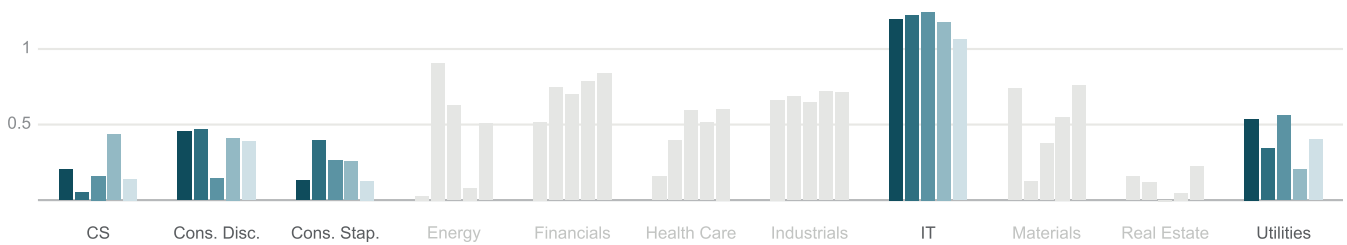
Earnings Quality L-S CAGR +1.4%



253 ● Change of Ownership Level - All Managers

Price Momentum

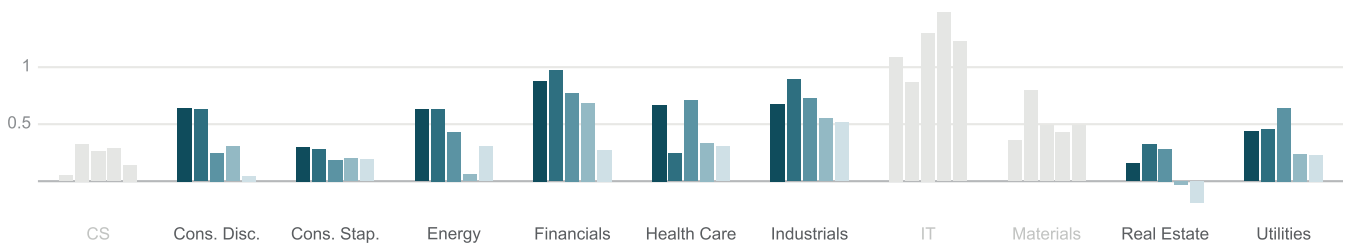
L-S CAGR +1.3%



254 ● 6M Chg in Target Price Gap

Analyst Expectations

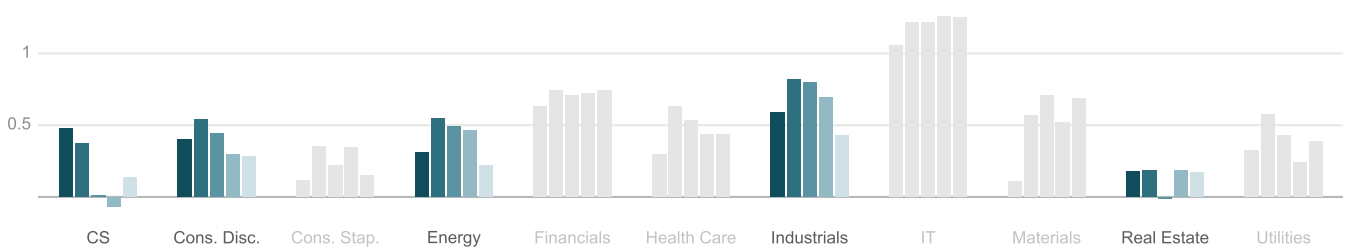
L-S CAGR +1.2%



255 ● 3Y Chg in Operating Margin

Historical Growth

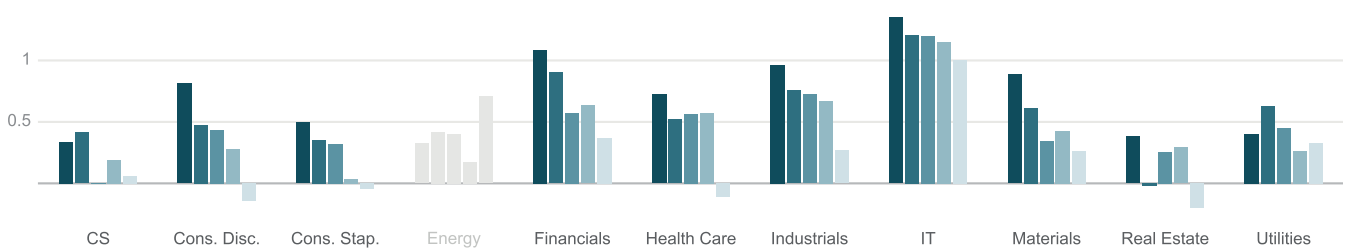
L-S CAGR +1.1%



256 ● 8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

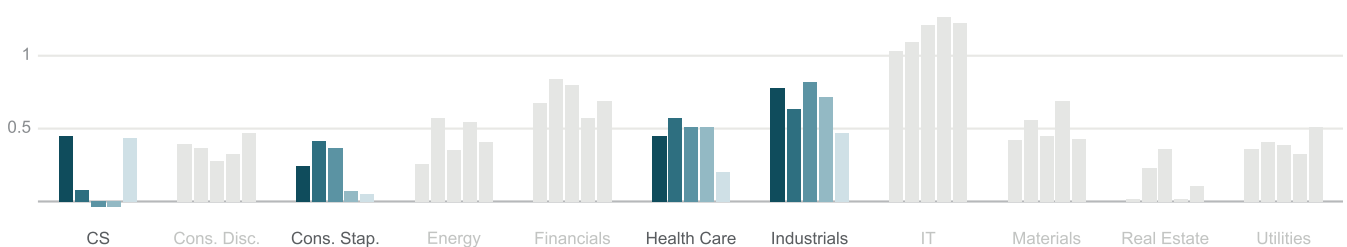
L-S CAGR +1.0%



257 ● Change in TTM Depr. to Capex

Earnings Quality

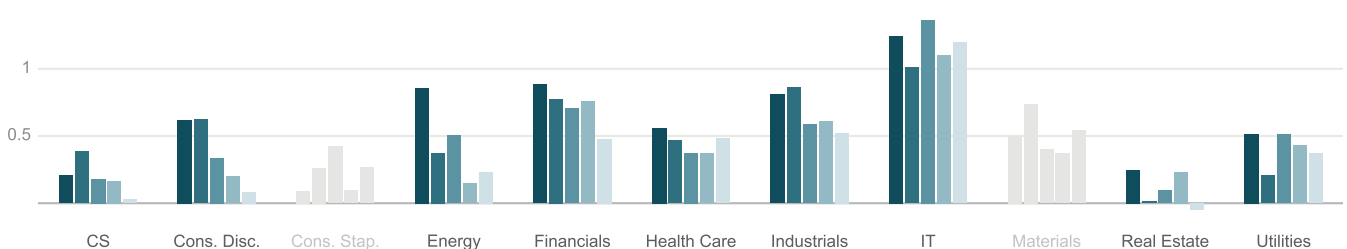
L-S CAGR +0.9%



258 ● 6M Avg Chg 1M Recommendation

Analyst Expectations

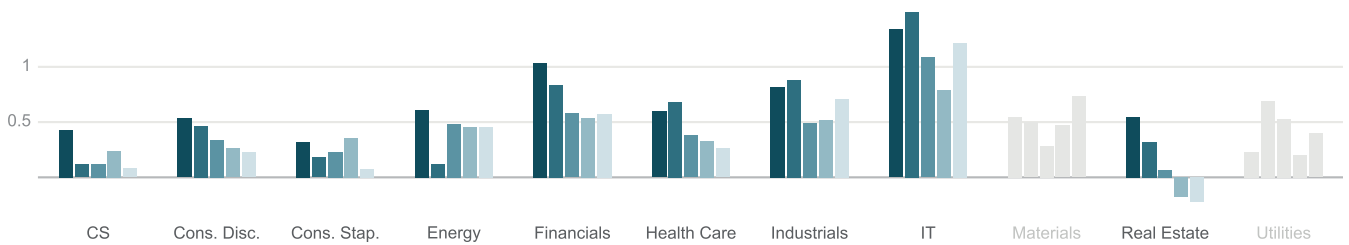
L-S CAGR +0.7%



259 ● 14-Day Relative Strength Index

Price Momentum

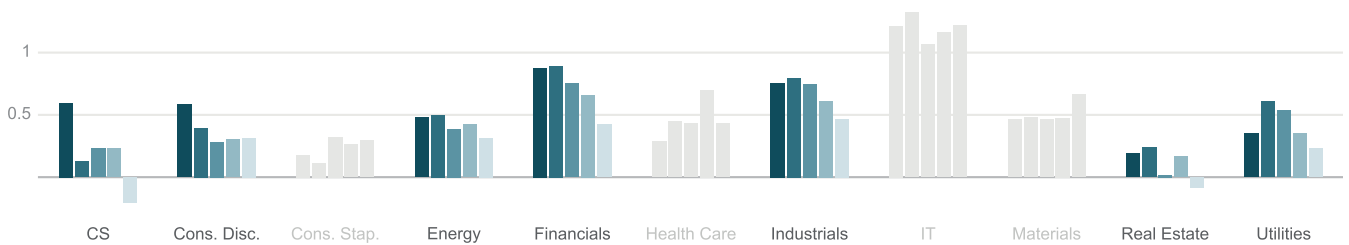
L-S CAGR +0.5%



260 ● Buy-to-Sell Recommendation Ratio Less 3M EMA

Analyst Expectations

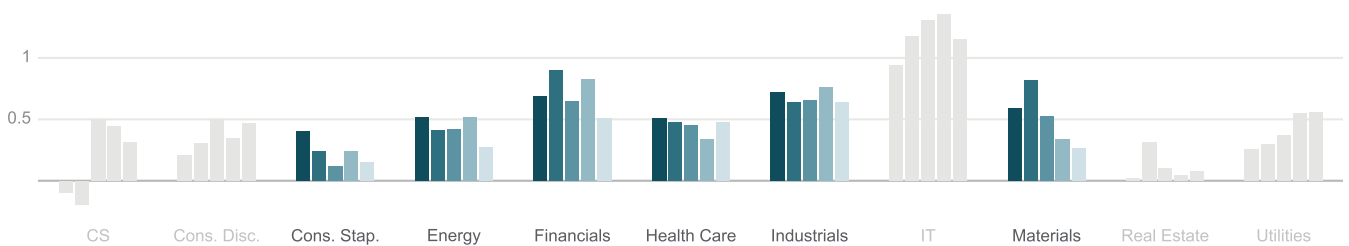
L-S CAGR +0.3%



261 ● 1Y Chg in Share Turnover

Price Momentum

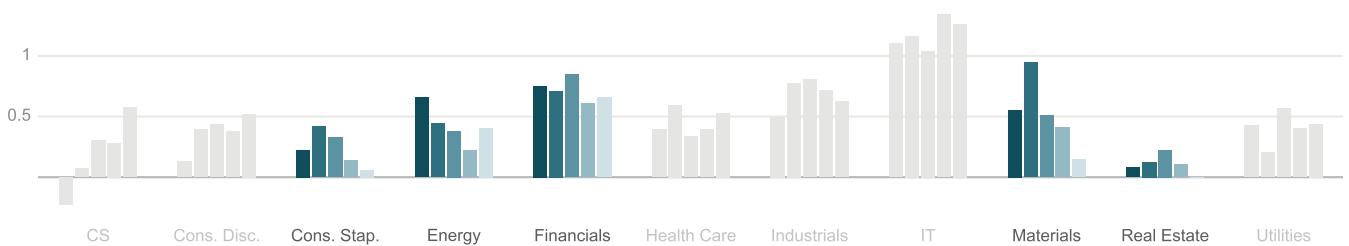
L-S CAGR +0.2%



262 ● 50 Day Volume Signal

Price Momentum

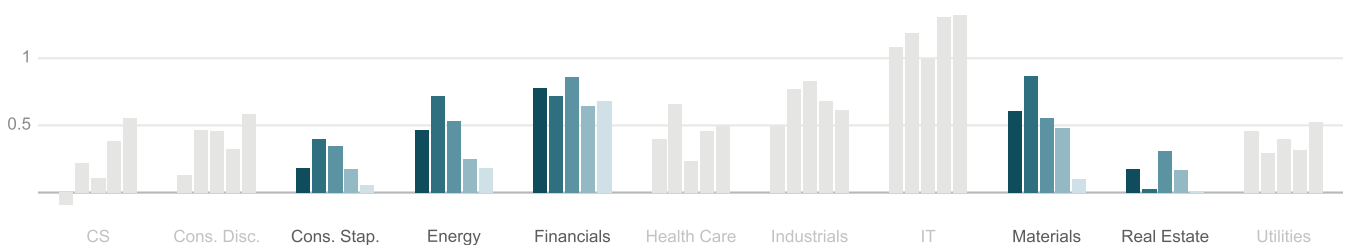
L-S CAGR -0.6%



263 ● Adj 50 Day Volume Signal

Volatility

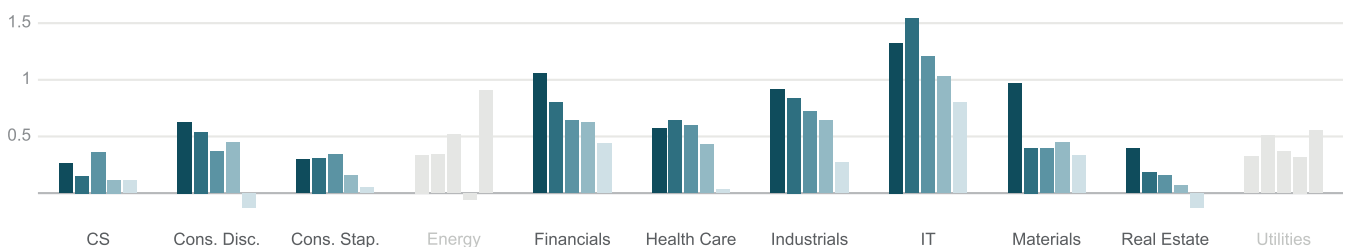
L-S CAGR -0.6%



264 ● 4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price

Valuation

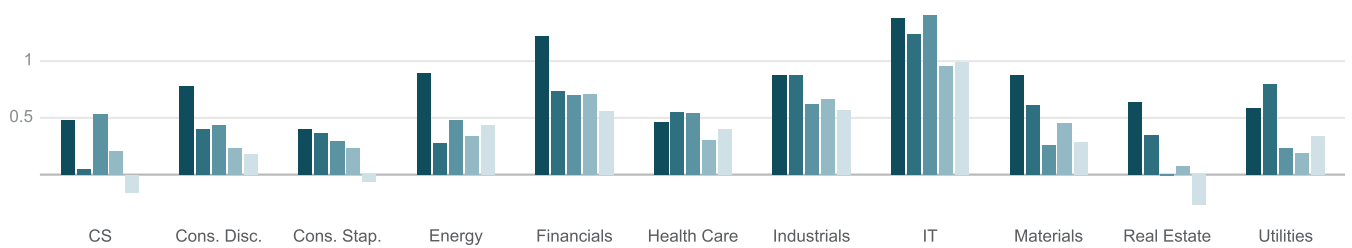
L-S CAGR -0.7%



265 ● 5-Day Money Flow/Volume

Price Momentum

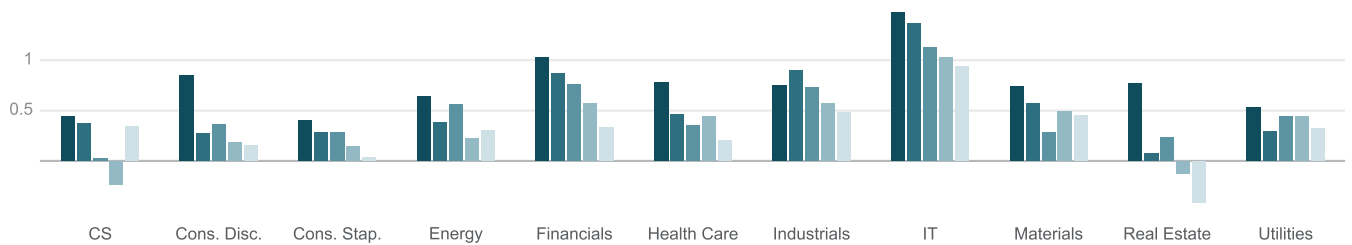
L-S CAGR -1.1%



266 ● 20-Day Lane's Stochastic Indicator

Price Momentum

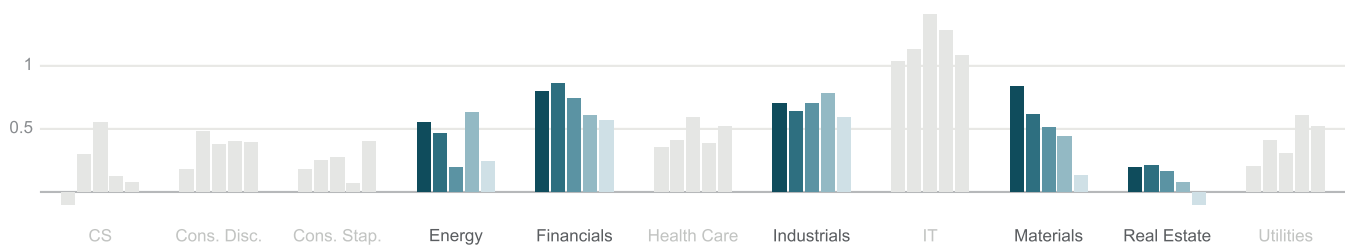
L-S CAGR -1.4%



267 ● 5 Day Volume Signal

Price Momentum

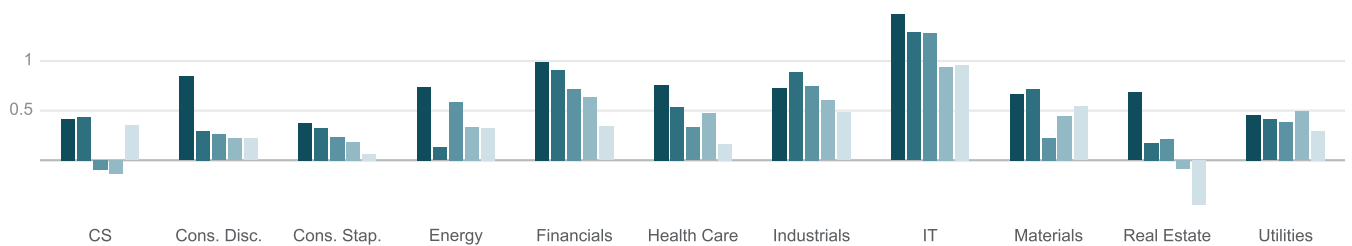
L-S CAGR -1.9%



268 ● 1M Price High - 1M Price Low

Price Momentum

L-S CAGR -1.9%



269 ● 1M Price Reversal

Price Momentum

L-S CAGR -4.0%

